al-Battānī, *al-Zīj al-Ṣābi'*

The tables of Part II of Nallino's edition, in English

C. A. Nallino prints al-Battānī's tables in Part II of *Al-Battānī sive Albatenii opus astronomicum* (Milan, 1907) from the Escorial codex, with the Arabic heads of the codex and his Latin translation of them. The tables that Part III also prints in Arabic (the chronological and the geographical tables, the circular diagrams, the star catalogue and the tables of the stars for the year 1211) are translated from the Arabic in the English *Zīj*; the spurious tables of pp. 297–307 are in English with Nallino's notes on them. This volume gives the other tables of Part II as the transcription of this repository reproduces them (S06), with English in place of Nallino's Latin. The numbers, the Arabic, the folio marks of the codex and Nallino's page numbers stand as printed; his square brackets mark what he supplies. The terms are those of the English *Zīj* and its glossary.

The weekday signs of the years and months	pp. 7–8
The concordance of the Hijra with the Seleucid era	pp. 9–18
Mean motions in the Arab and the Roman calendar	pp. 19–28, 72–77, 102–107
Mean syzygies	pp. 29–32, 84–87
Sines, declination, ascensions, the longest day, shadows and the equation of days	pp. 55–71
Equations of the Sun and the Moon, and the latitude of the Moon	pp. 78–83
Tables for computing eclipses	pp. 88–91
Parallaxes	pp. 93–101
Equations of the five planets	pp. 108–137
Stations of the planets	pp. 138–139
Latitudes of the planets	pp. 140–141
Elongations for the apparitions and occultations of the planets	pp. 142–143
Revolutions of the years, altitudes and directions of the hours	pp. 187–188

The English: AI-translated (Claude), 2026. No person has proofread it. Dedicated to the public domain (CC0 1.0). Nallino's edition is in the public domain.

f. 157,v.

جدول استخراج اوائل سني العرب وشهورهم بعلامات مختلفة ومعنى واحد

Table for finding the beginnings of the Arab years and months by different signs, which [nevertheless] indicate one and the same thing.

المجموعة Collected years [from the era].		المبسوطة Single years.			المجموعة Collected years [from the era].		المبسوطة Single years.		
عدد السنين Number of the years.	العلامات Signs.	عدد السنين No. of years.	العلامات Signs.		عدد السنين Number of the years.	العلامات Signs.	عدد السنين No. of years.	العلامات Signs.	الكبائس Leap years.
30	5	1	5		30	5	1	7	
60	3	2	2		60	3	2	4	B
90	1	3	7		90	1	3	2	
120	6	4	4		120	6	4	6	
150	4	5	1		150	4	5	3	B
180	2	6	6		180	2	6	1	
210	7	7	3		210	7	7	5	B
		8	1				8	3	
		9	5				9	7	
		10	2				10	4	B
الشهور العربية Months of the Arabs.		11	7		الشهور العربية Months of the Arabs.		11	2	
		12	4				12	6	
		13	1				13	3	B
		14	6				14	1	
		15	3				15	5	
اسماء الشهور Names of the months.	علاماتها Their signs.	16	7	عدد ايام الشهور Number of the days of the month.	اسماء الشهور Names of the months.	علاماتها Their signs.	16	2	B
		17	5				17	7	
		18	2				18	4	B
al–muḥarram	0	19	7	30	al–muḥarram	5	19	2	
ṣafar	2	20	4	29	ṣafar	7	20	6	
rabīʿ I	3	21	1	30	rabīʿ I	1	21	3	B
rabīʿ II	5	22	6	29	rabīʿ II	3	22	1	
ǵumādà I	6	23	3	30	ǵumādà I	4	23	5	
ǵumādà II	1	24	7	29	ǵumādà II	6	24	2	B
raǵab	2	25	5	30	raǵab	7	25	7	
shaʿbān	4	26	2	29	shaʿbān	2	26	4	B
ramaḍān	5	27	7	30	ramaḍān	3	27	2	
shawwāl	0	28	4	29	shawwāl	5	28	6	
dhū ’l–qaʿdah	1	29	1	30	dhū ’l–qaʿdah	6	29	3	B
dhū ’l–ḥiǵǵah	3	30	6	29	dhū ’l–ḥiǵǵah	1	30	1	

f. 158,r.

علامات شهور الروم بتاريخ ذي القرنين المقسوم على كح بزيادة سنة **The signs of the Roman months according to the era of Dhū 'l-qarnayn, which is to be divided by 28 and increased by one year.**													
سنو الروم البسوطة Single years of the Romans.	سبتمبر September. Aylūl. 30 d.	اكتوبر October. Tishrīn I. 31 d.	نوفمبر November. Tishrīn II. 30 d.	دجنبر December. Kānūn I. 31 d.	يناير January. Kānūn II. 30 d.	فبراير February. Subāṭ. 28 d.	سطور الكبائس Leap years.	مارس March. Ādhār. 31 d.	ابريل April. Nīsān. 30 d.	مايه May. Ayyār. 31 d.	يونيه June. Hazirān. 30 d.	يليه July. Tammūz. 31 d.	اغشت August. Āb. 31 d.
1	6	1	4	6	2	5	15	5	1	3	6	1	4
2	7	2	5	7	3	6	30	6	2	4	7	2	5
3	1	3	6	1	4	7	45	7	3	5	1	3	6
4	2	4	7	2	5	1	B	2	5	7	3	5	1
5	4	6	2	4	7	3	15	3	6	1	4	6	2
6	5	7	3	5	1	4	30	4	7	2	5	7	3
7	6	1	4	6	2	5	45	5	1	3	6	1	4
8	7	2	5	7	3	6	B	7	3	5	1	3	6
9	2	4	7	2	5	1	15	1	4	6	2	4	7
10	3	5	1	3	6	2	30	2	5	7	3	5	1
11	4	6	2	4	7	3	45	3	6	1	4	6	2
12	5	7	3	5	1	4	B	5	1	3	6	1	4
13	7	2	5	7	3	6	15	6	2	4	7	2	5
14	1	3	6	1	4	7	30	7	3	5	1	3	6
15	2	4	7	2	5	1	45	1	4	6	2	4	7
16	3	5	1	3	6	2	B	3	6	1	4	6	2
17	5	7	3	5	1	4	15	4	7	2	5	7	3
18	6	1	4	6	2	5	30	5	1	3	6	1	4
19	7	2	5	7	3	6	45	6	2	4	7	2	5
20	1	3	6	1	4	7	B	1	4	6	2	4	7
21	3	5	1	3	6	2	15	2	5	7	3	5	1
22	4	6	2	4	7	3	30	3	6	1	4	6	2
23	5	7	3	5	1	4	45	4	7	2	5	7	3
24	6	1	4	6	2	5	B	6	2	4	7	2	5
25	1	3	6	1	4	7	15	7	3	5	1	3	6
26	2	4	7	2	5	1	30	1	4	6	2	4	7
27	3	5	1	3	6	2	45	2	5	7	3	5	1
28	4	6	2	4	7	3	B	4	7	2	5	7	3

f. 158,v.

الجدول الاول من استخراج تاريخ العرب من تاريخ الروم وتاريخ الروم من تاريخ العرب

Table I for finding the era of the Arabs from the era of the Romans and that of the Romans from that of the Arabs.

سنو الهجرة Years of the Hijra.	اسم اول يوم من محرم Name of the first day of the month muḥarram.	سنو ذي القرنين Years of the era of Dhū 'l-qarnayn.	الايام الماضية من الشهر الرومي الذي يقع اول المحرم فيه Days elapsed of the Roman month in which the beginning of muḥarram falls.	Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.
1	5	933	15 tammūz	31	3	962	23 āb
2	2	934	4 »	32	7	963	11 »
3	7	935	23 ḥazīrān	33	5	964	1 »
4	4	936	12 »	34	2	965	21 tammūz
5	1	937	1 »	35	6	966	10 »
6	6	938	22 ayyār	36	4	967	29 ḥazīrān
7	3	939	10 »	37	1	968	18 »
8	1	940	30 nīsān	38	6	969	8 »
9	5	941	19 »	39	3	970	28 ayyār
10	2	942	8 »	40	7	971	16 »
11	7	943	28 ādhār	41	5	972	6 »
12	4	944	17 »	42	2	973	25 nīsān
13	1	945	6 »	43	6	974	14 »
14	6	946	24 subāṭ	44	4	975	3 »
15	3	947	13 »	45	1	976	23 ādhār
16	7	948	1 »	46	5	977	12 »
17	5	949	22 kānūn II	47	3	978	2 »
18	2	950	11 »	48	7	979	19 subāṭ
19	7	951	1 »	49	5	980	8 »
20	4	952	20 kānūn I	50	2	981	28 kānūn II
21	1	953	9 »	51	6	982	17 »
22	6	954	29 tishrīn II	52	4	983	7 »
23	3	955	18 »	53	1	984	26 kānūn I
24	7	956	6 »	54	5	985	15 »
25	5	957	27 tishrīn I	55	3	986	5 »
26	2	958	16 »	56	7	987	24 tishrīn II
27	7	959	6 »	57	5	988	13 »
28	4	960	24 aylūl	58	2	989	2 »
29	1	961	13 »	59	6	990	22 tishrīn I
30	6	962	3 »	60	4	991	12 »

f. 159,r.

الجدول الثاني من استخراج تاريخ العرب من تاريخ الروم وتاريخ الروم من تاريخ العرب

Table II for finding the era of the Arabs from the era of the Romans and that of the Romans from that of the Arabs.

Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.	Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls
61	1	992	31 aylūl	91	6	1021	8 tishrīn II
62	5	993	19 »	92	3	1022	28 tishrīn I
63	3	994	9 »	93	1	1023	18 »
64	7	994	29 āb	94	5	1024	6 »
65	4	995	17 »	95	2	1025	25 aylūl
66	2	996	7 »	96	7	1026	15 »
67	6	997	27 tammūz	97	4	1027	4 »
68	4	998	17 »	98	2	1027	24 āb
69	1	999	5 »	99	6	1028	13 »
70	5	1000	24 ḥazīrān	100	3	1029	2 »
71	3	1001	14 »	101	1	1030	23 tammūz
72	7	1002	3 »	102	5	1031	11 »
73	4	1003	22 ayyār	103	2	1032	30 ḥazīrān
74	2	1004	12 »	104	7	1033	20 »
75	6	1005	1 »	105	4	1034	9 »
76	3	1006	20 nīsān	106	1	1035	28 ayyār
77	1	1007	9 »	107	6	1036	18 »
78	5	1008	29 ādhār	108	3	1037	7 »
79	3	1009	19 »	109	1	1038	27 nīsān
80	7	1010	8 »	110	5	1039	15 »
81	4	1011	25 subāṭ	111	2	1040	4 »
82	2	1012	14 »	112	7	1041	25 ādhār
83	6	1013	3 »	113	4	1042	14 »
84	3	1014	23 kānūn II	114	1	1043	2 »
85	1	1015	13 »	115	6	1044	20 subāṭ
86	5	1016	1 »	116	3	1045	9 »
87	3	1017	22 kānūn I	117	1	1046	30 kānūn II
88	7	1018	11 »	118	5	1047	19 »
89	4	1019	30 tishrīn II	119	2	1048	7 »
90	2	1020	19 »	120	7	1049	28 kānūn I

f. 159,v.

الجدول الثالث من استخراج تاريخ العرب من تاريخ الروم وتاريخ الروم من تاريخ العرب

Table III for finding the era of the Arabs from the era of the Romans and that of the Romans from that of the Arabs.

Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.	Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.
121	4	1050	17 kānūn I	151	2	1079	25 kānūn II
122	1	1051	6 »	152	6	1080	13 »
123	6	1052	25 tishrīn II	153	4	1081	3 »
124	3	1053	14 »	154	1	1082	23 kānūn I
125	7	1054	3 »	155	5	1083	12 »
126	5	1055	24 tishrīn I	156	3	1084	1 »
127	2	1056	12 »	157	7	1085	20 tishrīn II
128	7	1057	2 »	158	5	1086	10 »
129	4	1058	21 aylūl	159	2	1087	30 tishrīn I
130	1	1059	10 »	160	6	1088	18 »
131	6	1059	30 āb	161	4	1089	8 »
132	3	1060	19 »	162	1	1090	27 aylūl
133	7	1061	8 »	163	5	1091	16 »
134	5	1062	29 tammūz	164	3	1092	5 »
135	2	1063	17 »	165	7	1092	25 āb
136	6	1064	6 »	166	4	1093	14 »
137	4	1065	26 ḥazīrān	167	2	1094	4 »
138	1	1066	15 »	168	6	1095	23 tammūz
139	6	1067	4 »	169	4	1096	13 »
140	3	1068	24 ayyār	170	1	1097	2 »
141	7	1069	13 »	171	5	1098	21 ḥazīrān
142	5	1070	3 »	172	3	1099	10 »
143	2	1071	21 nīsān	173	7	1100	30 ayyār
144	6	1072	10 »	174	4	1101	19 »
145	5	1073	31 ādhār	175	2	1102	9 »
146	1	1074	20 »	176	6	1103	27 nīsān
147	6	1075	9 »	177	4	1104	17 »
148	3	1076	26 subāṭ	178	1	1105	6 »
149	7	1077	15 »	179	5	1106	26 ādhār
150	5	1078	5 »	180	3	1107	15 »

f. 160,r.

الجدول الرابع من استخراج تاريخ العرب من تاريخ الروم وتاريخ الروم من تاريخ العرب

Table IV for finding the era of the Arabs from the era of the Romans and that of the Romans from that of the Arabs.

Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.	Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.
181	7	1108	4 ādhār	211	5	1137	12 nīsān
182	4	1109	21 subāṭ	212	2	1138	1 »
183	2	1110	11 »	213	7	1139	21 ādhār
184	6	1111	31 kānūn II	214	4	1140	10 »
185	3	1112	19 »	215	1	1141	27 subāṭ
186	1	1113	9 »	216	6	1142	17 »
187	5	1114	29 kānūn I	217	3	1143	6 »
188	3	1115	19 »	218	1	1144	26 kānūn II
189	7	1116	7 »	219	5	1145	15 »
190	4	1117	26 tishrīn II	220	2	1146	4 »
191	2	1118	16 »	221	7	1147	25 kānūn I
192	6	1119	5 »	222	4	1148	13 »
193	3	1120	24 tishrīn I	223	1	1149	2 »
194	1	1121	14 »	224	6	1150	22 tishrīn II
195	5	1122	3 »	225	3	1151	11 »
196	2	1123	22 aylūl	226	7	1152	30 tishrīn I
197	7	1124	11 »	227	5	1153	20 »
198	4	1125	31 āb	228	2	1154	9 »
199	2	1125	21 »	229	7	1155	29 aylūl
200	6	1126	10 »	230	4	1156	17 »
201	3	1127	29 tammūz	231	1	1157	6 »
202	1	1128	19 »	232	6	1157	27 āb
203	5	1129	8 »	233	3	1158	16 »
204	2	1130	27 ḥazīrān	234	7	1159	4 »
205	7	1131	16 »	235	5	1160	25 tammūz
206	4	1132	5 »	236	2	1161	14 »
207	2	1133	26 ayyār	237	7	1162	4 »
208	6	1134	15 »	238	4	1163	22 ḥazīrān
209	3	1135	3 »	239	1	1164	11 »
210	1	1136	23 nīsān	240	6	1165	1 »

f. 160,v.

الجدول الخامس من استخراج تاريخ العرب من تاريخ الروم وتاريخ الروم من تاريخ العرب

Table V for finding the era of the Arabs from the era of the Romans and that of the Romans from that of the Arabs.

Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.	Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls
241	3	1166	21 ayyār	271	1	1195	28 ḥazīrān
242	7	1167	9 »	272	5	1196	17 »
243	5	1168	29 nīsān	273	3	1197	7 »
244	2	1169	18 »	274	7	1198	27 ayyār
245	6	1170	7 »	275	4	1199	15 »
246	4	1171	27 ādhār	276	2	1200	5 »
247	1	1172	16 »	277	6	1201	24 nīsān
248	6	1173	6 »	278	4	1202	14 »
249	3	1174	23 subāṭ	279	1	1203	2 »
250	7	1175	12 »	280	5	1204	22 ādhār
251	5	1176	1 »	281	3	1205	12 »
252	2	1177	21 kānūn II	282	7	1206	1 »
253	6	1178	10 »	283	4	1207	18 subāṭ
254	4	1179	31 kānūn I	284	2	1208	7 »
255	1	1180	19 »	285	6	1209	27 kānūn II
256	5	1181	8 »	286	3	1210	16 »
257	3	1182	28 tishrīn II	287	1	1211	6 »
258	7	1183	17 »	288	5	1212	25 kānūn I
259	5	1184	6 »	289	3	1213	15 »
260	2	1185	26 tishrīn I	290	7	1214	4 »
261	6	1186	15 »	291	4	1215	23 tishrīn II
262	4	1187	5 »	292	2	1216	12 »
263	1	1188	23 aylūl	293	6	1217	1 »
264	5	1189	12 »	294	3	1218	21 tishrīn I
265	3	1190	2 »	295	1	1219	11 »
266	7	1190	22 āb	296	5	1220	29 aylūl
267	5	1191	11 »	297	3	1221	19 »
268	2	1192	31 tammūz	298	7	1222	8 »
269	6	1193	20 »	299	4	1222	28 āb
270	4	1194	10 »	300	2	1223	17 »

f. 161,r.

الجدول السادس من استخراج تاريخ العرب من تاريخ الروم وتاريخ الروم من تاريخ العرب

Table VI for finding the era of the Arabs from the era of the Romans and that of the Romans from that of the Arabs.

Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.	Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.
301	6	1224	6 āb	331	4	1254	14 aylūl
302	3	1225	26 tammūz	332	1	1255	3 »
303	1	1226	16 »	333	6	1255	23 āb
304	5	1227	4 »	334	3	1256	12 »
305	2	1228	23 ḥazīrān	335	7	1257	1 »
306	7	1229	13 »	336	5	1258	22 tammūz
307	4	1230	2 »	337	2	1259	10 »
308	2	1231	22 ayyār	338	7	1260	30 ḥazīrān
309	6	1232	11 »	339	4	1261	19 »
310	3	1233	30 nīsān	340	1	1262	8 »
311	1	1234	20 »	341	6	1263	28 ayyār
312	5	1235	8 »	342	3	1264	17 »
313	2	1236	28 ādhār	343	7	1265	6 »
314	7	1237	18 »	344	5	1266	26 nīsān
315	4	1238	7 »	345	2	1267	14 »
316	1	1239	24 subāṭ	346	6	1268	3 »
317	6	1240	13 »	347	4	1269	24 ādhār
318	3	1241	3 »	348	1	1270	13 »
319	1	1242	23 kānūn II	349	6	1271	2 »
320	5	1243	12 »	350	3	1272	19 subāṭ
321	2	1244	31 kānūn I	351	7	1273	8 »
322	7	1245	21 »	352	5	1274	29 kānūn II
323	4	1246	10 »	353	2	1275	18 »
324	1	1247	29 tishrīn II	354	6	1276	6 »
325	6	1248	18 »	355	4	1277	27 kānūn I
326	3	1249	7 »	356	1	1278	16 »
327	1	1250	28 tishrīn I	357	6	1279	6 »
328	5	1251	17 »	358	3	1280	24 tishrīn II
329	2	1252	5 »	359	7	1281	13 »
330	7	1253	25 aylūl	360	5	1282	3 »

f. 161,v.

الجدول السابع من استخراج تاريخ العرب من تاريخ الروم وتاريخ الروم من تاريخ العرب

Table VII for finding the era of the Arabs from the era of the Romans and that of the Romans from that of the Arabs.

Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.	Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.
361	2	1283	23 tishrīn I	391	7	1312	30 tishrīn II
362	6	1284	11 »	392	4	1313	19 »
363	4	1285	1 »	393	2	1314	9 »
364	1	1286	20 aylūl	394	6	1315	29 tishrīn I
365	5	1287	9 »	395	3	1316	17 »
366	3	1287	29 āb	396	1	1317	7 »
367	7	1288	18 »	397	5	1318	26 aylūl
368	5	1289	8 »	398	3	1319	16 »
369	2	1290	28 tammūz	399	7	1320	4 »
370	6	1291	16 »	400	4	1320	24 āb
371	4	1292	6 »	401	2	1321	14 »
372	1	1293	25 ḥazīrān	402	6	1322	3 »
373	5	1294	14 »	403	3	1323	22 tammūz
374	3	1295	3 »	404	1	1324	12 »
375	7	1296	23 ayyār	405	5	1325	1 »
376	4	1297	12 »	406	2	1326	20 ḥazīrān
377	2	4298	2 »	407	7	1327	9 »
378	6	1299	20 nīsān	408	4	1328	29 ayyār
379	4	1300	10 »	409	2	1329	19 »
380	1	1301	30 ādhār	410	6	1330	8 »
381	5	1302	19 »	411	3	1331	26 nīsān
382	3	1303	8 »	412	1	1332	16 »
383	7	1304	25 subāṭ	413	5	1333	5 »
384	4	1305	14 »	414	2	1334	25 ādhār
385	2	1306	4 »	415	7	1335	14 »
386	6	1307	24 kānūn II	416	4	1336	3 »
387	4	1308	13 »	417	2	1337	21 subāṭ
388	1	1309	2 »	418	6	1338	10 »
389	5	1310	22 kānūn I	419	3	1339	30 kānūn II
390	3	1311	12 »	420	1	1340	19 »

f. 162,r.

الجدول الثامن من استخراج تاريخ العرب من تاريخ الروم وتاريخ الروم من تاريخ العرب

Table VIII for finding the era of the Arabs from the era of the Romans and that of the Romans from that of the Arabs.

Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.	Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.
421	5	1341	8 kānūn II	451	3	1370	16 subāṭ
422	2	1342	28 kānūn I	452	7	1371	5 »
423	7	1343	18 »	453	5	1372	25 kānūn II
424	4	1344	6 »	454	2	1373	14 »
425	1	1345	25 tishrīn II	455	6	1374	4 »
426	6	1346	15 »	456	4	1375	24 kānūn I
427	3	1347	4 »	457	1	1376	12 »
428	1	1348	24 tishrīn I	458	6	1377	2 »
429	5	1349	13 »	459	3	1278	21 tishrīn II
430	2	1350	2 »	460	7	1379	10 »
431	7	1351	22 aylūl	461	5	1380	30 tishrīn I
432	4	1352	10 »	462	2	1381	19 »
433	1	1352	30 āb	463	6	1382	8 »
434	6	1353	20 »	464	4	1383	28 aylūl
435	3	1354	9 »	465	1	1384	16 »
436	7	1355	28 tammūz	466	5	1385	5 »
437	5	1356	18 »	467	3	1385	26 āb
438	2	1357	7 »	468	7	1386	15 »
439	7	1358	27 ḥazīrān	469	5	1387	4 »
440	4	1359	15 »	470	2	1388	24 tammūz
441	1	1360	4 »	471	6	1389	13 »
442	6	1361	25 ayyār	472	4	1390	3 »
443	3	1362	14 »	473	1	1391	21 ḥazīrān
444	7	1363	2 »	474	5	1392	10 »
445	5	1364	22 nīsān	475	3	1393	31 ayyār
446	2	1365	11 »	476	7	1394	20 »
447	7	1366	1 »	477	5	1395	9 »
448	4	1367	20 ādhār	478	2	1396	28 nīsān
449	1	1368	9 »	479	6	1397	17 »
450	6	1369	27 subāṭ	480	4	1398	7 »

f. 162,v.

الجدول التاسع من استخراج تاريخ العرب من تاريخ الروم وتاريخ الروم من تاريخ العرب

Table IX for finding the era of the Arabs from the era of the Romans and that of the Romans from that of the Arabs.

Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.	Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.
481	1	1399	26 ādhār	511	6	1428	4 ayyār
482	5	1400	15 »	512	3	1429	23 nīsān
483	3	1401	5 »	513	1	1430	13 »
484	7	1402	22 subāṭ	514	5	1431	1 »
485	4	1403	11 »	515	2	1432	21 ādhār
486	2	1404	31 kānūn II	516	7	1433	11 »
487	6	1405	20 »	517	4	1434	28 subāṭ
488	4	1406	10 »	518	2	1435	18 »
489	1	1407	30 kānūn I	519	6	1436	6 »
490	5	1408	18 »	520	3	1437	26 kānūn II
491	3	1409	8 »	521	1	1438	16 »
492	7	1410	27 tishrīn II	522	5	1439	5 »
493	4	1411	16 »	523	2	1440	24 kānūn I
494	2	1412	5 »	524	7	1441	14 »
495	6	1413	25 tishrīn I	525	4	1442	3 »
496	3	1414	14 »	526	1	1443	22 tishrīn II
497	1	1415	4 »	527	6	1444	11 »
498	5	1416	22 aylūl	528	3	1445	31 tishrīn I
499	3	1417	12 »	529	1	1446	21 »
500	7	1418	1 »	530	5	1447	10 »
501	4	1418	21 āb	531	2	1448	28 aylūl
502	2	1419	10 »	532	7	1449	18 »
503	6	1420	30 tammūz	533	4	1450	7 »
504	3	1421	19 »	534	1	1450	26 āb
505	1	1422	9 »	535	6	1451	16 »
506	5	1423	27 ḥazīrān	536	3	1452	5 »
507	3	1424	17 »	537	1	1453	26 tammūz
508	7	1425	6 »	538	5	1454	15 »
509	4	1426	26 ayyār	539	2	1455	3 »
510	2	1427	15 »	540	7	1456	23 ḥazīrān

f. 163,r.

الجدول العاشر من استخراج تاريخ العرب من تاريخ الروم وتاريخ الروم من تاريخ العرب

Table X for finding the era of the Arabs from the era of the Romans and that of the Romans from that of the Arabs.

Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.	Years of the Hijra.	Name of the first day of the month muḥarram.	Years of the era of Dhū 'l-qarnayn.	Days elapsed of the Roman month in which the beginning of muḥarram falls.
541	4	1457	12 ḥazīrān	571	2	1486	21 tammūz
542	1	1458	1 »	572	6	1487	9 »
543	6	1459	21 ayyār	573	4	1488	29 ḥazīrān
544	3	1460	10 »	574	1	1489	18 »
545	7	1461	29 nīsān	575	5	1490	7 »
546	5	1462	19 »	576	3	1491	27 ayyār
547	2	1463	7 »	577	7	1492	16 »
548	7	1464	28 ādhār	578	5	1493	6 »
549	4	1465	17 »	579	2	1494	25 nīsān
550	1	1466	6 »	580	6	1495	13 »
551	6	1467	24 subāṭ	581	4	1496	3 »
552	3	1468	12 »	582	1	1497	23 ādhār
553	7	1469	1 »	583	5	1498	12 »
554	5	1470	22 kānūn II	584	3	1499	1 »
555	2	1471	11 »	585	7	1500	18 subāṭ
556	6	1472	30 kānūn I	586	4	1501	7 »
557	4	1473	20 »	587	2	1502	28 kānūn II
558	1	1474	9 »	588	6	1503	17 »
559	6	1475	29 tishrīn II	589	4	1504	6 »
560	3	1476	17 »	590	1	1505	26 kānūn I
561	7	1477	6 »	591	5	1506	15 »
562	5	1478	27 tishrīn I	592	3	1507	5 »
563	2	1479	16 »	593	7	1508	23 tishrīn II
564	6	1480	4 »	594	4	1509	12 »
565	4	1481	24 aylūl	595	2	1510	2 »
566	1	1482	13 »	596	6	1511	22 tishrīn I
567	6	1483	3 »	597	4	1512	11 »
568	3	1483	22 āb	598	1	1513	30 aylūl
569	7	1484	11 »	599	5	1514	19 »
570	5	1485	1 »	600	3	1515	9 »

f. 164,v.

جدول حركات الشمس والقمر وحاصّة القمر والعقد الشمالي في السنين العربية المجموعة
Table of the motions of the sun, the moon, the anomaly of the moon and the northern node in collected Arab years.

سنو العرب المجموعة Collected Arab years [from the beginning of the Hijra].	مسير الشمس الاوسط Mean longitude of the sun.	مسير القمر الاوسط Mean longitude of the moon.	حركة حاصّة القمر Anomaly of the moon.	حركة العقد الشمالي Longitude of the northern node.
1	113° 58′ 4″	119° 43′ 46″	106° 30′ 40″	233° 45′ 18″
31	152 25 54	158 1 27	40 20 38	76 40 35
61	190 53 44	196 19 8	334 10 36	279 35 52
91	229 21 34	234 36 48	268 0 34	122 31 9
121	267 49 24	272 54 29	201 50 33	325 26 26
151	306 17 14	311 12 10	135 40 31	168 21 43
181	344 45 5	349 29 51	69 30 29	11 17 0
211	23 12 55	27 47 32	3 20 27	214 12 17
241	61 40 45	66 5 13	297 10 25	57 7 34
271	100 8 35	104 22 54	231 0 24	260 2 51
301	138 36 25	142 40 34	164 50 23	102 58 7
331	177 4 15	180 58 15	98 40 21	305 53 24
361	215 32 5	219 15 56	32 30 19	148 48 41
391	253 59 55	257 33 36	326 20 18	351 43 58
421	292 27 45	295 51 17	260 10 16	194 39 15
451	330 55 36	334 8 58	194 0 14	37 34 32
481	9 23 26	12 26 39	127 50 13	240 29 49
511	47 51 16	50 44 20	61 40 11	83 25 6
541	86 19 6	89 2 1	355 30 9	286 20 23
571	124 46 56	127 19 42	289 20 7	129 15 40
601	163 14 46	165 37 23	223 10 6	332 10 57
631	201 42 36	203 55 3	157 0 4	175 6 14
661	240 10 26	242 12 44	90 50 2	18 1 31
691	278 38 16	280 30 25	24 40 0	220 56 48
721	317 6 6	318 48 6	318 29 59	63 52 5
751	355 33 56	357 5 47	252 19 57	266 47 22
781	34 1 46	35 23 28	186 9 55	109 42 39
811	72 29 36	73 41 8	119 59 53	312 37 56
841	110 57 26	111 58 49	53 49 52	155 33 13
871	149 25 16	150 16 30	347 39 50	358 28 30

f. 165,r.

جدول حركات الشمس والقمر وحاصة القمر والعقد الشمالي في السنين العربية المبسوطة

Table of the motions of the sun, the moon, the anomaly of the moon and the northern node in single Arab years.

سنو العرب المبسوطة Single Arab years.	مسير الشمس الاوسط Mean course of the sun.	مسير القمر الاوسط Mean course of the moon.	حركة حاصة القمر Motion of the anomaly of the moon.	حركة العقد الشمالي Motion of the northern node.
1	348° 55′ 15″	344° 26′ 43″	305° 0′ 14″	18° 44′ 41″
2	338 49 37	342 4 0	263 4 22	37 32 32
3	327 44 52	326 30 43	208 4 36	56 17 12
4	316 40 7	310 57 25	153 4 50	75 1 52
5	306 34 30	308 34 43	111 8 59	93 49 43
6	295 29 44	293 1 25	56 9 13	112 34 24
7	285 24 7	290 38 43	14 13 21	131 22 15
8	274 19 22	275 5 25	319 13 35	150 6 56
9	263 14 37	259 32 8	264 13 49	168 51 36
10	253 8 59	257 9 25	222 17 57	187 39 27
11	242 4 14	241 36 8	167 18 11	206 24 8
12	230 59 29	226 2 50	112 18 25	225 8 49
13	220 53 52	223 40 8	70 22 33	243 56 39
14	209 49 6	208 6 50	15 22 47	262 41 20
15	198 44 21	192 33 33	320 23 2	281 26 2
16	188 38 44	190 10 50	278 27 10	300 13 53
17	177 33 58	174 37 33	223 27 24	318 58 34
18	167 28 20	172 14 51	181 31 32	337 46 25
19	156 23 35	156 41 34	126 31 46	356 31 6
20	145 18 50	141 8 17	71 32 0	15 15 46
21	135 13 13	138 45 34	29 36 8	34 3 36
22	124 8 28	123 12 17	334 36 23	52 48 17
23	113 3 43	107 38 59	279 36 37	71 32 57
24	102 58 6	105 16 16	237 40 45	90 20 48
25	91 53 20	89 42 59	182 40 59	109 5 28
26	81 47 45	87 20 16	140 45 8	127 53 9
27	70 42 58	71 46 59	85 45 22	146 38 2
28	59 38 13	56 13 41	30 45 36	165 22 43
29	49 32 35	53 50 59	348 49 44	184 10 34
30	38 27 50	38 17 41	293 49 58	202 55 14

f. 165,v.

جدول حركة الشمس والقمر وحاصة القمر والعقد الشمالي في الشهور العربية

Table of the motions of the sun, the moon, the anomaly of the moon and the northern node in the Arab months.

اسماء شهور العرب Names of the Arab months.	مسير الشمس الاوسط Mean course of the sun.	مسير القمر الاوسط Mean course of the moon.	حركة حاصة القمر Motion of the anomaly of the moon.	حركة العقد الشمالي Motion of the northern node.
al–muḥarram.	29° 34′ 10″	35° 17′ 31″	31° 56′ 58″	1° 35′ 19″
ṣafar.	58 9 12	57 24 27	50 50 2	3 7 27
rabīʿ I.	87 43 23	92 41 58	82 47 0	4 42 45
rabīʿ II.	116 18 25	114 48 54	101 40 4	6 14 54
ǵumādà I.	145 52 36	150 6 25	133 37 2	7 50 13
ǵumādà II.	174 27 38	172 13 22	152 30 6	9 22 21
raǵab.	204 1 48	207 30 53	184 27 5	10 57 40
shaʿbān.	232 36 50	229 37 49	203 20 9	12 29 48
ramaḍān.	262 11 0	264 55 20	235 17 7	14 5 7
shawwāl.	290 46 2	287 2 16	254 10 11	15 37 15
dhū ’l-qaʿdah.	320 20 13	322 19 47	286 7 9	17 12 33
dhū ’l–ḥiǵǵah				
common.	348 55 15	344 26 43	305 0 13	18 44 41
leap.	349 54 22	357 37 17	318 4 8	18 47 51

f. 166,r.

جدول حركة الشمس والقمر وحاصة القمر والعقد الشمالي في ايام الشهر القمري

Table of the motions of the sun, the moon, the anomaly of the moon and the northern node in the days of the lunar month.

عدد ايام الشهر Number of days.	مسير الشمس الاوسط Mean course of the sun.	مسير القمر الاوسط Mean course of the moon.	حركة حاصة القمر Motion of the anomaly of the moon.	حركة العقد الشمالي Motion of the northern node.
1	0° 59′ 8″	13° 10′ 35″	13° 3′ 54″	0° 3′ 11″
2	1 58 17	26 21 10	26 7 48	0 6 21
3	2 57 25	39 31 45	39 11 42	0 9 32
4	3 56 33	52 42 20	52 15 36	0 12 43
5	4 55 42	65 52 55	65 19 30	0 15 53
6	5 54 50	79 3 30	78 23 24	0 19 4
7	6 53 58	92 14 5	91 27 18	0 22 15
8	7 53 7	105 24 40	104 31 12	0 25 25
9	8 52 16	118 35 15	117 35 6	0 28 36
10	9 51 25	131 45 50	130 39 0	0 31 47
11	10 50 32	144 56 25	143 42 53	0 34 57
12	11 49 40	158 7 0	156 46 47	0 38 8
13	12 48 48	171 17 36	169 50 41	0 41 19
14	13 47 56	184 28 11	182 54 35	0 44 29
15	14 47 5	197 38 46	195 58 29	0 47 40
16	15 46 13	210 49 21	209 2 23	0 50 51
17	16 45 21	223 59 56	222 6 17	0 54 2
18	17 44 30	237 10 31	235 10 11	0 57 12
19	18 43 38	250 21 6	248 14 5	1 0 23
20	19 42 46	263 31 41	261 17 59	1 3 34
21	20 41 55	276 42 16	274 21 53	1 6 45
22	21 41 3	289 52 51	287 25 47	1 9 55
23	22 40 11	303 3 26	300 29 41	1 13 5
24	23 39 20	316 14 1	313 33 34	1 16 16
25	24 38 28	329 24 36	326 37 28	1 19 26
26	25 37 36	342 35 11	339 41 22	1 22 37
27	26 36 44	355 45 46	352 45 16	1 25 48
28	27 35 53	8 56 21	5 49 10	1 28 59
29	28 35 1	22 6 56	18 53 4	1 32 9
30	29 34 10	35 17 31	31 56 58	1 35 19

f 166,v.

جدول حركات الشمس والقمر وحاصة القمر والعقد الشمالي في الساعات المعتدلة

Table of the motions of the sun, the moon, the anomaly of the moon and the northern node in equal hours.

عدد الساعات المعتدلة Number of equal hours.	مسير الشمس الاوسط Mean course of the sun.	مسير القمر الاوسط Mean course of the moon.	حركة حاصة القمر Motion of the anomaly of the moon.	حركة العقد الشمالي Motion of the northern node.
1	0° 2′ 28″	0° 32′ 56″	0° 32′ 40″	0° 0′ 8″
2	0 4 56	1 5 53	1 5 19	0 0 16
3	0 7 25	1 38 49	1 37 59	0 0 24
4	0 9 51	2 11 46	2 10 39	0 0 32
5	0 12 19	2 44 42	2 43 18	0 0 40
6	0 14 47	3 17 39	3 15 59	0 0 48
7	0 17 15	3 50 35	3 48 38	0 0 56
8	0 19 43	4 23 32	4 21 18	0 1 4
9	0 22 10	4 56 28	4 53 58	0 1 12
10	0 24 38	5 29 25	5 26 38	0 1 20
11	0 27 6	6 2 21	5 59 17	0 1 27
12	0 29 34	6 35 17	6 31 57	0 1 35
13	0 32 2	7 8 14	7 4 37	0 1 43
14	0 34 30	7 41 10	7 37 16	0 1 51
15	0 36 58	8 14 7	8 9 56	0 1 59
16	0 39 25	8 47 3	8 42 36	0 2 7
17	0 41 53	9 20 0	9 15 16	0 2 15
18	0 44 21	9 52 56	9 47 55	0 2 23
19	0 46 49	10 25 53	10 20 36	0 2 31
20	0 49 17	10 58 49	10 53 15	0 2 39
21	0 51 45	11 31 46	11 25 55	0 2 47
22	0 54 12	12 4 42	11 58 35	0 2 55
23	0 56 40	12 37 38	12 31 13	0 3 3
24	0 59 8	13 10 35	13 3 54	0 3 11

f. 168,r.

جدول حركات الكواكب الخمسة المتحيرة في السنين العربية المجموعة
Table of the motions of the five planets in collected Arab years.

السنون العربية المجموعة Collected Arab years [from the beginning of the Hijra].	وسط زحل Mean longitude of Saturn.	وسط المشتري Mean longitude of Jupiter.	وسط المريخ Mean longitude of Mars.	حاصة الزهرة Anomaly of Venus.	حاصة عطارد Anomaly of Mercury.
1	116° 15′	332° 3′	211° 48′	45° 7′	73° 23′
31	112 23	135 51	23 14	119 24	340 44
61	108 31	299 39	194 41	193 41	248 5
91	104 38	103 26	6 7	267 57	155 26
121	100 46	267 14	177 34	342 14	62 47
151	96 54	71 2	349 0	56 31	330 8
181	93 2	234 50	160 27	130 48	237 29
211	89 10	38 38	331 53	205 5	144 50
241	85 17	202 25	143 20	279 21	52 11
271	81 25	6 13	314 46	353 38	319 32
301	77 33	170 1	126 13	67 55	226 53
331	73 41	333 49	297 39	142 12	134 14
361	69 49	137 37	109 6	216 29	41 35
391	65 56	301 25	280 32	290 45	308 56
421	62 4	105 12	91 59	5 2	216 17
451	58 12	269 0	263 25	79 19	123 38
481	54 20	72 48	74 52	153 36	30 59
511	50 28	236 36	246 18	227 53	298 20
541	46 35	40 23	57 45	302 9	205 41
571	42 43	204 11	229 11	16 26	113 2
601	38 51	7 59	40 38	90 43	20 23
631	34 59	171 47	212 4	165 0	287 44
661	31 6	335 35	23 31	239 17	195 5
691	27 14	139 22	194 57	313 33	102 26
721	23 22	303 10	6 24	27 50	9 47
751	19 30	106 58	177 50	102 7	277 8
781	15 38	270 46	349 17	176 24	184 29
811	11 46	73 34	160 43	250 41	91 50
841	7 53	237 21	332 10	324 57	359 11
871	4 1	41 9	143 36	39 14	266 32

f. 168,v.

جدول حركات الكواكب الخمسة المتحيرة في السنين العربية المبسوطة
Table of the motions of the five planets in single Arab years.

سنو العرب المبسوطة Single Arab years.	وسط زحل Mean motion of Saturn.	وسط المشتري Mean motion of Jupiter.	وسط المريخ Mean motion of Mars.	حاصة الزهرة Motion of the anomaly of Venus.	حاصة عطارد Motion of the anomaly of Mercury.
1	11° 52′	29° 26′	185° 31′	218° 15′	19° 46′
2	23 45	58 51	11 34	76 30	42 39
3	35 37	88 22	197 5	295 22	62 26
4	47 28	117 48	22 37	153 37	82 12
5	59 22	157 19	208 40	12 29	105 5
6	71 14	176 45	34 11	230 44	124 51
7	83 7	206 10	220 14	89 36	147 44
8	94 59	235 41	45 45	307 51	167 30
9	106 50	265 7	231 16	166 6	187 17
10	118 42	294 33	57 19	24 58	210 10
11	130 35	324 3	242 51	243 13	229 56
12	142 27	353 29	68 22	101 28	249 42
13	154 20	23 0	254 25	320 20	272 36
14	166 12	52 26	79 56	178 35	292 22
15	178 3	81 51	265 27	36 49	312 9
16	189 57	111 22	91 30	255 41	335 1
17	201 48	140 48	277 1	113 57	354 47
18	213 42	170 19	103 4	332 49	17 40
19	225 34	199 45	288 35	191 4	37 26
20	237 25	229 10	114 7	49 19	57 12
21	249 19	258 41	300 10	268 11	80 5
22	261 10	288 7	125 41	126 26	99 51
23	273 2	317 33	311 12	344 41	119 37
24	284 56	347 3	137 15	203 33	142 31
25	296 47	16 29	322 46	61 48	162 17
26	308 41	46 0	148 50	280 40	185 10
27	320 32	75 26	334 21	138 55	204 56
28	332 23	104 51	159 52	357 10	224 42
29	344 17	134 22	345 55	216 2	247 35
30	356 8	163 48	171 26	74 17	267 21

f. 169,r.

جدول حركات الكواكب الخمسة المتحيرة في الشهور العربية

Table of the motions of the five planets in the Arab months.

اسماء الشهور العربية Names of the Arab months.	وسط زحل Mean motion of Saturn.	وسط المشتري Mean motion of Jupiter.	وسط المريخ Mean motion of Mars.	حاصة الزهرة Motion of the anomaly of Venus.	حاصة عطارد Motion of the anomaly of Mercury.
al–muḥarram.	1° 0′	2° 30′	15° 43′	18° 30′	93° 12′
ṣafar.	1 58	4 54	30 56	36 22	183 18
rabīʿ I.	2 59	7 24	46 39	54 52	276 30
rabīʿ II.	3 57	9 48	61 50	72 45	6 35
ǵumādà I.	4 57	12 18	77 33	91 15	99 47
ǵumādà II.	5 56	14 42	92 46	109 7	189 53
raǵab.	6 56	17 12	108 29	127 37	283 5
shaʿbān.	7 54	19 37	123 40	145 29	13 10
ramaḍān.	8 54	22 6	139 23	163 59	106 22
shawwāl.	9 52	24 31	154 36	181 52	196 27
dhū ’l–qaʿdah.	10 52	27 0	170 19	200 21	289 39
dhū ’l–ḥiǵǵah.	11 51	29 25	185 30	218 14	19 45

f. 169,v.

حركات الكواكب الخمسة المتحيرة في ايام الشهر القمري
Motions of the five planets in the days of the lunar month.

عدد ايام الشهر Number of the days of the month.	مسير زحل Course of Saturn.	مسير المشتري Course of Jupiter.	مسير المريخ Course of Mars.	حاصة الزهرة Motion of the anomaly of Venus.	حاصة عطارد Motion of the anomaly of Mercury.
1	0° 2′	0° 5′	0° 31′	0° 37′	3° 6′
2	0 4	0 10	1 3	1 14	6 13
3	0 6	0 15	1 34	1 51	9 19
4	0 8	0 20	2 6	2 28	12 26
5	0 10	0 25	2 37	3 5	15 32
6	0 12	0 30	3 9	3 42	18 39
7	0 14	0 35	3 40	4 19	21 45
8	0 16	0 40	4 12	4 56	24 51
9	0 18	0 45	4 43	5 33	27 58
10	0 20	0 50	5 15	6 10	31 4
11	0 22	0 55	5 46	6 47	34 10
12	0 24	1 0	6 18	7 24	37 17
13	0 26	1 5	6 49	8 1	40 23
14	0 28	1 10	7 20	8 38	43 30
15	0 30	1 15	7 52	9 15	46 36
16	0 32	1 20	8 23	9 52	49 42
17	0 34	1 25	8 54	10 29	52 49
18	0 36	1 30	9 26	11 6	55 55
19	0 38	1 35	9 57	11 43	59 2
20	0 40	1 40	10 29	12 20	62 8
21	0 42	1 45	11 0	12 57	65 14
22	0 44	1 50	11 32	13 34	68 21
23	0 46	1 55	12 3	14 11	71 27
24	0 48	2 0	12 35	14 48	74 34
25	0 50	2 5	13 6	15 25	77 40
26	0 52	2 10	13 38	16 2	80 46
27	0 54	2 15	14 9	16 39	83 53
28	0 56	2 20	14 41	17 16	86 59
29	0 58	2 25	15 12	17 53	90 6
30	1 0	2 30	15 43	18 30	93 12

f. 170,r.

جدول حركات الكواكب الخمسة المتحيرة في الساعات المعتدلة
Table of the motions of the five planets in equal hours.

عدد الساعات المعتدلة Number of the hours of the day.	مسير زحل Course of Saturn.	مسير المشتري Course of Jupiter.	مسير المريخ Course of Mars.	حركة حاصة الزهرة Motion of the anomaly of Venus.	حركة حاصة عطارد Motion of the anomaly of Mercury.
1	0° 0′	0° 0′	0° 1′	0° 2′	0° 8′
2	0 0	0 0	0 3	0 3	0 16
3	0 0	0 1	0 4	0 5	0 23
4	0 0	0 1	0 5	0 6	0 31
5	0 0	0 1	0 6	0 8	0 39
6	0 0	0 1	0 8	0 9	0 47
7	0 1	0 1	0 9	0 11	0 54
8	0 1	0 2	0 10	0 12	1 2
9	0 1	0 2	0 12	0 14	1 10
10	0 1	0 2	0 13	0 15	1 18
11	0 1	0 2	0 14	0 17	1 25
12	0 1	0 2	0 16	0 18	1 33
13	0 1	0 3	0 17	0 20	1 41
14	0 1	0 3	0 18	0 22	1 49
15	0 1	0 3	0 20	0 23	1 57
16	0 1	0 3	0 21	0 25	2 4
17	0 1	0 4	0 22	0 26	2 12
18	0 1	0 4	0 24	0 28	2 20
19	0 2	0 4	0 25	0 29	2 28
20	0 2	0 4	0 26	0 31	2 35
21	0 2	0 4	0 28	0 32	2 43
22	0 2	0 5	0 29	0 34	2 50
23	0 2	0 5	0 30	0 36	2 58
24	0 2	0 5	0 31	0 37	3 6

f. 170,v.

جدول اجتماعات الشمس والقمر الوسطى في السنين المصرية المجموعة
Table of the mean conjunctions of the sun and the moon in collected Egyptian years.

السنين المجموعة Collected years [from the era].	ايام توت Days of the month Tōt.	وسطَي النيّرين Mean motions of both luminaries.	حاصة القمر Anomaly of the moon.	حركته في العرض Motion in latitude of the moon.
915	22^{d} 14′ 44″	26° 3′ 27″	75° 50′ 33″	247° 38′ 11″
940	22 11 57	20 5 2	133 12 17	4 50 17
965	22 9 10	14 6 37	190 34 1	122 2 23
990	22 6 23	8 8 12	247 55 45	239 14 29
1015	22 3 36	2 9 47	305 17 29	356 26 35
1040	22 0 48	356 11 23	2 39 13	113 38 41
1065	21 58 1	350 12 58	60 0 57	230 50 47
1090	21 55 14	344 14 33	117 22 41	348 2 53
1115	21 52 27	338 16 8	174 44 25	105 14 58
1140	21 49 40	332 17 43	232 6 9	222 27 4
1165	21 46 53	326 19 18	289 27 53	339 39 10
1190	21 44 6	320 20 53	346 49 37	96 51 16
1215	21 41 19	314 22 28	44 11 21	214 3 22
1240	21 38 31	308 24 3	101 33 5	331 15 28
1265	21 35 44	302 25 38	158 54 49	88 27 34
1290	21 32 57	296 27 13	216 16 33	205 39 39
1315	21 30 10	290 28 48	273 38 17	322 51 45
1340	21 27 23	284 30 23	331 0 1	80 3 51
1365	21 24 36	278 31 59	28 21 45	197 15 57
1390	21 21 49	272 33 34	85 43 29	314 28 3
1415	21 19 2	266 35 9	143 5 13	71 40 9
1440	21 16 15	260 36 44	200 26 57	188 52 15
1465	21 13 28	254 38 19	257 48 41	306 4 21
1490	21 10 41	248 39 54	315 10 25	63 16 26
1515	21 7 54	242 41 29	12 32 9	180 28 32
1540	21 5 6	236 43 4	69 53 53	297 40 38
1565	21 2 19	230 44 39	127 15 37	54 52 44
1590	20 59 32	224 46 15	184 37 21	172 4 50
1615	20 56 45	218 47 49	241 59 5	289 16 56
1640	20 53 58	212 49 25	299 20 49	46 29 2
1665	20 51 11	206 50 59	356 42 33	163 41 8
1690	20 48 24	200 52 35	54 4 17	280 53 14

f. 171,r.

جدول مقابلات الشمس والقمر الوسطى في السنين المصرية المجموعة
Table of the mean oppositions of the sun and the moon in collected Egyptian years.

السنون المجموعة Collected years [from the era].	ايام توت Days of the month Tōt.	وسط الشمس والقمر قبالته Mean motion of the sun, and of the moon at the opposite point.	حاصة القمر Anomaly of the moon.	حركته في العرض Motion in latitude of the moon.
915	7d 28′ 49″	11° 30′ 15″	242° 56′ 3″	52° 18′ 4″
940	7 26 2	5 31 50	300 17 47	169 30 10
965	7 23 15	359 33 25	357 39 31	286 42 16
990	7 20 28	353 35 0	55 1 15	43 54 22
1015	7 17 41	347 36 35	112 22 59	161 6 28
1040	7 14 53	341 38 10	169 44 43	278 18 34
1065	7 12 6	335 39 45	227 6 27	35 30 40
1090	7 9 19	329 41 20	284 28 11	152 42 46
1115	7 6 32	323 42 55	341 49 55	269 54 51
1140	7 3 45	317 44 31	39 11 39	27 6 57
1165	7 0 58	311 46 6	96 33 23	144 19 3
1190	6 58 11	305 47 41	155 55 7	261 31 9
1215	6 55 24	299 49 16	211 16 51	18 43 15
1240	6 52 36	293 50 51	268 38 35	135 55 20
1265	6 49 49	287 52 26	326 0 19	253 7 26
1290	6 47 2	281 54 1	23 22 3	10 19 32
1315	6 44 15	275 55 36	80 43 47	127 31 38
1340	6 41 28	269 57 11	138 5 31	244 43 44
1365	6 38 41	263 58 46	195 27 15	1 55 50
1390	6 35 54	258 0 22	252 48 59	119 7 56
1415	6 33 7	252 1 57	310 10 43	236 20 1
1440	6 30 20	246 3 32	7 32 27	353 32 7
1465	6 27 33	210 5 7	64 54 11	110 44 13
1490	6 24 46	234 6 42	122 15 55	227 56 19
1515	6 21 59	228 8 17	179 37 39	345 8 25
1540	6 19 12	222 9 52	236 59 23	102 20 31
1565	6 16 25	216 11 28	294 21 7	219 32 37
1590	6 13 37	210 13 3	351 42 51	336 44 43
1615	6 10 50	204 14 38	49 4 35	93 56 48
1640	6 8 3	198 16 13	106 26 19	111 8 54
1665	6 5 16	192 17 48	163 48 3	328 21 0
1690	6 2 29	186 19 23	221 9 47	85 33 6

f. 171,v.

جدول تجزئة ايام الشهور وشهور القبط وجملة سنين مفردة للزيادة او النقصان
Table of the division of the days and months of the Copts, and the intervals of years to be added or subtracted.

جدول ما يحتاج اليه من تجزئة ايام الشهر القبطي
Table of the division of the days of the Coptic month, which is to be used.

A sixth of the month.	4 55 18	4 51 4	64 18 10	65 6 42
A quarter of the month.	7 22 57	7 16 36	96 27 15	97 40 3
A third of the month.	9 50 37	9 42 8	128 36 20	130 13 25
Half of the month.	14 45 55	14 33 12	192 54 30	195 20 7

جدول الاجتماعات والمقابلات في الشهور القبطية
Table of the conjunctions and oppositions in the months of the Copts.

عدد الشهور Number of the months.	الايام القمرية Lunar days.	وسطي النيّرين Mean motions of both luminaries.	حاصة القمر Anomaly of the moon.	حركة العرض Motion in latitude of the moon.
1	29^{d} 31′ 50″	29° 6′ 25″	25° 49′ 0″	30° 40′ 14″
2	59 3 40	58 12 49	51 38 0	61 20 28
3	88 35 30	87 19 14	77 27 1	92 0 42
4	118 7 21	116 25 39	103 16 1	122 40 57
5	147 39 11	145 32 4	129 5 1	153 21 11
6	177 11 1	174 38 28	154 54 1	184 1 25
7	206 42 51	203 44 53	180 43 1	214 41 39
8	236 14 41	232 51 17	206 32 1	245 21 53
9	265 46 31	261 57 42	232 21 1	276 2 7
10	295 18 21	291 4 7	258 10 1	306 42 21
11	324 50 12	320 10 31	283 59 2	337 22 36
12	354 22 2	349 16 56	309 48 2	8 2 50
جملة سنين مفردة تزاد او تنقص Intervals of years that are added or subtracted.	تنقص مما يستانف وتزاد على ما تقدم It is subtracted from what is to come, and added to what has gone before.	تزاد على ما يستانف وتنقص مما تقدم It is added to what is to come, and subtracted from what has gone before.	تزاد على ما يستانف وتنقص مما تقدم It is added to what is to come, and subtracted from what has gone before.	تزاد على ما يستانف وتنقص مما تقدم It is added to what is to come, and subtracted from what has gone before.
50	5′ 34″ 10‴	348° 3′ 10″	114° 43′ 28″	234° 24′ 12″
100	11 8 20	336 6 20	229 26 56	108 48 24
200	22 16 40	312 12 41	98 53 52	217 36 47
300	33 25 0	288 19 1	328 20 48	326 25 11
400	44 33 20	264 25 22	197 47 44	75 13 34
500	55 41 40	240 31 42	67 14 40	184 1 58
600	66 50 0	216 38 3	296 41 36	292 50 20

f. 172,r.

جدول الاجتماعات والاستقبالات الوسطى في السنين المصرية المبسوطة
Table of the mean conjunctions and oppositions in single Egyptian years.

السنون المبسوطة Single years.	زيادات الايام Increments of days.	وسطي النيّرين Mean motions of both luminaries.	حاصة القمر Anomaly of the moon.	حركة العرض Motion in latitude of the moon.
1	18^{d} 53′ 52″	18° 23′ 22″	335° 37′ 2″	38° 43′ 3″
2	8 15 53	7 40 19	285 25 4	46 45 50
3	27 9 45	26 3 41	261 2 5	85 28 52
4	16 31 47	15 20 39	210 50 7	93 31 40
5	5 53 49	4 37 36	160 38 9	101 34 29
6	24 47 50	23 0 58	136 15 11	140 17 31
7	14 9 42	12 17 55	86 3 12	148 20 17
8	3 31 44	1 34 53	35 51 14	156 23 6
9	22 25 35	19 58 15	11 28 16	195 6 8
10	11 47 37	9 15 12	321 16 18	203 8 53
11	1 9 29	358 32 9	271 4 19	211 11 43
12	20 3 31	16 55 31	246 41 21	249 54 46
13	9 25 32	6 12 29	196 29 23	257 57 33
14	28 19 34	24 35 51	172 6 25	296 40 36
15	17 41 26	13 52 48	121 54 26	304 43 25
16	7 3 28	3 9 45	71 42 28	312 46 11
17	25 57 19	21 33 7	47 19 30	351 29 14
18	15 19 21	10 50 5	357 7 32	359 32 1
19	4 41 13	0 7 2	306 55 33	7 34 48
20	23 35 14	18 30 24	282 32 35	46 17 50
21	12 57 16	7 47 21	232 20 37	54 20 36
22	2 19 18	357 4 18	182 8 39	62 23 27
23	21 13 9	15 27 41	157 45 41	101 6 29
24	10 35 11	4 44 38	107 33 43	109 9 17

25 0^{d} 57′ 13″ 5^{III} 0^{IV} 354° 1′ 35″ 6^{III} 26^{IV} 57° 21′ 44″ 0^{III} 117° 12′ 5″ 51^{III}

The limits of an eclipse of the sun (حدود كسوف الشمس). — If the mean motion of latitude is from 159° 44′ to 191° 16′, from 0° to 20° 16′, and from 348° 44′ to 360°, the sun can be eclipsed; otherwise it cannot.

The limits of an eclipse of the moon (وحدود كسوف القمر). — If the mean motion of latitude is from 165° 13′ to 194° 47′, from 0° to 14° 47′, and from 345° 13′ to 360°, the moon can be eclipsed; otherwise it cannot.

f. 176,v.

جدول الاوتار المنصفة والقسي المتفاضلة بنصف جزء

Tables of the sines and the arcs, set out at intervals of half a degree.

القسي المتفاضلة بنصف جزء Arcs at intervals of half a degree.	الاوتار المنصفة Sines.	القسي المتفاضلة بنصف جزء Arcs at intervals of half a degree.	الاوتار المنصفة Sines.	القسي المتفاضلة بنصف جزء Arcs at intervals of half a degree.	لاوتار المنصفة Sines.
0°30′ 179°30′	0P31′25″	15°30′ 164°30′	16P 2′ 4″	30°30′ 149°30′	30P27′ 8″
1 0 179 0	1 2 50	16 0 164 0	16 32 18	31 0 149 0	30 54 8
1 30 178 30	1 34 14	16 30 163 30	17 2 27	31 30 148 30	31 21 0
2 0 178 0	2 5 38	17 0 163 0	17 32 33	32 0 148 0	31 47 43
2 30 177 30	2 37 2	17 30 162 30	18 2 32	32 30 147 30	32 14 17
3 0 177 0	3 8 25	18 0 162 0	18 32 27	33 0 147 0	32 40 42
3 30 176 30	3 39 46	18 30 161 30	19 2 18	33 30 146 30	33 6 58
4 0 176 0	4 11 7	19 0 161 0	19 32 3	34 0 146 0	33 33 6
4 30 175 30	4 42 27	19 30 160 30	20 1 42	34 30 145 30	33 59 3
5 0 175 0	5 13 46	20 0 160 0	20 31 16	35 0 145 0	34 24 53
5 30 174 30	5 45 3	20 30 159 30	21 0 45	35 30 144 30	34 50 32
6 0 174 0	6 16 18	21 0 159 0	21 30 7	36 0 144 0	35 16 2
6 30 173 30	6 47 32	21 30 158 30	21 59 25	36 30 143 30	35 41 22
7 0 173 0	7 18 44	22 0 158 0	22 28 35	37 0 143 0	36 6 32
7 30 172 30	7 49 54	22 30 157 30	22 57 39	37 30 142 30	36 31 32
8 0 172 0	8 21 2	23 0 157 0	23 26 38	38 0 142 0	36 56 23
8 30 171 30	8 52 7	23 30 156 30	23 55 30	38 30 141 30	37 21 3
9 0 171 0	9 23 10	24 0 156 0	24 24 15	39 0 141 0	37 45 34
9 30 170 30	9 54 10	24 30 155 30	24 52 54	39 30 140 30	38 9 53
10 0 170 0	10 25 8	25 0 155 0	25 21 25	40 0 140 0	38 34 2
10 30 169 30	10 56 3	25 30 154 30	25 49 50	40 30 139 30	38 58 1
11 0 169 0	11 26 55	26 0 154 0	26 18 8	41 0 139 0	39 21 49
11 30 168 30	11 57 44	26 30 153 30	26 46 19	41 30 138 30	39 45 26
12 0 168 0	12 28 29	27 0 153 0	27 14 22	42 0 138 0	40 8 52
12 30 167 30	12 59 11	27 30 152 30	27 42 18	42 30 137 30	40 32 8
13 0 167 0	13 29 49	28 0 152 0	28 10 6	43 0 137 0	40 55 12
13 30 166 30	14 0 25	28 30 151 30	28 37 46	43 30 136 30	41 18 4
14 0 166 0	14 30 55	29 0 151 0	29 5 19	44 0 136 0	41 40 46
14 30 165 30	15 1 22	29 30 150 30	29 32 45	44 30 135 30	42 3 16
15 0 165 0	15 31 45	30 0 150 0	30 0 0	45 0 135 0	42 25 35

f. 177,r.

تمام جداول الاوتار المنصفة والقسي المتفاضلة بنصف جزء

End of the tables of the sines and the arcs, set out at intervals of half a degree.

Arcs at intervals of half a degree.	Sines.	Arcs at intervals of half a degree.	Sines.	Arcs at intervals of half a degree.	Sines.
45°30′ 134°30′	42^{p}47′ 42″	60°30′ 119°30′	52^{p}13′ 17″	75°30′ 104°30′	58^{p} 5′ 20″
46 0 134 0	43 9 38	61 0 119 0	52 28 38	76 0 104 0	58 13 4
46 30 133 30	43 31 21	61 30 118 30	52 43 45	76 30 103 30	58 20 32
47 0 133 0	43 52 53	62 0 118 0	52 58 37	77 0 103 0	58 27 44
47 30 132 30	44 14 12	62 30 117 30	53 13 14	77 30 102 30	58 34 40
48 0 132 0	44 35 19	63 0 117 0	53 27 38	78 0 102 0	58 41 20
48 30 131 30	44 56 15	63 30 116 30	53 41 46	78 30 101 30	58 47 44
49 0 131 0	45 16 58	64 0 116 0	53 55 40	79 0 101 0	58 53 51
49 30 130 30	45 37 28	64 30 115 30	54 9 18	79 30 100 30	58 59 43
50 0 130 0	45 57 46	65 0 115 0	54 22 43	80 0 100 0	59 5 19
50 30 129 30	46 17 51	65 30 114 30	54 35 52	80 30 99 30	59 10 38
51 0 129 0	46 37 44	66 0 114 0	54 48 46	81 0 99 0	59 15 41
51 30 128 30	46 57 24	66 30 113 30	55 1 25	81 30 98 30	59 20 28
52 0 128 0	47 16 50	67 0 113 0	55 13 49	82 0 98 0	59 24 58
52 30 127 30	47 36 4	67 30 112 30	55 25 58	82 30 97 30	59 29 12
53 0 127 0	47 55 5	68 0 112 0	55 37 52	83 0 97 0	59 33 10
53 30 126 30	48 13 53	68 30 111 30	55 49 30	83 30 96 30	59 36 52
54 0 126 0	48 32 28	69 0 111 0	56 0 53	84 0 96 0	59 40 17
54 30 125 30	48 50 49	69 30 110 30	56 12 1	84 30 95 30	59 43 26
55 0 125 0	49 8 57	70 0 110 0	56 22 54	85 0 95 0	59 46 19
55 30 124 30	49 26 51	70 30 109 30	56 33 31	85 30 94 30	59 48 54
56 0 124 0	49 44 32	71 0 109 0	56 43 52	86 0 94 0	59 51 14
56 30 123 30	50 1 59	71 30 108 30	56 53 58	86 30 93 30	59 53 17
57 0 123 0	50 19 13	72 0 108 0	57 3 48	87 0 93 0	59 55 3
57 30 122 30	50 36 12	72 30 107 30	57 13 23	87 30 92 30	59 56 35
58 0 122 0	50 52 59	73 0 107 0	57 22 42	88 0 92 0	59 57 49
58 30 121 30	51 9 31	73 30 106 30	57 31 45	88 30 91 30	59 58 46
59 0 121 0	51 25 48	74 0 106 0	57 40 33	89 0 91 0	59 59 27
59 30 120 30	51 41 52	74 30 105 30	57 49 4	89 30 90 30	59 59 52
60 0 120 0	51 57 42	75 0 105 0	57 57 20	90 0 90 0	60 0 0

f. 177,v.

ابتداء جداول ميل الشمس عن فلك معدل النهار في دائرة نصف النهار
Beginning of the tables of the declination of the sun from the equator, in the meridian.

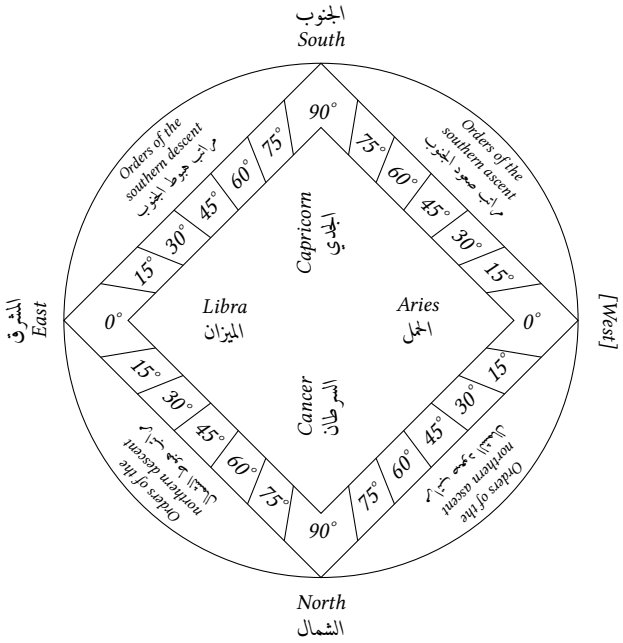
البعد من دائرة معدل النهار Declination.	سطور العدد المربع من نطاق البروج The fourfold numbers of the zodiac.				Declination.	The fourfold numbers of the zodiac.			
	حمل Aries.	سنبلة Virgo.	ميزان Libra.	حوت Pisces.		ثور Taurus.	اسد Leo.	عقرب Scorpio.	دلو Aquarius.
0° 24′ 0″	1°	179°	181°	359°	11° 53′ 29″	31°	149°	211°	329°
0 48 0	2	178	182	358	12 14 25	32	148	212	328
1 11 59	3	177	183	357	12 35 10	33	147	213	327
1 35 57	4	176	184	356	12 55 40	34	146	214	326
1 59 54	5	175	185	355	13 15 59	35	145	215	325
2 23 49	6	174	186	354	13 36 4	36	144	216	324
2 47 41	7	173	187	353	13 55 56	37	143	217	323
3 11 31	8	172	188	352	14 15 35	38	142	218	322
3 35 17	9	171	189	351	14 34 58	39	141	219	321
3 59 1	10	170	190	350	14 54 8	40	140	220	320
4 22 42	11	169	191	349	15 13 2	41	139	221	319
4 46 17	12	168	192	348	15 31 41	42	138	222	318
5 9 49	13	167	193	347	15 50 3	43	137	223	317
5 33 16	14	166	194	346	16 8 9	44	136	224	316
5 56 37	15	165	195	345	16 26 0	45	135	225	315
6 19 53	16	164	196	344	16 43 33	46	134	226	314
6 43 4	17	163	197	343	17 0 50	47	133	227	313
7 6 7	18	162	198	342	17 17 48	48	132	228	312
7 29 3	19	161	199	341	17 34 29	49	131	229	311
7 51 53	20	160	200	340	17 50 51	50	130	230	310
8 14 36	21	159	201	339	18 6 55	51	129	231	309
8 37 12	22	158	202	338	18 22 36	52	128	232	308
8 59 37	23	157	203	337	18 38 2	53	127	233	307
9 21 56	24	156	204	336	18 53 7	54	126	234	306
9 44 5	25	155	205	335	19 7 52	55	125	235	305
10 6 5	26	154	206	334	19 22 16	56	124	236	304
10 27 54	27	153	207	333	19 36 19	57	123	237	303
10 49 34	28	152	208	332	19 50 1	58	122	238	302
11 11 3	29	151	209	331	20 3 21	59	121	239	301
11 32 22	30	150	210	330	20 16 20	60	120	240	300

f. 178,r.

تمام جداول ميل الشمس ونطاق البروج
End of the tables of the declination of the sun and of the [arcs] of the zodiac.

Declination.	The fourfold numbers of the zodiac.			
	جوزاء Gemini.	سرطان Cancer.	قوس Sagittarius.	جدي Capricorn.
20° 28′ 57″	61°	119°	241°	299°
20 41 11	62	118	242	298
20 53 2	63	117	243	297
21 4 30	64	116	244	296
21 15 36	65	115	245	295
21 26 17	66	114	246	294
21 36 34	67	113	247	293
21 46 28	68	112	248	292
21 55 56	69	111	249	291
22 5 0	70	110	250	290
22 13 39	71	109	251	289
22 21 52	72	108	252	288
22 29 42	73	107	253	287
22 37 5	74	106	254	286
22 44 2	75	105	255	285
22 50 33	76	104	256	284
22 56 38	77	103	257	283
23 2 16	78	102	258	282
23 7 29	79	101	259	281
23 12 15	80	100	260	280
23 16 34	81	99	261	279
23 20 24	82	98	262	278
23 23 50	83	97	263	277
23 26 48	84	96	264	276
23 29 18	85	95	265	275
23 31 21	86	94	266	274
23 32 56	87	93	267	273
23 34 5	88	92	268	272
23 34 46	89	91	269	271
23 35 0	90	90	270	270

مراتب الميل في الصعود والهبوط وشكلها
The orders of the declination in the ascent and the descent and their figure.



عشرات العدد Tens of the numbers.	مطالع العشرات في الفلك المستقيم Ascensions of the tens in the right sphere.	اوتار المطالع المنصفة Sines of the ascensions.
10°	9° 10′ 46″	**9° 34′ 17″**
20	18 26 51	**18 57 10**
30	27 53 4	**28 3 41**
40	37 33 40	**36 34 36**
50	47 31 25	**44 15 12**
60	57 47 26	**50 45 56**
70	68 20 24	**55 45 50**
80	79 6 35	**58 55 12**
90	90 0 0	**60 0 0**

f. 178,v.

جدول لمعرفة نصف زيادة النهار الاطول من قبل ارتفاع القطب في كل بلد ومعرفة ارتفاع القطب من قبل زيادة النهار الاطول ان شاء الله

Table by which are known half the excess of the longest day in any place of the earth from the altitude of the pole, and the altitude of the pole from the excess of the longest day, God willing.

ارتفاع القطب Altitude of the pole.	نصف زيادة النهار الاطول Half the excess of the longest day.	ارتفاع القطب Altitude of the pole.	نصف زيادة النهار الاطول Half the excess of the longest day.	ارتفاع القطب Altitude of the pole.	نصف زيادة النهار الاطول Half the excess of the longest day.
0° 30′	0° 13′	20° 30′	9° 23′	40° 30′	21° 53′
1 0	0 26	21 0	9 39	41 0	22 18
1 30	0 39	21 30	9 55	41 30	22 43
2 0	0 52	22 0	10 11	42 0	23 9
2 30	1 6	22 30	10 27	42 30	23 35
3 0	1 19	23 0	10 43	43 0	24 2
3 30	1 32	23 30	10 59	43 30	24 29
4 0	1 45	24 0	11 16	44 0	24 56
4 30	1 58	24 30	11 33	44 30	25 24
5 0	2 12	25 0	11 51	45 0	25 53
5 30	2 26	25 30	12 8	45 30	26 22
6 0	2 40	26 0	12 24	46 0	26 52
6 30	2 54	26 30	12 41	46 30	27 23
7 0	3 8	27 0	12 57	47 0	27 55
7 30	3 22	27 30	13 14	47 30	28 27
8 0	3 36	28 0	13 30	48 0	29 0
8 30	3 49	28 30	13 47	48 30	29 32
9 0	4 1	29 0	14 3	49 0	30 6
9 30	4 14	29 30	14 20	49 30	30 43
10 0	4 26	30 0	14 36	50 0	31 22
10 30	4 39	30 30	14 54	50 30	32 0
11 0	4 51	31 0	15 13	51 0	32 39
11 30	5 5	31 30	15 32	51 30	33 18
12 0	5 19	32 0	15 50	52 0	33 57
12 30	5 33	32 30	16 8	52 30	34 39
13 0	5 47	33 0	16 28	53 0	35 24
13 30	6 1	33 30	16 47	53 30	36 9
14 0	6 15	34 0	17 7	54 0	36 53
14 30	6 29	34 30	17 28	54 30	37 43
15 0	6 43	35 0	17 49	55 0	38 34
15 30	6 57	35 30	18 9	55 30	39 26
16 0	7 11	36 0	18 30	56 0	40 20
16 30	7 25	36 30	18 51	56 30	41 17
17 0	7 40	37 0	19 13	57 0	42 14
17 30	7 54	37 30	19 35	57 30	43 15
18 0	8 9	38 0	19 57	58 0	44 19
18 30	8 23	38 30	20 19	58 30	45 25
19 0	8 38	39 0	20 40	59 0	46 35
19 30	8 53	39 30	21 3	59 30	47 49
20 0	9 8	40 0	21 29	60 0	49 7

f. 179,r.

جدول لمعرفة الظل من قبل الارتفاع بالمقدار الذي تكون به اجزاء المقياس يب جزء ومعرفة الارتفاع
من قبل هذا الظل المبسوط وبهذه الجداول وعرف احدهما من قبل الآخر

Table by which the shadow is known from the altitude of the pole, on the reckoning by which the gnomon is divided into 12 parts, and the altitude of the pole from this horizontal shadow.
In these tables each is found from the other.

قوس الارتفاع Arc of the altitude.	اصابع الظل Digits of the shadow.	قوس الارتفاع Arc of the altitude.	اصابع الظل Digits of the shadow.	قوس الارتفاع Arc of the altitude.	اصابع الظل Digits of the shadow.
1	687 dig. 29′	31	19 dig. 58′	61	6 dig. 39′
2	343 38	32	19 12	62	6 23
3	228 58	33	18 29	63	6 7
4	171 36	34	17 47	64	5 51
5	137 10	35	17 8	65	5 36
6	114 10	36	16 31	66	5 21
7	97 44	37	15 55	67	5 6
8	85 23	38	15 21	68	4 51
9	75 46	39	14 49	69	4 36
10	68 3	40	14 18	70	4 22
11	61 44	41	13 48	71	4 8
12	56 27	42	13 20	72	3 54
13	51 59	43	12 52	73	3 40
14	48 8	44	12 26	74	3 26
15	44 47	45	12 0	75	3 13
16	41 51	46	11 35	76	2 59
17	39 15	47	11 11	77	2 46
18	36 56	48	10 48	78	2 32
19	34 51	49	10 26	79	2 19
20	32 58	50	10 4	80	2 6
21	31 16	51	9 43	81	1 54
22	29 42	52	9 22	82	1 41
23	28 16	53	9 3	83	1 28
24	26 57	54	8 43	84	1 15
25	25 44	55	8 24	85	1 2
26	24 36	56	8 6	86	0 50
27	23 33	57	7 48	87	0 37
28	22 34	58	7 30	88	0 25
29	21 40	59	7 13	89	0 12
30	20 47	60	6 56	90	0 0

f. 179,v.

ابتداء جداول مطالع البروج في الفلك المستقيم وتعَديل الايام بلياليها

Beginning of the tables of the ascensions of the signs in the right sphere and of the equation of the days with their nights.

العدد المشترك Numbers common [to all].	الجدي درج المطالع Capricorn. Degrees of the ascensions.	تعديل الايام بلياليها Equation of the days with their nights.	الدلو درج المطالع Aquarius. Degrees of the ascensions.	Equation of the days with their nights.	الحوت درج المطالع Pisces. Degrees of the ascensions.	Equation of the days with their nights.
1	1° 5′	3° 41′	33° 15′	0° 37′	63° 4′	0° 12′
2	2 11	3 33	34 17	0 33	64 1	0 15
3	3 17	3 25	35 19	0 30	64 58	0 17
4	4 22	3 18	36 21	0 26	65 55	0 20
5	5 28	3 11	37 23	0 23	66 52	0 22
6	6 33	3 4	38 25	0 20	67 48	0 25
7	7 38	2 57	39 26	0 18	68 45	0 28
8	8 43	2 50	40 27	0 15	69 41	0 32
9	9 48	2 43	41 28	0 13	70 37	0 35
10	10 53	2 37	42 29	0 10	71 34	0 38
11	11 58	2 31	43 29	0 8	72 29	0 42
12	13 3	2 24	44 30	0 6	73 25	0 45
13	14 8	2 17	45 30	0 5	74 21	0 49
14	15 13	2 10	46 30	0 4	75 17	0 54
15	16 18	2 3	47 30	0 3	76 12	0 58
16	17 23	1 56	48 30	0 2	77 8	1 3
17	18 27	1 49	49 29	0 1	78 3	1 7
18	19 31	1 43	50 28	0 0	78 59	1 12
19	20 35	1 37	51 27	0 0	79 54	1 16
20	21 39	1 31	52 26	0 1	80 49	1 21
21	22 43	1 25	53 25	0 1	81 44	1 25
22	23 47	1 19	54 24	0 1	82 40	1 30
23	24 51	1 14	55 22	0 2	83 35	1 34
24	25 55	1 9	56 20	0 2	84 30	1 39
25	26 58	1 4	57 19	0 3	85 25	1 44
26	28 1	0 59	58 17	0 4	86 20	1 49
27	29 4	0 54	59 14	0 6	87 15	1 54
28	30 7	0 49	60 12	0 7	88 10	2 0
29	31 10	0 45	61 10	0 8	89 5	2 5
30	32 13	0 41	62 7	0 10	90 0	2 10

f. 180,r.

من جداول مطالع البروج في الفلك المستقيم وتعديل الايام بلياليها

From the tables of the ascensions of the signs in the right sphere and of the equation of the days with their nights.

Numbers common [to all].	الحمل درج المطالع Aries. Degrees of the ascensions.	Equation of the days with their nights.	الثور درج المطالع Taurus. Degrees of the ascensions.	Equation of the days with their nights.	الجوزاء درج المطالع Gemini. Degrees of the ascensions.	Equation of the days with their nights.
1	90° 55′	2° 15′	118° 50′	4° 41′	148° 50′	5° 33′
2	91 50	2 20	119 48	4 45	149 53	5 33
3	92 45	2 25	120 46	4 48	150 56	5 33
4	93 40	2 30	121 43	4 52	151 59	5 33
5	94 35	2 35	122 41	4 55	153 2	5 33
6	95 30	2 41	123 40	4 59	154 5	5 32
7	96 25	2 46	124 38	5 2	155 9	5 29
8	97 20	2 52	125 37	5 5	156 13	5 26
9	98 16	2 57	126 35	5 8	157 17	5 23
10	99 11	3 3	127 34	5 10	158 21	5 20
11	100 6	3 8	128 33	5 13	159 25	5 17
12	101 1	3 14	129 32	5 16	160 29	5 15
13	101 57	3 19	130 31	5 18	161 33	5 12
14	102 52	3 24	131 30	5 20	162 37	5 10
15	103 48	3 29	132 30	5 22	163 42	5 8
16	104 43	3 34	133 30	5 23	164 47	5 5
17	105 39	3 39	134 30	5 25	165 52	5 0
18	106 35	3 44	135 30	5 27	166 57	4 58
19	107 31	3 49	136 31	5 28	168 2	4 56
20	108 27	3 55	137 31	5 29	169 7	4 53
21	109 23	4 0	138 32	5 30	170 12	4 49
22	110 19	4 5	139 33	5 31	171 17	4 46
23	111 15	4 11	140 34	5 32	172 22	4 43
24	112 12	4 16	141 36	5 32	173 27	4 40
25	113 8	4 20	142 37	5 32	174 32	4 36
26	114 5	4 23	143 39	5 32	175 38	4 33
27	115 2	4 27	144 41	5 32	176 43	4 30
28	115 59	4 31	145 43	5 33	177 49	4 27
29	116 56	4 34	146 45	5 33	178 54	4 24
30	117 53	4 38	147 47	5 33	180 0	4 20

f. 180,v.

من جداول مطالع البروج في الفلك المستقيم وتعديل الايام بلياليها
From the tables of the ascensions of the signs in the right sphere, and of the equation of the days with their nights.

Numbers common [to all].	السرطان درج المطالع Cancer. Degrees of the ascensions.	Equation of the days with their nights.	الاسد درج المطالع Leo. Degrees of the ascensions.	Equation of the days with their nights.	السنبلة درج المطالع Virgo. Degrees of the ascensions.	Equation of the days with their nights.
1	181° 5′	4° 17′	213° 15′	3° 4′	243° 4′	3° 48′
2	182 10	4 14	214 17	3 4	244 1	3 51
3	183 16	4 10	215 19	3 4	244 58	3 54
4	184 22	4 7	216 21	3 4	245 55	3 57
5	185 28	4 3	217 23	3 4	246 52	4 1
6	186 33	4 0	218 25	3 4	247 48	4 6
7	187 38	3 57	219 26	3 4	248 45	4 11
8	188 43	3 55	220 27	3 4	249 41	4 16
9	189 48	3 51	221 28	3 4	250 37	4 21
10	190 53	3 48	222 29	3 5	251 33	4 26
11	191 58	3 45	223 29	3 6	252 29	4 31
12	193 3	3 42	224 30	3 7	253 25	4 36
13	194 8	3 39	225 30	3 8	254 21	4 41
14	195 13	3 36	226 30	3 9	255 17	4 46
15	196 18	3 33	227 30	3 10	256 12	4 51
16	197 23	3 30	228 30	3 11	257 8	4 56
17	198 27	3 27	229 29	3 13	258 3	5 1
18	199 31	3 25	230 28	3 14	258 59	5 6
19	200 35	3 23	231 27	3 16	259 54	5 11
20	201 39	3 21	232 26	3 17	260 49	5 16
21	202 43	3 19	233 25	3 19	261 44	5 21
22	203 47	3 17	234 24	3 21	262 40	5 26
23	204 51	3 15	235 22	3 24	263 35	5 31
24	205 55	3 13	236 20	3 27	264 30	5 36
25	206 58	3 11	237 19	3 30	265 25	5 41
26	208 1	3 9	238 17	3 33	266 20	5 45
27	209 4	3 8	239 14	3 36	267 15	5 50
28	210 7	3 7	240 12	3 39	268 10	5 55
29	211 10	3 6	241 10	3 42	269 5	5 59
30	212 13	3 5	242 7	3 45	270 0	6 4

f. 181,r.

من جداول مطالع البروج في الفلك المستقيم وتعديل الايام بلياليها

End of the tables of the ascensions of the signs in the right sphere, and of the equation of the days with their nights.

Numbers common [to all].	الميزان درج المطالع Libra. Degrees of the ascensions.	Equation of the days with their nights.	العقرب درج المطالع Scorpio. Degrees of the ascensions.	Equation of the days with their nights.	القوس درج المطالع Sagittarius. Degrees of the ascensions.	Equation of the days with their nights.
1	270° 55′	6° 9′	298° 50′	7° 49′	328° 50′	6° 59′
2	271 50	6 13	299 48	7 50	329 53	6 55
3	272 45	6 18	300 46	7 51	330 56	6 50
4	273 40	6 22	301 43	7 52	331 59	6 43
5	274 35	6 27	302 41	7 52	333 2	6 38
6	275 30	6 32	303 40	7 53	334 5	6 32
7	276 25	6 36	304 38	7 53	335 9	6 27
8	277 20	6 41	305 36	7 54	336 13	6 22
9	278 16	6 45	306 35	7 54	337 17	6 17
10	279 11	6 49	307 34	7 54	338 21	6 12
11	280 6	6 53	308 33	7 53	339 25	6 6
12	281 1	6 57	309 32	7 52	340 29	5 59
13	281 57	7 1	310 31	7 51	341 33	5 53
14	282 52	7 6	311 30	7 50	342 37	5 46
15	283 48	7 10	312 30	7 49	343 42	5 39
16	284 43	7 15	313 30	7 49	344 47	5 32
17	285 39	7 19	314 30	7 47	345 52	5 25
18	286 35	7 23	315 30	7 45	346 57	5 18
19	287 31	7 27	316 31	7 43	348 2	5 11
20	288 27	7 31	317 31	7 41	349 7	5 4
21	289 23	7 34	318 32	7 39	350 12	4 57
22	290 19	7 36	319 33	7 37	351 17	4 49
23	291 15	7 38	320 35	7 34	352 22	4 42
24	292 12	7 39	321 36	7 31	353 27	4 34
25	293 8	7 41	322 37	7 27	354 32	4 27
26	294 5	7 42	323 39	7 23	355 38	4 19
27	295 2	7 44	324 41	7 20	356 43	4 12
28	295 59	7 45	325 43	7 16	357 49	4 4
29	296 56	7 47	326 45	7 12	358 54	3 56
30	297 53	7 48	327 47	7 7	360 0	3 49

f. 181,v.

جدول مطالع البروج في الاقاليم بتفاضل ربع ساعة وعشرة اجزاء في العدد

Tables of the ascensions of the signs, by tens of degrees, in the climes set out at intervals of a quarter of an hour.

اسماء البروج Names of the signs.	عدد الاجزاء Tens of degrees.	Clime I, lat. 16° 39′. Hours 13h. المطالع Degrees of the ascensions.	Lat. 20° 28′. Hours 13h 15m. Degrees of the ascensions.	Clime II, lat. 24° 5′. Hours 13h 30m. Degrees of the ascensions.	Lat. 27° 28′. Hours 13h 45m. Degrees of the ascensions.	Clime III, lat. 30° 40′. Hours 14h. Degrees of the ascensions.	Lat. 33° 37′. Hours 14h 15m. Degrees of the ascensions.
Aries.	10	7° 59′	7° 42′	7° 24′	7° 6′	6° 49′	6° 32′
	20	16 5	15 30	14 55	14 20	13 45	13 11
	30	24 23	23 31	22 39	21 47	20 56	20 5
Taurus.	40	33 0	31 52	30 44	29 36	28 29	27 22
	50	42 3	40 38	39 15	37 53	36 31	35 10
	60	51 27	49 52	48 17	46 43	45 8	43 35
Gemini.	70	61 23	59 38	57 55	56 10	54 25	52 42
	80	71 45	69 55	68 4	66 13	64 24	62 34
	90	82 30	80 37	78 45	76 52	75 0	73 7
Cancer.	100	93 31	91 41	89 51	88 0	86 10	84 19
	110	104 41	102 57	101 13	99 28	97 44	96 0
	120	115 52	114 17	112 43	111 8	109 34	108 0
Leo.	130	127 0	125 35	124 12	122 50	121 28	120 7
	140	137 52	136 44	135 36	134 29	133 22	132 15
	150	148 37	147 45	146 53	146 1	145 10	144 19
Virgo.	160	159 11	158 36	158 1	157 26	156 51	156 17
	170	169 38	169 20	169 2	168 45	168 27	168 10
	180	180 0	180 0	180 0	180 0	180 0	180 0
Libra.	190	190 22	190 40	190 58	191 15	191 33	191 50
	200	200 49	201 24	201 59	202 34	203 9	203 43
	210	211 23	212 21	213 7	213 59	214 50	215 41
Scorpio.	220	222 8	223 16	224 24	225 31	226 38	227 45
	230	233 4	234 26	235 48	237 10	238 32	239 53
	240	244 8	245 43	247 17	248 52	250 26	252 0
Sagittarius.	250	255 19	257 3	258 47	267 32	260 16	264 0
	260	266 29	268 19	270 9	272 0	273 50	275 41
	270	277 30	279 23	281 15	283 8	285 0	286 52
Capricorn.	280	288 15	290 5	291 56	293 47	295 36	297 27
	290	298 37	300 22	302 6	303 50	305 35	307 18
	300	308 33	310 8	311 43	313 17	314 52	316 25
Aquarius.	310	317 59	319 22	320 45	322 7	323 29	324 50
	320	327 0	328 8	329 16	330 24	331 31	332 38
	330	335 37	336 30	337 21	338 13	339 4	339 55
Pisces.	340	343 55	344 30	345 5	345 40	346 15	346 49
	350	352 1	352 18	352 36	352 54	353 11	353 28
	360	360 0	360 0	360 0	360 0	360 0	360 0

f. 182,r.

جدول مطالع البروج في الاقاليم بتفاضل ربع ساعة وعشرة اجزاء في العدد

Tables of the ascensions of the signs, by tens of degrees, in the climes set out at intervals of a quarter of an hour.

Names of the signs.	Tens of degrees.	Clime IV, lat. 36° 22′. Hours 14^h 30^m. Degrees of the ascens.	Lat. 38° 54′. Hours 14^h 45^m. Degrees of the ascens.	Clime V, lat. 41° 15′. Hours 15^h. Degrees of the ascens.	Lat. 43° 25′. Hours 15^h 15^m. Degrees of the ascens.	Clime VI, lat. 45° 22′. Hours 15^h 30^m. Degrees of the ascens.	Lat. 47° 12′. Hours 15^h 45^m. Degrees of the ascens.	Clime VII, lat. 48° 53′. Hours 16^h. Degrees of the ascens.
Aries.	10	6° 14′	5° 57′	5° 40′	5° 24′	5° 8′	4° 52′	4° 36′
	20	12 37	12 3	11 29	10 57	10 24	9 52	9 21
	30	19 14	18 24	17 35	16 46	15 57	15 9	14 22
Taurus.	40	26 16	25 10	24 4	23 0	21 55	20 52	19 49
	50	33 49	32 28	31 8	29 48	28 29	27 11	25 53
	60	42 0	40 27	38 54	37 21	35 49	34 17	32 46
Gemini.	70	50 59	49 15	47 31	45 47	44 4	42 22	40 39
	80	60 43	58 53	57 4	55 14	53 23	51 32	49 42
	90	71 15	69 22	67 30	65 37	63 45	61 52	60 0
Cancer.	100	82 29	80 39	78 50	76 59	75 9	73 19	71 29
	110	94 16	92 33	90 50	89 6	87 24	85 40	83 58
	120	106 26	104 52	103 19	101 46	100 14	98 42	97 11
Leo.	130	118 46	117 25	116 4	114 45	113 26	112 8	110 50
	140	131 8	130 2	128 56	127 52	126 48	125 44	124 42
	150	143 28	142 38	141 48	140 59	140 11	139 23	138 36
Virgo.	160	155 43	155 9	154 35	154 3	153 30	152 58	152 26
	170	167 53	167 36	167 19	167 3	166 46	166 30	166 15
	180	180 0	180 0	180 0	180 0	180 0	180 0	180 0
Libra.	190	192 7	192 24	192 41	192 57	193 14	193 30	193 45
	200	204 17	204 51	205 25	205 57	206 30	207 2	207 36
	210	216 32	217 22	218 12	219 1	219 49	220 37	221 24
Scorpio.	220	228 52	229 58	231 3	232 8	233 12	234 14	235 18
	230	241 14	242 35	243 57	245 15	246 34	247 52	249 10
	240	253 34	255 8	256 41	258 14	259 46	261 18	262 49
Sagittarius.	250	265 44	267 27	269 10	270 54	272 37	274 20	276 2
	260	277 31	279 21	281 10	283 1	284 51	286 41	288 31
	270	288 45	290 37	292 30	294 23	296 15	298 7	300 0
Capricorn.	280	299 17	301 7	302 56	304 46	306 37	308 28	310 18
	290	309 1	310 46	312 29	314 13	315 56	317 38	319 21
	300	318 0	319 33	321 6	322 39	324 11	325 43	327 14
Aquarius.	310	326 11	327 32	328 52	330 12	331 31	332 49	334 7
	320	333 44	334 50	335 56	337 0	338 5	339 8	340 11
	330	340 46	341 36	342 26	343 14	344 3	344 51	345 38
Pisces.	340	347 23	347 57	348 31	349 3	349 36	350 8	350 39
	350	353 46	354 3	354 20	354 36	354 52	355 8	355 24
	360	360 0	360 0	360 0	360 0	360 0	360 0	360 0

f. 182,v.

جدول مطالع البروج وازمان الساعات بمكة وبغداد وحران

The ascensions of the signs and the times of the hours in the cities of Mekkah, Baghdād and Ḥarrān.

Names of the signs.	Tens of degrees.	درج المطالع مكة Ascensions Mekkah, lat. 21° 40′.	ازمان الساعات Times of the hours $13^h\ 20^m$.	درج المطالع بغداد Ascensions Baghdād, lat. 33° 9′.	ازمان الساعات Times of the hours $14^h\ 12^m\ 32^s$.	درج المطالع حران Ascensions Ḥarrān, lat. 36° 40′.	ازمان الساعات Times of the hours $14^h\ 32^m$.
Aries.	10	7° 36′	15° 16′	6° 34′	15° 26′	6° 12′	15° 30′
	20	15 18	15 31	13 16	15 52	12 32	15 59
	30	23 14	15 47	20 13	16 17	19 7	16 26
Taurus.	40	31 30	16 1	27 33	16 40	26 6	16 55
	50	40 11	16 13	35 23	17 1	33 37	17 19
	60	49 21	16 24	43 50	17 20	41 47	17 40
Gemini.	70	59 4	16 33	52 58	17 34	50 43	17 56
	80	69 19	16 38	62 51	17 43	60 27	18 7
	90	80 1	16 40	73 26	17 46	70 59	18 10
Cancer.	100	91 5	16 38	84 37	17 43	82 13	18 7
	110	102 23	16 33	96 17	17 34	94 2	17 56
	120	113 46	16 24	108 15	17 20	106 12	17 40
Leo.	130	125 8	16 13	120 20	17 1	118 34	17 19
	140	136 22	16 1	132 26	16 40	130 59	16 55
	150	147 28	15 47	144 27	16 17	143 21	16 26
Virgo.	160	158 24	15 31	156 23	15 52	155 38	15 59
	170	169 14	15 16	168 13	15 26	167 51	15 30
	180	180 0	15 0	180 0	15 0	180 0	15 0
Libra.	190	190 46	14 44	191 47	14 34	192 9	14 30
	200	201 36	14 29	203 37	14 8	204 22	14 1
	210	212 32	14 13	215 33	13 43	216 39	13 34
Scorpio.	220	223 38	13 59	227 34	13 20	229 1	13 5
	230	234 52	13 47	239 40	12 59	241 26	12 41
	240	246 14	13 36	251 45	12 40	253 48	12 20
Sagittarius.	250	257 37	13 26	263 43	12 26	265 58	12 3
	260	268 55	13 22	275 23	12 17	277 47	11 53
	270	279 59	13 20	286 34	12 14	289 1	11 50
Capricorn.	280	290 41	13 22	297 9	12 17	299 33	11 53
	290	300 56	13 26	307 2	12 26	309 17	12 3
	300	310 39	13 36	316 10	12 40	318 13	12 20
Aquarius.	310	319 49	13 47	324 37	12 59	326 23	12 41
	320	328 30	13 59	332 27	12 20	333 54	13 5
	330	336 46	14 13	339 47	12 43	340 53	13 34
Pisces.	340	344 42	14 29	346 44	14 8	347 28	14 1
	350	352 24	14 44	353 26	14 34	353 48	14 30
	360	360 0	15 0	360 0	15 0	360 0	15 0

f. 183,v.-184,r.

اول جداول مطالع البروج بمدينة الرقة والعرض لو والساعات يد كع

Beginning of the tables of the ascensions of the signs in the city of ar-Raqqah, whose latitude is 36° 0′ and whose hours are 14^{h} 28^{m}.

العدد المشترك Common numbers.	الحمل — المطالع Ascensions of Aries.	ازمان الساعات Times of the hours.	الثور — المطالع Ascensions of Taurus.	Times of the hours.	الجوزاء — المطالع Ascensions of Gemini.	Times of the hours.
1	0° 37′	15° 3′	20° 3′	16° 28′	43° 4′	17° 38′
2	1 15	15 6	20 44	16 31	43 56	17 40
3	1 53	15 9	21 26	16 33	44 49	17 41
4	2 31	15 12	22 8	16 36	45 42	17 43
5	3 8	15 15	22 50	16 38	46 36	17 44
6	3 46	15 17	23 33	16 41	47 30	17 46
7	4 24	15 20	24 16	16 44	48 24	17 47
8	5 2	15 23	24 59	16 46	49 19	17 49
9	5 40	15 26	25 43	16 49	50 15	17 50
10	6 18	15 29	26 26	16 51	51 11	17 52
11	6 56	15 32	27 9	16 54	52 8	17 53
12	7 34	15 34	27 53	16 56	53 5	17 54
13	8 12	15 37	28 37	16 59	54 2	17 55
14	8 50	15 40	29 22	17 1	55 0	17 56
15	9 28	15 43	30 7	17 4	55 58	17 57
16	10 6	15 46	30 52	17 6	56 57	17 58
17	10 44	15 49	31 38	17 9	57 56	17 59
18	11 23	15 52	32 24	17 11	58 56	18 0
19	12 2	15 55	33 11	17 13	59 57	18 1
20	12 41	15 58	33 58	17 16	60 58	18 2
21	13 20	16 1	34 45	17 18	61 59	18 2
22	13 59	16 3	35 33	17 20	63 0	18 3
23	14 38	16 6	36 22	17 22	64 3	18 3
24	15 18	16 9	37 11	17 24	65 5	18 4
25	15 57	16 12	38 0	17 26	66 8	18 4
26	16 37	16 15	38 50	17 28	67 11	18 4
27	17 18	16 17	39 40	17 30	68 15	18 5
28	17 59	16 20	40 31	17 32	69 20	18 5
29	18 40	16 23	41 22	17 34	70 25	18 5
30	19 21	16 25	42 13	17 36	71 30	18 5

f. 184,r.-v.

من جداول مطالع البروج بمدينة الرقة
From the tables of the ascensions of the signs in the city of ar-Raqqah.

Common numbers.	السرطان — المطالع Ascensions of Cancer.	Times of the hours.	الاسد — المطالع Ascensions of Leo.	Times of the hours.	السنبلة — المطالع Ascensions of Virgo.	Times of the hours.
1	72° 36′	18° 5′	107° 51′	17° 34′	144° 48′	16° 23′
2	73 42	18 5	109 5	17 32	146 1	16 20
3	74 48	18 5	110 18	17 30	147 14	16 17
4	75 55	18 4	111 32	17 28	148 27	16 15
5	77 2	18 4	112 45	17 26	149 41	16 12
6	78 9	18 4	113 59	17 24	150 54	16 9
7	79 17	18 3	115 13	17 22	152 8	16 6
8	80 26	18 3	116 28	17 20	153 21	16 3
9	81 35	18 2	117 43	17 18	154 34	16 1
10	82 44	18 2	118 58	17 16	155 47	15 58
11	83 53	18 1	120 12	17 13	157 0	15 55
12	85 3	18 0	121 26	17 11	158 13	15 52
13	86 13	17 59	122 40	17 9	159 27	15 49
14	87 24	17 58	123 54	17 6	160 40	15 46
15	88 34	17 57	125 8	17 4	161 53	15 43
16	89 45	17 56	126 21	17 1	163 6	15 40
17	90 56	17 55	127 35	16 59	164 19	15 37
18	92 7	17 54	128 49	16 56	165 32	15 34
19	93 18	17 53	130 4	16 54	166 44	15 32
20	94 30	17 52	131 18	16 51	167 56	15 29
21	95 42	17 50	132 33	16 49	169 9	15 26
22	96 54	17 49	133 47	16 46	170 22	15 23
23	98 7	17 47	135 1	16 44	171 34	15 20
24	99 20	17 46	136 14	16 41	172 46	15 17
25	100 32	17 44	137 28	16 38	173 58	15 15
26	101 44	17 43	138 42	16 36	175 11	15 12
27	102 57	17 41	139 55	16 33	176 23	15 9
28	104 10	17 40	141 8	16 31	177 35	15 6
29	105 23	17 38	142 22	16 28	178 47	15 3
30	106 37	17 36	143 35	16 25	180 0	15 0

f. 185,r.-v.

من جداول مطالع البروج بمدينة الرقة
From the tables of the ascensions of the signs in the city of ar-Raqqah.

Common numbers.	الميزان — المطالع Ascensions of Libra.	Times of the hours.	العقرب — المطالع Ascensions of Scorpio.	Times of the hours.	القوس — المطالع Ascensions of Sagittarius.	Times of the hours.
1	181° 13′	14° 57′	217° 38′	13° 32′	254° 37′	12° 22′
2	182 25	14 54	218 52	13 29	255 50	12 20
3	183 37	14 51	220 5	13 27	257 3	12 19
4	184 49	14 48	221 18	13 24	258 16	12 17
5	186 2	14 45	222 32	13 22	259 28	12 16
6	187 14	14 43	223 46	13 19	260 40	12 14
7	188 26	14 40	224 59	13 16	261 53	12 13
8	189 38	14 37	226 13	13 14	263 6	12 11
9	190 51	14 34	227 27	13 11	264 18	12 10
10	192 3	14 31	228 42	13 9	265 30	12 8
11	193 16	14 28	229 56	13 6	266 42	12 7
12	194 28	14 26	231 11	13 3	267 53	12 6
13	195 41	14 23	232 25	13 1	269 4	12 5
14	196 54	14 20	233 39	12 59	270 15	12 4
15	198 7	14 17	234 53	12 56	271 26	12 3
16	199 20	14 14	236 6	12 54	272 36	12 2
17	200 33	14 11	237 20	12 51	273 47	12 1
18	201 47	14 8	238 34	12 49	274 57	12 0
19	203 0	14 5	239 47	12 47	276 7	11 59
20	204 13	14 2	241 2	12 44	277 16	11 58
21	205 26	13 59	242 17	12 42	278 25	11 58
22	206 39	13 57	243 32	12 40	279 34	11 57
23	207 52	13 54	244 46	12 38	280 43	11 57
24	209 6	13 51	246 1	12 36	281 51	11 56
25	210 19	13 48	247 15	12 34	282 58	11 56
26	211 33	13 45	248 28	12 32	284 5	11 56
27	212 46	13 43	249 42	12 30	285 12	11 55
28	213 59	13 40	250 55	12 28	286 18	11 55
29	215 12	13 36	252 9	12 26	287 24	11 55
30	216 25	13 34	253 23	12 24	288 30	11 55

f. 185,v.-186,r.

تمام جداول مطالع البروج بمدينة الرقة ولعرض لو وساعات يد كح

End of the tables of the ascensions of the signs in the city of ar-Raqqah, whose latitude is 36° 0′ and whose hours are 14^{h} 28^{m}.

Common numbers.	الجدي — المطالع Ascensions of Capricorn.	Times of the hours.	الدلو — المطالع Ascensions of Aquarius.	Times of the hours.	الحوت — المطالع Ascensions of Pisces.	Times of the hours.
1	289° 35′	11° 55′	318° 38′	12° 26′	341° 20′	13° 37′
2	290 40	11 55	319 29	12 28	342 1	13 40
3	291 45	11 55	320 20	12 30	342 42	13 43
4	292 49	11 56	321 10	12 32	343 23	13 45
5	293 52	11 56	322 0	12 34	344 3	13 48
6	294 55	11 56	322 49	12 36	344 42	13 51
7	295 57	11 57	323 38	12 38	345 22	13 54
8	297 0	11 57	324 27	12 40	346 1	13 57
9	298 1	11 58	325 15	12 42	346 40	13 59
10	299 2	11 58	326 2	12 44	347 19	14 2
11	300 3	11 59	326 49	12 47	347 58	14 5
12	301 4	12 0	327 36	12 49	348 37	14 8
13	302 4	12 1	328 22	12 51	349 15	14 11
14	303 3	12 2	329 8	12 54	349 54	14 14
15	304 2	12 3	329 53	12 56	350 32	14 17
16	305 0	12 4	330 38	12 59	351 10	14 20
17	305 58	12 5	331 23	13 1	351 48	14 23
18	306 55	12 6	332 7	13 4	352 26	14 26
19	307 52	12 7	332 51	13 6	353 4	14 28
20	308 49	12 8	333 34	13 9	353 42	14 31
21	309 45	12 10	334 17	13 11	354 20	14 34
22	310 41	12 11	335 1	13 14	354 58	14 37
23	311 36	12 13	335 44	13 16	355 36	14 40
24	312 30	12 14	336 27	13 19	356 14	14 43
25	313 24	12 16	337 10	13 22	356 52	14 45
26	314 18	12 17	337 52	13 24	357 29	14 48
27	315 11	12 19	338 34	13 27	358 7	14 51
28	316 4	12 20	339 16	13 29	358 44	14 54
29	316 56	12 22	339 57	13 32	359 22	14 57
30	317 47	12 24	340 39	13 33	360 0	15 0

f. 186,v.

جدول حركة الشمس والقمر وحاصة القمر في السنين الرومية المجموعة

Table of the motion of the sun, the moon, the anomaly of the moon [and the northern node] in collected years of the Romans [from the era].

سنو الروم المجموعة Collected Roman years.	وسط الشمس Mean longitude of the sun.	وسط القمر Mean longitude of the moon.	حاصة القمر Anomaly of the moon.	العقد الشمالي Longitude of the northern node.
931	340° 23′ 37″	228° 58′ 5″	312° 14′ 10″	187° 53′ 56″
951	340 34 48	2 33 38	351 56 9	214 42 19
971	340 45 58	136 9 12	31 38 8	241 30 43
991	340 57 8	269 44 45	71 20 8	268 19 6
1011	341 8 18	43 20 18	111 2 7	295 7 29
1031	341 19 28	176 55 51	150 44 6	321 55 53
1051	341 30 39	310 31 25	190 26 5	348 44 16
1071	341 41 49	84 6 58	230 8 5	15 32 40
1091	341 52 59	217 42 31	269 50 4	42 21 3
1111	342 4 9	351 18 4	309 32 3	69 9 26
1131	342 15 19	124 53 37	349 14 2	95 57 50
1151	342 26 30	258 29 10	28 56 2	122 46 13
1171	342 37 40	32 4 44	68 38 1	140 34 37
1191	342 48 50	165 40 17	108 20 0	176 23 0
1211	343 0 0	299 15 50	148 1 59	203 11 23
1231	343 11 10	72 51 23	187 43 58	229 59 47
1251	343 22 21	206 26 56	227 25 58	256 48 10
1271	343 33 31	340 2 30	267 7 57	283 36 34
1291	343 44 41	113 38 3	306 49 56	310 24 57
1311	343 55 51	247 13 36	346 31 55	337 13 20
1331	344 7 1	20 49 9	26 13 55	4 1 44
1351	344 18 12	154 24 42	65 55 54	30 50 7
1371	344 29 22	288 0 16	105 37 53	57 38 31
1391	344 40 32	61 35 49	145 19 52	84 26 54
1411	344 51 42	195 11 22	185 1 51	111 15 17
1431	345 2 52	328 46 55	224 43 51	138 3 41
1451	345 14 3	102 22 28	264 25 50	164 52 4
1471	345 25 13	235 58 2	304 7 49	191 40 28
1491	345 36 23	9 33 35	343 49 49	218 28 51
1511	345 47 33	143 9 8	23 31 48	245 17 14
1531	345 58 43	276 44 41	63 13 47	272 5 38
1551	346 9 54	50 20 15	102 55 46	298 54 1
1571	346 21 4	183 55 48	142 37 46	325 42 25
1591	346 32 14	317 31 21	182 19 45	352 30 48
1611	346 43 24	91 6 54	222 1 44	19 19 11
1631	346 54 34	224 42 28	261 43 43	46 7 35

f. 187,r.

جدول حركات الشمس والقمر وحاصة القمر والعقد الشمالي في السنين الرومية المبسوطة

Table of the motions of the sun, the moon, the anomaly of the moon and the northern node in single years of the Romans.

سنو الروم المبسوطة Single Roman years.	وسط الشمس Mean motion of the sun.	وسط القمر Mean motion of the moon.	حاصة القمر Motion of the anomaly of the moon.	العقد الشمالي Motion of the northern node.
1	359° 45′ 46″	129° 23′ 8″	88° 43′ 7″	19° 19′ 38″
2	359 31 33	258 46 16	177 26 14	38 39 15
3	359 17 19	28 9 24	266 9 22	57 58 53
4	0 2 14	170 43 7	7 56 23	77 21 41
5	359 48 0	300 6 15	96 39 30	96 41 18
6	359 33 47	69 29 22	185 22 38	116 0 56
7	359 19 33	198 52 30	274 5 46	135 20 33
8	0 4 28	341 26 13	15 52 48	154 43 21
9	359 50 15	110 49 21	104 35 55	174 2 58
10	359 36 1	240 12 29	193 19 2	193 22 36
11	359 21 47	9 35 37	282 2 11	212 42 13
12	0 6 42	152 9 20	23 49 12	232 5 2
13	359 52 29	281 32 28	112 32 19	251 24 39
14	359 38 15	50 55 36	201 15 26	270 44 18
15	359 24 1	180 18 44	289 58 34	290 3 55
16	0 8 56	322 52 27	31 45 35	309 26 43
17	359 54 43	92 15 34	120 28 43	328 46 20
18	359 40 29	221 38 42	209 11 50	348 5 58
19	359 26 15	351 1 50	297 54 57	7 25 35
20	0 11 10	133 35 33	39 41 59	26 48 23

f. 187,v.

جدول حركات الشمس والقمر وحاصة القمر والعقد الشمالي في الشهور الرومية

Table of the motions of the sun, the moon, the anomaly of the moon and the northern node in the Roman months.

اسماء الشهور الرومية Names of the Roman months.	Mean motion of the sun.	Mean motion of the moon.	Motion of the anomaly of the moon.	Motion of the northern node.
ādhār [March].	30° 33′ 19″	48° 28′ 6″	45° 0′ 52″	1° 38′ 30″
nīsān [April].	60 7 29	83 45 37	76 57 50	3 13 49
ayyār [May].	90 40 48	132 13 43	121 58 42	4 52 18
ḥazīrān [June].	120 14 58	167 31 14	153 55 40	6 27 37
tammūz [July].	150 48 17	215 59 20	198 56 33	8 6 6
āb [August].	181 21 36	264 27 27	243 57 25	9 44 34
aylūl [September].	210 55 46	299 44 58	275 54 23	11 19 53
tishrīn I [October].	241 29 5	348 13 4	320 55 15	12 58 22
tishrīn II [November].	271 3 15	23 30 35	352 52 13	14 33 41
kānūn I [December].	301 36 34	71 58 41	37 53 5	16 12 10
kānūn II [January].	332 9 53	120 26 47	82 53 57	17 50 39
subāṭ [February].				
common.	359 45 46	129 23 8	88 43 7	19 19 38
leap.	360 44 54	142 33 43	101 47 1	19 22 48

f. 188,r.

حركات الشمس والقمر وحاصة القمر والعقد الشمالي في الايام

Motion of the sun, the moon, the anomaly of the moon and the northern node in days.

عدد الايام Number of days.	Mean motion of the sun.	Mean motion of the moon.	Motion of the anomaly of the moon.	Motion of the northern node.
1	0° 59′ 8″	13° 10′ 35″	13° 3′ 54″	0° 3′ 11″
2	1 58 17	26 21 10	26 7 48	0 6 21
3	2 57 25	39 31 45	39 11 42	0 9 32
4	3 56 33	52 42 20	52 15 36	0 12 43
5	4 55 42	65 52 55	65 19 30	0 15 53
6	5 54 50	79 3 30	78 23 24	0 19 4
7	6 53 58	92 14 5	91 27 18	0 22 15
8	7 53 7	105 24 40	104 31 12	0 25 25
9	8 52 15	118 35 15	117 35 6	0 28 36
10	9 51 23	131 45 50	130 38 59	0 31 47
11	10 50 32	144 56 25	143 42 53	0 34 57
12	11 49 40	158 7 0	156 46 47	0 38 8
13	12 48 48	171 17 36	169 50 41	0 41 19
14	13 47 56	184 28 11	182 54 35	0 44 29
15	14 47 5	197 38 46	195 58 29	0 47 40
16	15 46 13	210 49 21	209 2 23	0 50 51
17	16 45 21	223 59 56	222 6 17	0 54 2
18	17 44 30	237 10 31	235 10 11	0 57 12
19	18 43 38	250 21 6	248 14 5	1 0 23
20	19 42 46	263 31 41	261 17 59	1 3 34
21	20 41 55	276 42 16	274 21 53	1 6 44
22	21 41 3	289 52 51	287 25 47	1 9 55
23	22 40 11	303 3 26	300 29 41	1 13 5
24	23 39 20	316 14 1	313 33 35	1 16 16
25	24 38 28	329 24 36	326 37 28	1 19 26
26	25 37 36	342 35 11	339 41 22	1 22 37
27	26 36 44	355 45 46	352 45 16	1 25 48
28	27 35 53	8 56 21	5 49 10	1 28 59
29	28 35 1	22 6 56	18 53 4	1 32 9
30	29 34 10	35 17 31	31 56 58	1 35 19

f. 188,v.

جدول حركة الشمس والقمر وحاصة القمر والعقد الشمالي في الساعات

Table of the motion of the sun, the moon, the anomaly of the moon and the northern node in hours.

عدد الساعات المستوية Number of equal hours.	Mean motion of the sun.	Mean motion of the moon.	Motion of the anomaly of the moon.	Motion of the northern node.
1	0° 2′ 28″	0° 32′ 56″	0° 32′ 40″	0° 0′ 8″
2	0 4 56	1 5 53	1 5 19	0 0 16
3	0 7 24	1 38 49	1 37 59	0 0 24
4	0 9 51	2 11 46	2 10 39	0 0 32
5	0 12 19	2 44 42	2 43 18	0 0 40
6	0 14 47	3 17 39	3 15 59	0 0 48
7	0 17 15	3 50 35	3 48 38	0 0 56
8	0 19 43	4 23 32	4 21 18	0 1 4
9	0 22 10	4 56 28	4 53 58	0 1 12
10	0 24 38	5 29 25	5 26 37	0 1 20
11	0 27 6	6 2 21	5 59 17	0 1 27
12	0 29 34	6 35 17	6 31 57	0 1 35
13	0 32 2	7 8 14	7 4 37	0 1 43
14	0 34 30	7 41 11	7 37 16	0 1 51
15	0 36 58	8 14 7	8 9 56	0 1 59
16	0 39 25	8 47 3	8 42 36	0 2 7
17	0 41 53	9 20 0	9 15 16	0 2 15
18	0 44 21	9 52 56	9 47 55	0 2 23
19	0 46 49	10 25 53	10 20 36	0 2 31
20	0 49 17	10 58 49	10 53 15	0 2 39
21	0 51 45	11 31 46	11 25 55	0 2 47
22	0 54 12	12 4 42	11 58 35	0 2 55
23	0 56 40	12 37 38	12 31 14	0 3 3
24	0 59 8	13 10 35	13 3 54	0 3 11

f. 189,r.

جدول حركات الشمس والقمر وحاصة القمر والعقد الشمالي في السنين الرومية المفردة

Table of the motions of the sun, the moon, the anomaly of the moon and the northern node in intervals of Roman years.

سنون رومية مفردة Intervals of Roman years.	Mean motion of the sun. $0^\circ\ 11'\ 10''\ 14'''\ 35^{IV}\ 31^{V}\ 30^{VI}$.	Mean motion of the moon. $133^\circ\ 35'\ 33''\ 11'''\ 4^{IV}\ 59^{V}\ 45^{VI}$.	Motion of the anomaly of the moon. $39^\circ\ 41'\ 59''\ 15'''\ 5^{IV}\ 38^{V}\ 55^{VI}$	Motion of the northern node.
20	0° 11′ 10″	133° 35′ 33″	39° 41′ 59″	26° 48′ 23″
40	0 22 20	267 11 6	79 23 59	53 36 47
60	0 33 31	40 46 40	119 5 58	80 25 10
80	0 44 41	174 22 13	158 47 57	107 13 34
100	0 55 51	307 57 46	198 29 57	134 1 57
200	1 51 42	255 55 32	36 59 53	268 3 54
300	2 47 33	203 53 18	235 29 49	42 5 51
400	3 43 25	151 51 4	73 59 45	176 7 48
500	4 39 16	99 48 50	272 29 41	310 9 44
600	5 35 7	47 46 36	110 59 38	84 11 41

f. 189,v.

ابتداء جداول تعديل الشمس وتعديل القمر المجموع الكلي بحصص البعدين وعرضه

Beginning of the tables of the equation of the sun, of the equation of the moon (to which are always added the shares of both distances), and of the latitude of the moon.

سطور العدد Columns of numbers.		تعديل الشمس Equation of the sun.	تعديل القمر المفرد Single equation of the moon.	تعديل الحاصة Equation of the anomaly.	حصص البعد Shares of the distances.	البعد الاقرب At the nearest distance.	عرض القمر Latitude of the moon.
مستوي Direct order.	منكوس Inverse order.				دقائق تزيد Minutes to be added.		
1	359	0° 2′ 1″	0° 4′ 50″	0° 9′	0′	0° 3′	0° 5′ 13″
2	358	0 4 1	0 9 40	0 18	0	0 5	0 10 27
3	357	0 6 1	0 14 29	0 27	0	0 7	0 15 40
4	356	0 8 2	0 19 18	0 36	0	0 10	0 20 53
5	355	0 10 2	0 24 7	0 45	0	0 12	0 26 7
6	354	0 12 2	0 28 56	0 53	0	0 14	0 31 19
7	353	0 14 1	0 33 45	1 2	0	0 17	0 36 31
8	352	0 16 1	0 38 32	1 11	0	0 19	0 41 42
9	351	0 18 0	0 43 19	1 20	0	0 21	0 46 52
10	350	0 19 59	0 48 5	1 29	0	0 24	0 52 1
11	349	0 21 58	0 52 51	1 38	0	0 26	0 57 9
12	348	0 23 57	0 57 36	1 46	0	0 28	1 2 16
13	347	0 25 55	1 2 20	1 55	0	0 31	1 7 23
14	346	0 27 52	1 7 4	2 4	1	0 33	1 12 30
15	345	0 29 49	1 11 47	2 13	1	0 35	1 17 33
16	344	0 31 46	1 16 28	2 22	1	0 38	1 22 35
17	343	0 33 42	1 21 8	2 31	1	0 40	1 27 33
18	342	0 35 38	1 25 47	2 39	1	0 42	1 32 31
19	341	0 37 33	1 31 25	2 48	1	0 45	1 37 29
20	340	0 39 27	1 35 3	2 57	2	0 47	1 42 27
21	339	0 41 20	1 39 39	3 5	2	0 49	1 47 23
22	338	0 43 13	1 44 13	3 14	2	0 52	1 52 17
23	337	0 45 6	1 48 45	3 23	2	0 54	1 57 8
24	336	0 46 58	1 53 17	3 31	2	0 56	2 1 56
25	335	0 48 49	1 57 48	3 40	2	0 59	2 6 37
26	334	0 50 38	2 2 17	3 49	3	1 1	2 11 22
27	333	0 52 26	2 6 43	3 57	3	1 3	2 16 2
28	332	0 54 14	2 11 5	4 6	3	1 6	2 20 39
29	331	0 56 2	2 15 26	4 15	3	1 8	2 25 17
30	330	0 57 49	2 19 45	4 23	3	1 10	2 29 52

f. 190,r.

من جداول تعديل الشمس وتعديل القمر المجموع الكلي بحصص البعدين وعرضه

From the tables of the equation of the sun, of the equation of the moon (to which are always added the shares of both distances) and of the latitude of the moon.

Columns of numbers.		Equation of the sun.	Single equation of the moon.	Equation of the anomaly.	Shares of the distances.	At the nearest distance.	Latitude of the moon.
Direct order.	Inverse order.				Minutes to be added.		
31	329	0° 59′ 35″	2° 24′ 3″	4° 32′	3′	1° 12′	2° 34′ 24″
32	328	1 1 19	2 28 20	4 41	4	1 14	2 38 52
33	327	1 3 3	2 32 34	4 49	4	1 16	2 43 17
34	326	1 4 46	2 36 43	4 58	4	1 19	2 47 39
35	325	1 6 28	2 40 12	5 7	4	1 21	2 51 56
36	324	1 8 8	2 45 17	5 15	4	1 23	2 56 11
37	323	1 9 47	2 49 42	5 24	5	1 25	3 0 21
38	322	1 11 25	2 53 6	5 33	5	1 27	3 4 29
39	321	1 13 0	2 57 7	5 41	5	1 29	3 8 35
40	320	1 14 35	3 1 4	5 50	5	1 31	3 12 39
41	319	1 16 9	3 4 17	5 59	6	1 33	3 16 39
42	318	1 17 41	3 8 48	6 7	6	1 35	3 20 35
43	317	1 19 12	3 12 37	6 16	6	1 37	3 24 27
44	316	1 20 43	3 16 23	6 25	7	1 39	3 28 15
45	315	1 22 12	3 20 6	6 33	7	1 40	3 32 0
46	314	1 23 40	3 23 46	6 42	7	1 42	3 35 41
47	313	1 25 6	3 27 22	6 50	8	1 44	3 39 17
48	312	1 26 30	3 30 55	6 58	8	1 45	3 42 49
49	311	1 27 53	3 34 25	7 7	8	1 47	3 46 17
50	310	1 29 15	3 37 52	7 15	9	1 48	3 49 40
51	309	1 30 36	3 41 16	7 23	9	1 49	3 53 0
52	308	1 31 55	3 44 37	7 32	9	1 51	3 56 16
53	307	1 33 13	3 47 54	7 40	10	1 53	3 59 28
54	306	1 34 28	3 51 6	7 48	10	1 55	4 2 35
55	305	1 35 41	3 54 16	7 56	10	1 56	4 5 38
56	304	1 36 51	3 57 23	8 4	11	1 58	4 8 37
57	303	1 37 59	4 0 25	8 12	11	1 59	4 11 30
58	302	1 39 5	4 3 22	8 20	11	2 1	4 14 22
59	301	1 40 9	4 6 16	8 28	12	2 2	4 17 7
60	300	1 41 14	4 9 6	8 36	12	2 3	4 19 47

f. 190,v.

من جداول تعديل الشمس وتعديل القمر المجموع الكلي بحصص البعدين وعرضه

From the tables of the equation of the sun, of the equation of the moon (to which are always added the shares of both distances) and of the latitude of the moon.

Columns of numbers.		Equation of the sun.	Single equation of the moon.	Equation of the anomaly.	Shares of the distances.	At the nearest distance.	Latitude of the moon.
Direct order.	Inverse order.				Minutes to be added.		
61	299	1° 42′ 18″	4° 11′ 33″	8° 44′	12′	2° 5′	4° 22′ 20″
62	298	1 43 21	4 14 36	8 52	13	2 6	4 24 49
63	297	1 44 23	4 17 8	8 59	13	2 7	4 26 14
64	296	1 45 23	4 19 54	9 7	14	2 9	4 29 34
65	295	1 46 21	4 22 24	9 15	14	2 10	4 31 49
66	294	1 47 17	4 24 46	9 22	15	2 12	4 33 59
67	293	1 48 10	4 27 7	9 30	15	2 13	4 36 4
68	292	1 49 0	4 29 26	9 37	16	2 14	4 38 4
69	291	1 49 48	4 31 43	9 44	16	2 15	4 40 0
70	290	1 50 35	4 33 55	9 52	17	2 16	4 41 52
71	289	1 51 20	4 35 59	9 59	17	2 17	4 43 38
72	288	1 52 5	4 37 57	10 6	18	2 18	4 45 18
73	287	1 52 45	4 38 52	10 13	18	2 19	4 46 52
74	286	1 53 23	4 41 43	10 20	19	2 20	4 48 20
75	285	1 53 59	4 43 29	10 27	19	2 21	4 49 44
76	284	1 54 34	4 45 10	10 34	20	2 22	4 51 3
77	283	1 55 7	4 46 46	10 41	20	2 23	4 52 17
78	282	1 55 39	4 48 17	10 48	21	2 24	4 53 25
79	281	1 56 9	4 49 43	10 55	21	2 25	4 54 28
80	280	1 56 35	4 51 5	11 2	22	2 26	4 55 25
81	279	1 56 59	4 52 23	11 8	22	2 27	4 56 17
82	278	1 57 21	4 53 36	11 15	22	2 29	4 57 4
83	277	1 57 42	4 54 43	11 21	23	2 30	4 57 45
84	276	1 58 1	4 55 43	11 27	23	2 31	4 58 21
85	275	1 58 17	4 56 39	11 33	23	2 32	4 58 51
86	274	1 58 30	4 57 30	11 39	24	2 33	4 59 15
87	273	1 58 41	4 58 16	11 44	24	2 34	4 59 35
88	272	1 58 50	4 59 0	11 50	25	2 35	4 59 47
89	271	1 58 57	4 59 36	11 55	25	2 36	4 59 56
90	270	1 59 3	5 0 2	12 0	26	2 37	5 0 0

f. 191,r.

من جداول تعديل الشمس وتعديل القمر المجموع الكلي بحصص البعدين وعرضه

From the tables of the equation of the sun, of the equation of the moon (to which are always added the shares of both distances), and of the latitude of the moon.

Columns of numbers.		Equation of the sun.	Single equation of the moon.	Equation of the anomaly.	Shares of the distances.	At the nearest distance.	Latitude of the moon.
Direct order.	Inverse order.				Minutes to be added.		
91	269	1° 59′ 8″	5° 0′ 26″	12° 5′	26′	2° 37′	4° 59′ 56″
92	268	1 59 10	5 0 44	12 10	27	2 37	4 59 47
93	267	1 59 8	5 0 55	12 15	27	2 38	4 59 35
94	266	1 59 3	5 0 59	12 20	28	2 38	4 59 15
95	265	1 58 57	5 1 0	12 24	29	2 38	4 58 51
96	264	1 58 50	5 0 57	12 28	30	2 38	4 58 21
97	263	1 58 40	5 0 45	12 32	30	2 38	4 57 45
98	262	1 58 28	5 0 37	12 36	31	2 39	4 57 4
99	261	1 58 13	5 0 21	12 39	31	2 39	4 56 17
100	260	1 57 56	5 0 1	12 42	32	2 39	4 55 25
101	259	1 57 38	4 59 32	12 45	32	2 39	4 54 27
102	258	1 57 20	4 58 55	12 48	33	2 39	4 53 25
103	257	1 56 59	4 58 11	12 51	33	2 40	4 52 16
104	256	1 56 38	4 57 24	12 54	34	2 40	4 51 3
105	255	1 56 0	4 56 33	12 56	34	2 40	4 49 44
106	254	1 55 28	4 55 38	12 58	35	2 40	4 48 20
107	253	1 54 58	4 54 34	13 0	35	2 40	4 46 52
108	252	1 54 30	4 53 30	13 2	36	2 40	4 45 18
109	251	1 53 54	4 52 10	13 4	37	2 40	4 43 38
110	250	1 53 14	4 50 43	13 5	37	2 39	4 41 52
111	249	1 52 30	4 49 12	13 6	38	2 39	4 40 0
112	248	1 51 47	4 47 32	13 7	38	2 38	4 38 4
113	247	1 51 3	4 45 47	13 8	39	2 38	4 36 4
114	246	1 50 20	4 44 6	13 9	39	2 37	4 33 59
115	245	1 49 33	4 42 24	13 9	40	2 36	4 31 49
116	244	1 48 39	4 40 31	13 8	40	2 35	4 29 34
117	243	1 47 45	4 38 31	13 7	41	2 35	4 27 14
118	242	1 46 50	4 36 23	13 6	41	2 34	4 24 49
119	241	1 45 55	4 34 9	13 5	42	2 33	4 22 20
120	240	1 44 56	4 31 50	13 4	43	2 31	4 19 46

f. 191,v.

من جداول تعديل الشمس وتعديل القمر المجموع الكلي بحصص البعدين وعرضه

From the tables of the equation of the sun, of the equation of the moon (to which are always added the shares of both distances) and of the latitude of the moon.

Columns of numbers.		Equation of the sun.	Single equation of the moon.	Equation of the anomaly.	Shares of the distances.	At the nearest distance.	Latitude of the moon.
Direct order.	Inverse order.				Minutes to be added.		
121	239	1° 43′ 53″	4° 29′ 26″	13° 3′	43′	2° 30′	4° 17′ 7″
122	238	1 42 50	4 26 56	13 1	44	2 29	4 14 22
123	237	1 41 47	4 24 23	12 59	44	2 27	4 11 30
124	236	1 40 43	4 21 44	12 56	45	2 26	4 8 36
125	235	1 39 35	4 18 58	12 53	45	2 25	4 5 38
126	234	1 38 23	4 16 6	12 50	46	2 23	4 2 35
127	233	1 37 6	4 13 6	12 46	46	2 22	3 59 28
128	232	1 35 49	4 10 3	12 41	47	2 21	3 56 16
129	231	1 34 32	4 6 58	12 36	47	2 19	3 53 0
130	230	1 33 15	4 3 48	12 30	48	2 18	3 49 40
131	229	1 31 55	4 0 32	12 23	48	2 17	3 46 20
132	228	1 30 33	3 57 10	12 16	48	2 15	3 42 51
133	227	1 29 7	3 53 42	12 9	49	2 14	3 39 20
134	226	1 27 41	3 50 6	12 2	49	2 12	3 35 51
135	225	1 26 14	3 46 23	11 54	50	2 10	3 32 10
136	224	1 24 48	3 42 33	11 46	50	2 9	3 28 35
137	223	1 23 18	3 38 42	11 38	51	2 7	3 24 47
138	222	1 21 45	3 34 50	11 29	51	2 5	3 20 55
139	221	1 20 10	3 30 56	11 20	52	2 3	3 16 39
140	220	1 18 34	3 26 58	11 11	52	2 1	3 12 39
141	219	1 16 58	3 22 52	11 2	52	1 58	3 8 35
142	218	1 15 22	3 18 41	10 53	53	1 56	3 4 29
143	217	1 13 42	3 14 25	10 43	53	1 54	3 0 21
144	216	1 12 1	3 10 5	10 33	53	1 51	2 56 10
145	215	1 10 17	3 5 42	10 22	54	1 49	2 51 56
146	214	1 8 32	3 1 13	10 11	54	1 46	2 47 39
147	213	1 6 46	2 56 39	10 0	54	1 43	2 43 17
148	212	1 5 0	2 52 1	9 48	55	1 41	2 38 52
149	211	1 3 13	2 47 20	9 35	55	1 38	2 34 25
150	210	1 1 24	2 42 36	9 22	55	1 35	2 29 52

f. 192,r.

تمام جداول تعديل الشمس وتعديل القمر المجموع الكلي بحصص البعدين وعرضه

End of the tables of the equation of the sun, of the equation of the moon (to which are always added the shares of both distances) and of the latitude of the moon.

Columns of numbers.		Equation of the sun.	Single equation of the moon.	Equation of the anomaly.	Shares of the distances.	At the nearest distance.	Latitude of the moon.
Direct order.	Inverse order.				Minutes to be added.		
151	209	0° 59′ 29″	2° 37′ 49″	9° 8′	55′	1° 32′	2° 25′ 17″
152	208	0 57 34	2 32 57	8 53	56	1 29	2 20 40
153	207	0 55 38	2 28 1	8 38	56	1 26	2 16 2
154	206	0 53 41	2 23 2	8 22	56	1 24	2 11 22
155	205	0 51 43	2 18 0	8 5	56	1 21	2 6 40
156	204	0 49 46	2 12 55	7 48	57	1 18	2 1 56
157	203	0 47 48	2 7 43	7 32	57	1 16	1 57 8
158	202	0 45 49	2 2 37	7 14	57	1 13	1 52 17
159	201	0 43 51	1 57 22	6 56	57	1 10	1 47 23
160	200	0 41 52	1 52 4	6 39	58	1 8	1 42 27
161	199	0 39 53	1 46 45	6 21	58	1 5	1 37 29
162	198	0 37 53	1 41 21	6 3	58	1 2	1 32 31
163	197	0 35 52	1 35 56	5 45	58	0 59	1 27 33
164	196	0 33 50	1 30 29	5 27	58	0 56	1 22 34
165	195	0 31 48	1 25 0	5 8	58	0 52	1 17 34
166	194	0 29 46	1 19 29	4 49	59	0 49	1 12 30
167	193	0 27 43	1 13 56	4 30	59	0 46	1 7 23
168	192	0 25 38	1 8 23	4 11	59	0 42	1 2 16
169	191	0 23 31	1 2 48	3 52	59	0 39	0 57 9
170	190	0 21 24	0 57 11	3 32	59	0 36	0 52 1
171	189	0 19 17	0 51 32	3 12	59	0 32	0 46 52
172	188	0 17 9	0 45 51	2 52	60	0 29	0 41 42
173	187	0 15 1	0 40 9	2 32	60	0 25	0 36 31
174	186	0 12 53	0 34 26	2 11	60	0 21	0 31 19
175	185	0 10 45	0 28 42	1 50	60	0 18	0 26 7
176	184	0 8 36	0 22 58	1 29	60	0 15	0 20 53
177	183	0 6 27	0 17 14	1 7	60	0 11	0 15 40
178	182	0 4 18	0 11 30	0 45	60	0 8	0 10 27
179	181	0 2 9	0 5 45	0 23	60	0 4	0 5 13
180	180	0 0 0	0 0 0	0 0	60	0 0	0 0 0

f. 192,v.

جداول اجتماعات الشمس والقمر الوسطى في السنين الرومية المجموعة
Tables of the mean conjunctions of the sun and the moon in collected Roman years [from the era].

سنو الروم المجموعة Collected years of the Romans [from the era].	ايام اذار للاجتماعات Days of the month ādhār [March] up to the conjunctions.	وسطي النيرين Mean motion of both luminaries.	حاصة القمر Anomaly of the moon.	حركة العرض Motion in latitude of the moon.
879	13*days* 58′ 4″	353° 40′ 15″	31° 23′ 22″	256° 37′ 3″
903	18 33 15	358 24 53	138 57 5	5 46 20
927	23 8 26	3 9 31	246 30 48	114 55 38
951	27 43 37	7 54 9	354 4 31	224 4 55
975	2 46 58	343 32 22	75 49 14	302 33 58
999	7 22 9	348 17 0	183 22 57	51 43 16
1023	11 57 21	353 1 38	290 56 40	160 52 33
1047	16 32 32	357 46 16	38 30 23	270 1 50
1071	21 7 43	2 30 54	146 4 6	19 11 8
1095	25 42 54	7 15 32	253 37 49	128 20 26
1119	0 46 15	342 53 45	335 22 32	206 49 29
1143	5 21 26	347 38 23	82 56 15	315 58 47
1167	9 56 38	352 23 1	190 29 58	65 8 4
1191	14 31 49	357 7 39	298 3 41	174 17 21
1215	19 7 0	1 52 17	45 37 24	283 26 38
1239	23 42 11	6 36 55	153 11 7	32 35 56
1263	28 17 22	11 21 33	260 44 50	141 45 13
1287	3 20 44	346 59 49	342 29 33	220 14 16
1311	7 55 55	351 44 27	90 3 16	329 23 33
1335	12 31 6	356 29 5	197 36 59	78 32 51
1359	17 6 17	1 13 43	305 10 42	187 42 8
1383	21 41 28	5 58 21	52 44 25	296 51 25
1407	26 16 39	10 42 59	160 18 8	46 0 43
1431	1 20 1	346 21 14	242 2 51	124 29 46
1455	5 55 13	351 5 52	349 36 34	233 39 3
1479	10 30 24	355 50 30	97 10 17	342 48 21
1503	15 5 35	0 35 8	204 44 0	91 57 38
1527	19 40 46	5 19 46	312 17 43	201 6 55
1551	24 15 57	10 4 24	59 51 26	310 16 13
1575	28 51 8	14 49 2	167 25 9	59 25 30
1599	3 54 29	350 27 15	249 9 52	137 54 33
1623	8 29 40	355 11 53	356 43 35	247 3 0

f. 193,r.

جداول مقابلات الشمس والقمر الوسطى في السنين الرومية المجموعة
Tables of the mean oppositions of the sun and the moon in collected Roman years [from the era].

Collected years of the Romans [from the era].	ايام اذار لمعرفة الاستقبالات Days of the month ādhār for finding the oppositions.	وسط الشمس قبالة القمر Mean motion of the sun at the point opposite the moon.	حاصة القمر Anomaly of the moon.	حركته في العرض Motion in latitude of the moon.
879	28 *days* 43′ 59″	8° 13′ 27″	224° 17′ 52″	91° 57′ 11″
903	3 47 20	343 51 41	306 2 35	170 26 14
927	8 22 31	348 36 19	53 36 18	279 35 31
951	12 57 42	353 20 57	161 10 1	28 44 49
975	17 32 53	358 5 35	268 43 44	137 54 6
999	22 8 4	2 50 13	16 17 27	247 3 23
1023	26 43 16	7 34 51	123 51 10	356 12 41
1047	1 46 37	343 13 4	205 35 53	74 41 44
1071	6 21 48	347 57 42	313 9 36	183 51 1
1095	10 56 59	352 42 20	60 43 19	293 0 19
1119	15 32 10	357 26 58	168 17 2	42 9 36
1143	20 7 21	2 11 36	275 50 45	151 18 53
1167	24 42 33	6 56 14	23 24 28	260 28 11
1191	29 17 44	11 40 52	130 58 11	9 37 28
1215	4 21 5	347 19 5	212 42 54	88 6 31
1239	8 56 16	352 3 43	320 16 37	197 15 49
1263	13 31 27	356 48 21	67 50 20	306 25 6
1287	18 6 39	1 32 59	175 24 3	55 34 23
1311	22 41 50	6 17 37	282 57 46	164 43 40
1335	27 17 1	11 2 15	30 31 29	273 52 58
1359	2 20 22	346 40 30	112 16 12	352 22 1
1383	6 55 33	351 25 8	219 49 55	101 31 19
1407	11 30 44	356 9 46	327 23 38	210 40 36
1431	16 5 56	0 54 24	74 57 21	319 49 54
1455	20 41 7	5 39 2	182 31 4	68 59 11
1479	25 16 18	10 23 40	290 4 47	178 8 28
1503	0 19 40	346 1 55	11 49 30	256 37 32
1527	4 54 51	350 46 33	119 23 13	5 46 49
1551	9 30 2	355 31 11	226 56 56	114 56 6
1575	14 5 13	0 15 50	334 30 39	224 5 24
1599	18 40 24	5 0 28	82 4 22	333 14 41
1623	22 15 35	9 45 6	189 38 5	82 23 58

f. 193,v.

جداول الاجتماعات والاستقبالات في السنين الرومية المبسوطة
Tables of the conjunctions and oppositions in single years of the Romans.

سنو الروم المبسوطة Years of the Romans one by one.	ايام اذار للاجتماعات والاستقبالات Days of the month ādhār to the syzygies.	وسطي الشمس والقمر Mean motion of the sun and the moon.	حاصة القمر Anomaly of the moon.	حركته في العرض Motion of the moon in latitude.
1	18*days* 53′ 52″	18° 23′ 22″	335° 37′ 2″	38° 43′ 3″
2	8 15 53	7 40 19	285 25 4	46 45 50
3	27 9 45	26 3 41	261 2 5	85 28 52
4	15 31 47	15 20 39	210 50 7	93 31 40
5	4 53 49	4 37 36	160 38 9	101 34 29
6	23 47 40	23 1 0	136 15 11	140 17 31
7	13 9 42	12 17 58	86 3 12	148 20 17
8	1 31 44	1 34 56	35 51 14	156 23 6
9	20 25 35	19 58 18	11 28 16	195 6 8
10	9 47 37	9 15 17	321 16 18	203 8 53
11	28 41 29	27 38 39	296 53 19	241 51 55
12	17 3 31	16 55 38	246 41 21	249 54 46
13	6 25 32	6 12 27	196 29 23	257 57 33
14	25 19 24	24 35 49	172 6 25	296 40 36
15	14 41 26	13 52 46	121 54 26	304 43 25
16	3 3 28	3 9 44	71 42 28	312 46 11
17	21 57 19	21 33 6	47 19 30	351 29 14
18	11 19 21	10 50 3	357 7 32	359 32 1
19	0 41 23	0 7 0	306 55 33	7 34 48
20	18 35 14	18 30 23	282 32 35	46 17 50
21	7 57 16	7 47 21	232 20 37	54 20 36
22	26 51 8	26 10 43	207 57 39	93 3 38
23	16 13 9	15 27 40	157 45 41	101 6 29
24	4 35 11	4 44 38	107 33 43	109 9 17

f. 194,r.

جداول تجزئة ايام الشهر للاجتماعات والاستقبالات الاوسطين

Tables of the division of the month for [finding] the mean conjunctions and oppositions.

Half of the month.	14^{days} 45′ 55″	14° 33′ 12″	192° 54′ 30″	195° 20′ 7″
A third of the month.	9 50 37	9 42 8	128 36 20	130 13 25
A quarter of the month.	7 22 58	7 16 36	96 27 15	97 40 3
A sixth of the month.	4 55 18	4 51 4	64 18 10	65 6 42

جداول الشهور الرومية للاجتماعات والاستقبالات الاوسطين

Tables of the Roman months for [finding] the mean conjunctions and oppositions.

شهور الروم Months of the Romans.	جملة ايام مدة الشهور Sum of the days of the months.	ايام الشهور القمرية Days of the lunar months.	وسطي النيرين Mean motion of both luminaries.	حاصة القمر Anomaly of the moon.	حركته في العرض Motion of the moon in latitude.
ādhār.	31	29^{days} 31′ 50″	29° 6′ 25″	25° 49′ 0″	30° 40′ 14″
nīsān.	61	59 3 40	58 12 49	51 38 0	61 20 28
ayyār.	92	88 35 30	87 19 14	77 27 0	92 0 42
ḥazīrān.	122	118 7 21	116 25 39	103 16 1	122 40 57
tammūz.	153	147 39 11	145 32 4	129 5 1	153 21 11
āb.	184	177 11 1	174 38 28	154 54 1	184 1 25
aylūl.	214	206 42 51	203 44 53	180 43 1	214 41 39
tishrīn I.	245	236 14 41	232 51 17	206 32 1	245 21 53
tishrīn II.	275	265 46 31	261 57 42	232 21 1	276 2 7
kānūn I.	306	295 18 21	291 4 7	258 10 1	306 42 21
kānūn II.	337	324 50 12	320 10 31	283 59 2	337 22 36
subāṭ.	365	354 22 2	349 16 56	309 48 2	8 2 50

Intervals of years (سنون مفردة).

48	9^{d} 10′ 22″	9° 29′ 16″	215° 7′ 26″	218° 18′ 35″
72	13 45 34	14 13 54	322 41 9	327 27 54
96	18 20 45	18 58 32	70 14 52	76 37 9
192	7 9 40	8 50 39	114 40 44	153 14 18
288	25 30 25	27 49 11	189 55 36	229 56 27
384	14 19 19	17 41 19	229 21 28	306 28 36
480	3 8 14	7 33 26	273 47 20	23 5 45

f. 194,v.

جدول حركتي الشمس والقمر المختلفة في الساعة المعتدلة في اوقات الاجتماعات والمقابلات

Table of the unequal motions of the sun and the moon in an equal hour at the times of the conjunctions and oppositions.

سطري العدد المتفاضلين و و Columns of numbers at intervals of 6 degrees.		حركة الشمس المختلفة في الساعة Unequal hourly motion of the sun.	حركة القمر المختلفة في الساعة Unequal hourly motion of the moon.
0	360	2′ 23″	30′ 18″
6	354	2 23	30 19
12	348	2 23	30 21
18	342	2 23	30 24
24	336	2 23	30 28
30	330	2 24	30 35
36	324	2 24	30 43
42	318	2 24	30 51
48	312	2 25	31 1
54	306	2 25	31 12
60	300	2 25	31 25
66	294	2 26	31 38
72	288	2 26	31 53
78	282	2 27	32 8
84	276	2 27	32 24
90	270	2 28	32 41
96	264	2 28	32 59
102	258	2 29	33 17
108	252	2 29	33 36
114	246	2 29	33 55
120	240	2 30	34 14
126	234	2 30	34 32
132	228	2 31	34 49
138	222	2 31	35 4
144	216	2 32	35 18
150	210	2 32	35 31
156	204	2 33	35 43
162	198	2 33	35 52
168	192	2 33	35 58
174	186	2 33	36 2
180	180	2 33	36 4

فضل ما بين الشمس والقمر الاوسط Mean difference [in longitude] between the sun and the moon.	الزيادة والنقصان في حركة الشمس والقمر في الساعة Increase and decrease of the hourly motions of the sun and the moon.
1°	0′ 1″
2	0 2
3	0 3
4	0 4
5	0 5
6	0 6
7	0 7

اما حدود كسوف الشمس اذا كانت حركة العرض الوسطى من قنط مد الى قضا يو ومن ٥ الى ك يو ومن سمح مد الى تمام سص فانه يمكن ان تنكسف الشمس والا فلا تنكسف

واما حدود كسوف القمر اذا كانت حركة العرض الوسطى من قصه يج الى قضد مز ومن ٥ الى يد مز ومن سمه يج الى تمام سص فانه يمكن ان ينكسف القمر والا فلا تنكسف

The limits of an eclipse of the sun. — If the mean motion of latitude is from 159° 44′ to 191° 16′, and from 0° to 20° 16′, and from 348° 44′ to 360°, the sun can be eclipsed; otherwise it cannot.

The limits of an eclipse of the moon. — If the mean motion of latitude is from 165° 13′ to 194° 47′, and from 0° to 14° 47′, and from 345° 13′ to 360°, the moon can be eclipsed, otherwise it cannot.

f. 195,r.

جدول لتقويم
Table of correction.

سطرا العدد المتفاضلان ستة اجزاء Columns of numbers at intervals of 6°.		حصص ابعاد القمر Shares of the distances of the moon.	فلك التدوير Epicycle.	القلك الخارج Eccentric.
6	354	0′ 21″	0′	0′
12	348	0 42	0	0
18	342	1 22	0	1
24	336	2 42	1	2
30	330	4 1	1	3
36	324	5 21	1	4
42	318	7 18	2	5
48	312	9 15	2	6
54	306	11 37	2	8
60	300	14 0	3	9
66	294	16 48	3	11
72	288	19 36	4	13
78	282	22 36	4	14
84	276	25 36	5	16
90	270	28 42	6	17
96	264	31 48	6	19
102	258	34 54	7	21
108	252	38 0	8	22
114	246	41 0	8	24
120	240	44 0	9	25
126	234	46 45	9	27
132	228	49 30	10	28
138	222	51 39	10	29
144	216	53 48	11	30
150	210	55 32	11	30
156	204	57 15	11	31
162	198	58 18	12	31
168	192	59 21	12	32
174	186	59 41	12	32
180	180	60 0	12	32

وقدر اصابع الكسوف وانحراف القمر في الظلم في البدء والانجلا
And [the table] of the quantity of the digits of the eclipse and of the inclination of the moon in the darkness, at the beginning [of the immersion] and [at the end] of the clearing.

عدد اصابع الكسوف من قطري النيرين Digits of the diameters of the sun and the moon that are eclipsed.	قدر ما ينكسف من دائرة الشمس بالمقدار الذي به يكون تكسير دائرتها التي عشرة اصبعا The quantity of the solar disc that is eclipsed, the area of the disc being divided into twelve digits.	قدر ما ينكسف من دائرة القمر بالمقدار الذي به يكون تكسير دائرته اثني عشر جزءا The quantity of the lunar disc that is eclipsed, the area of the disc being divided into twelve parts.	عدد اصابع الكسوف من اقطار النيرين Digits of the diameters of the sun and the moon that are eclipsed.	اول كسوف الشمس واخر انجلائها Beginning of the eclipse of the sun and end of the clearing.	اول كسوف القمر واخر انجلائه Beginning of the immersion of the moon and end of the clearing.	اخر كسوف القمر واول انجلائه End of the immersion of the moon and beginning of the clearing.
1	0dig. 20′	0dig. 30′	0dig.	90°	90°	0°
			1	67	73	0
2	1 0	1 10	2	57	65	0
			3	49	59	0
3	1 45	2 5	4	43	54	0
			5	37	50	0
4	2 40	3 10	6	31	46	0
			7	26	43	0
5	3 40	4 20	8	21	39	0
			9	16	36	0
6	4 40	5 30	10	11	32	0
			11	6	29	0
7	5 50	6 45	12	2	26	90
			13	0	23	64
8	7 0	8 0	14	0	21	53
			15	0	18	43
9	8 20	9 10	16	0	15	36
			17	0	12	29
10	9 40	10 20	18	0	10	22
			19	0	7	16
11	10 50	11 20	20	0	4	10
			21	0	2	4
12	12 0	12 0	total.	0	0	0

f. 195,v.

الكسوفات القمرية في البعد الاقرب **Eclipses of the moon at the nearest distance.**			
True latitude of the moon.	Quantities of the fall.	Quantity of [half] the stay.	Digits of the lunar diameter that are eclipsed.
63′ 36″	0′ 0″	0′ 0″	0
60 39	19 9	0 0	1
57 43	26 45	0 0	2
54 46	32 20	0 0	3
51 49	36 53	0 0	4
48 53	40 42	0 0	5
45 56	43 59	0 0	6
42 59	46 53	0 0	7
40 3	49 25	0 0	8
37 6	51 40	0 0	9
34 9	53 39	0 0	10
31 13	55 25	0 0	11
28 16	56 59	0 0	12
25 19	45 47	12 35	13
22 23	42 15	17 16	14
19 26	40 2	20 32	15
16 29	38 27	22 58	16
13 33	37 20	24 48	17
10 36	36 31	26 12	18
7 40	35 56	27 42	19
4 43	35 35	28 12	20
1 46	35 22	28 53	21
0 0	35 20	28 56	21^{d} 36′

تفاضله ه ب نو ل
Its [i. e. the latitude's] interval 0° 2′ 56″ 30‴

الكسوفات القمرية في البعد الابعد **Eclipses of the moon at the farthest distance.**			
عرض القمر الحقيقي True latitude of the moon.	مقادير السقوط Quantities of the fall.	مقدار المكث Quantity of [half] the stay.	عدد الاصابع المنكسفة من قطر القمر Digits of the lunar diameter that are eclipsed.
53′ 0″	0′ 0″	0′ 0″	0
50 33	15 56	0 0	1
48 5	22 19	0 0	2
45 38	26 7	0 0	3
43 11	29 45	0 0	4
40 43	33 16	0 0	5
38 15	36 42	0 0	6
35 48	39 5	0 0	7
33 20	41 12	0 0	8
30 53	43 5	0 0	9
28 25	44 44	0 0	10
25 58	46 12	0 0	11
23 30	47 7	0 0	12
21 3	38 11	10 26	13
18 35	35 19	14 23	14
16 8	33 24	17 5	15
13 40	32 5	19 7	16
11 13	31 9	20 39	17
8 45	30 27	21 49	18
6 18	29 58	22 39	19
3 50	29 41	23 11	20
1 23	29 31	23 28	21
0 0	29 30	23 30	21^{d} 31′ 30″

تفاضله ه ب كز ل
Its [i. e. the latitude's] interval 0° 2′ 27″ 30‴

f. 196,r.

اقدار كسوف الشمس في بعد القمر الاقرب **Eclipses of the sun at the nearest distance of the moon.**		
Visible latitude of the moon.	Digits of the solar diameter that are eclipsed.	Quantities of the fall.
34′ 0″	0	0′ 0″
31 18	1	13 16
28 35	2	18 25
25 53	3	22 2
23 10	4	24 50
20 28	5	27 9
17 45	6	29 0
15 3	7	30 30
12 20	8	31 40
9 38	9	32 37
6 55	10	33 16
4 13	11	33 44
1 30	12	33 58
0 0	12^{d} 33′	34 0

تفاضله ه ب مب ل
Its [i. e. the latitude's] interval
0° 2′ 42″ 30‴

اقدار كسوف الشمس في بعد القمر الابعد **Eclipses of the sun at the farthest distance of the moon.**		
عرض القمر المرئي Visible latitude of the moon.	الاصابع المنكسفة من قطر الشمس Digits of the solar diameter that are eclipsed.	مقادير السقوط Quantities of the fall.
31′ 0″	0	0′ 0″
28 18	1	12 39
25 35	2	17 30
22 53	3	20 55
20 10	4	23 33
17 28	5	25 47
14 45	6	27 16
12 3	7	28 34
9 20	8	29 33
6 38	9	30 17
3 55	10	30 49
1 13	11	30 59
0 0	11^{d} 23′ 30″	31 0

تفاضله ه ب مب ل
Its [i. e. the latitude's] interval
0° 2′ 42″ 30‴

f. 197,v.

جداول اختلاف منظر القمر في دائرة الارتفاع في حدود الابعاد الاربعة من قبل القسي والزوايا
وهي لبطليموس من كتاب المجسطي ان شاء الله

Tables for finding, by arcs and angles, the parallaxes of the moon in the circle of altitude, at four limits. From Ptolemy's Almagest, God willing.

سطر العدد المتفاضل بدرجتين درجتين Numbers, at intervals of two degrees.	اختلاف منظر الشمس Parallax of the sun.	[Parallax of the moon.]				[Sixtieth parts.]		
		الحد الاول First limit.	الحد الثاني Second limit. [Difference].	الحد الثالث Third limit.	الحد الرابع Fourth limit. [Difference].	فلك التدوير [At the apogee] of the epicycle.	فلك التدوير [At the perigee] of the epicycle.	الفلك الخارج In the eccentric.
2	0′ 7″	1′ 54″	0′ 23″	0° 2′ 40″	0′ 50″	0′ 14″	0′ 11″	0′ 15″
4	0 13	3 48	0 45	0 5 25	1 40	0 28	0 22	0 30
6	0 19	5 41	1 7	0 8 10	2 30	0 42	0 33	0 45
8	0 25	7 34	1 29	0 10 55	3 20	1 22	1 7	1 33
10	0 31	9 27	1 51	0 13 40	4 10	2 2	1 41	2 21
12	0 37	11 19	2 12	0 16 20	5 0	2 42	2 15	3 9
14	0 42	13 10	2 33	0 19 0	5 50	3 35	3 13	4 22
16	0 48	15 0	2 54	0 21 40	6 40	4 28	4 11	5 35
18	0 53	16 49	3 15	0 24 20	7 30	5 21	5 9	6 48
20	0 58	18 36	3 36	0 27 0	8 20	6 39	6 25	8 25
22	1 4	20 22	3 57	0 29 30	9 10	7 57	7 41	10 2
24	1 9	22 6	4 18	0 32 0	10 0	9 15	8 57	11 39
26	1 14	23 49	4 39	0 34 30	10 50	10 50	10 29	13 32
28	1 20	25 30	4 59	0 37 0	11 40	12 25	12 1	15 25
30	1 25	27 9	5 18	0 39 30	12 30	14 0	13 33	17 18
32	1 30	28 46	5 37	0 42 0	13 20	15 52	15 22	19 23
34	1 35	30 21	5 55	0 44 21	14 0	17 44	17 11	21 28
36	1 40	31 54	6 13	0 46 40	14 40	19 36	19 0	23 33
38	1 44	33 24	6 30	0 49 0	15 20	21 36	20 59	25 40
40	1 49	34 51	6 47	0 51 20	16 0	23 36	22 58	27 47
42	1 54	36 14	7 4	0 53 33	16 40	25 36	24 57	29 54
44	1 58	37 37	7 20	0 55 32	17 20	27 40	27 1	32 0

f. 198,r.

بقية جداول بطاميوس ذات التسعة الصفوف المتخذة لاختلاف منظر القمر في دائرة الارتفاع
في حدود الابعاد الاربعة من قبل القسي والزوايا التامة ان شاء الله

Here end Ptolemy's tables, furnished with nine columns, to be used for finding, by arcs and angles, the parallaxes of the moon in the circle of altitude, at the four limits of the distances, God willing.

Numbers, at intervals of two degrees.	Parallax of the sun.	[Parallax of the moon.]				[Sixtieth parts.]		
		First limit.	Second limit. [Difference].	Third limit.	Fourth limit. [Difference].	[At the apogee] of the epicycle.	[At the perigee] of the epicycle.	In the eccentric.
46	2′ 3″	38′ 57″	7′ 35″	0° 57′ 30″	18′ 0″	29′ 44″	29′ 5″	34′ 6″
48	2 8	40 14	7 49	0 59 20	18 40	31 43	31 9	36 12
50	2 12	41 28	8 3	1 1 0	19 10	33 52	33 11	38 9
52	2 16	42 39	8 16	1 2 40	19 40	35 56	35 19	40 6
54	2 20	43 45	8 29	1 4 10	20 10	38 0	37 24	42 3
56	2 23	44 48	8 42	1 5 40	20 40	40 0	39 24	43 39
58	2 26	45 48	8 53	1 7 10	21 10	42 0	41 24	45 35
60	2 29	46 46	9 3	1 8 20	21 40	44 0	43 24	47 21
62	2 32	47 40	9 13	1 9 30	22 10	45 50	45 13	48 49
64	2 34	48 30	9 22	1 10 40	22 35	47 40	47 2	50 17
66	2 36	49 15	9 31	1 11 50	23 0	49 30	48 51	51 45
68	2 38	49 57	9 39	1 13 0	23 20	50 56	50 24	52 57
70	2 40	50 36	9 46	1 14 0	23 35	52 22	51 57	54 9
72	2 42	51 11	9 53	1 15 0	23 50	53 48	53 30	55 41
74	2 44	51 44	9 59	1 15 40	24 3	54 57	54 41	56 12
76	2 46	52 12	10 4	1 16 20	24 16	56 6	55 52	57 3
78	2 47	52 34	10 8	1 17 0	24 26	57 15	56 3	57 54
80	2 48	52 53	10 11	1 17 20	24 36	57 57	57 47	58 26
82	2 49	53 9	10 14	1 17 40	24 45	58 39	58 31	58 58
84	2 50	53 21	10 16	1 18 0	24 52	59 21	59 15	59 30
86	2 50	53 29	10 16	1 18 20	24 57	59 34	59 30	59 40
88	2 51	53 33	10 17	1 18 40	24 59	59 47	59 45	59 50
90	2 51	53 34	10 17	1 19 0	25 0	60 0	60 0	60 0

f. 198,v.

جداول اختلاف منظر القمر في الاقليم الاول وعرضه يو لب وساعاته يج

Tables of the parallaxes of the moon in longitude and latitude, in the first clime, whose latitude is 16° 32′ and whose hours are 13^{h}.

ساعات السرطان Hours of Cancer.	اختلاف منظر القمر Parallax of the moon. Long. Lat.	ساعات الاسد Hours of Leo.	Parallax of the moon. Long. Lat.	ساعات السنبلة Hours of Virgo.	Parallax of the moon. Long. Lat.	ساعات الميزان Hours of Libra.	Parallax of the moon. Long. Lat.	ساعات العقرب Hours of Scorpio.	Parallax of the moon. Long. Lat.	ساعات القوس Hours of Sagittarius.	Parallax of the moon. Long. Lat.
6^{h} 30′	49′ 16′ south	6^{h} 25′	51′ 5′ south	6^{h} 15′	51′ +4′ north	6^{h}	51′ +6′ north	5^{h} 45′	51′ +4′ north	5^{h} 35′	51′ 5′ south
6	50 13	6	51 3	6	51 +5	5	49 +6	5	49 +4	5	51 7
5	49 8	5	48 +2 north	5	49 +6	4	45 +4	4	47 3 south	4	46 12
4	42 4	4	43 +5	4	43 +6	3	35 +1	3	40 8	3	39 18
3	35 +1 north	3	34 +6	3	34 +6	2	28 6 south	2	31 12	2	31 23
2	25 +4	2	24 +7	2	25 +4	1	17 8	1	20 19	1	19 27
1	13 +6	1	12 +6	1	14 0 south						
culm.	0 +7	culm.	1 +4	culm.	2 4	culm.	9 13	culm.	9 22	culm.	7 30
1	13 +6	1	13 0 south	1	10 9						
2	25 +4	2	25 4	2	21 14	1	6 18	1	3 27	1	6 32
3	35 +1	3	34 9	3	29 19	2	17 23	2	15 30	2	18 34
4	42 4 south	4	41 14	4	36 24	3	26 27	3	25 32	3	28 33
5	49 8	5	44 18	5	40 28	4	33 30	4	33 33	4	37 31
6	50 13	6	45 24	6	41 30	5	37 32	5	37 32	5	42 28
6^{h} 30′	49 16	6^{h} 25′	44 25	6^{h} 15′	40 31	6	39 32	5^{h} 45′	40 31	5^{h} 35′	44 25

f. 199,r.

ساعات الجدي Hours of Capricorn.	Parallax of the moon. Long. Lat.	ساعات الدلو Hours of Aquarius.	Parallax of the moon. Long. Lat.	ساعات الحوت Hours of Pisces.	Parallax of the moon. Long. Lat.	ساعات الحمل Hours of Aries.	Parallax of the moon. Long. Lat.	ساعات الثور Hours of Taurus.	Parallax of the moon. Long. Lat.	ساعات الجوزاء Hours of Gemini.	Parallax of the moon. Long. Lat.
5^{h} 30′	49′ 16′ south	5^{h} 35′	44′ 25′ south	5^{h} 45′	40′ 31′ south	6^{h}	39′ 32′ south	6^{h} 15′	40′ 31′ south	6^{h} 25′	44′ 25′ south
5	47 18	5	42 28	5	37 32	5	37 32	6	41 30	6	45 24
4	42 23	4	37 31	4	33 33	4	33 30	5	40 28	5	44 18
3	35 27	3	28 33	3	25 32	3	26 27	4	36 24	4	41 14
2	24 30	2	18 34	2	15 30	2	17 23	3	29 19	3	34 9
1	13 33	1	6 32	1	3 27	1	6 18	2	21 14	2	25 4
								1	10 9	1	13 0
culm.	0 33	culm.	7 30	culm.	9 22	culm.	9 13	culm.	2 4	culm.	1 +4 north
								1	14 0	1	12 +6
1	13 33	1	19 27	1	20 19	1	17 8	2	25 +4 north	2	24 +7
2	24 30	2	31 23	2	31 12	2	28 6	3	34 +6	3	34 +6
3	35 27	3	39 18	3	40 8	3	35 +1 north	4	43 +6	4	43 +5
4	42 23	4	46 12	4	47 3	4	45 +4	5	49 +6	5	48 +2
5	47 18	5	51 7	5	49 +4 north	5	49 +6	6	51 +5	6	51 3 south
5^{h} 30′	49 16	5^{h} 35′	51 5	5^{h} 45′	51 +4	6	51 +6	6^{h} 15′	51 +4	6^{h} 25′	51 5

f. 199,v.

جداول اختلاف منظر القمر في الطول والعرض للاقليم الثاني وعرضه كد وساعاته يج ل

Tables of the parallaxes of the moon in longitude and latitude, in the second clime, whose latitude is 24° and whose hours are 13^h 30^m.

Hours of Cancer.	Parallax of the moon. Long.	Lat. south.	Hours of Leo.	Parallax of the moon. Long.	Lat. south.	Hours of Virgo.	Parallax of the moon. Long.	Lat. south.	Hours of Libra.	Parallax of the moon. Long.	Lat. south.	Hours of Scorpio.	Parallax of the moon. Long.	Lat. south.	Hours of Sagittarius.	Parallax of the moon. Long.	Lat. south.
6^h 45′	46′	23′	6^h 38′	50′	11′	6^h 21′	51′	3′	6^h	51′	0′	5^h 39′	51′	3′	5^h 22′	50′	11′
6	46	19	6	49	10	6	51	2	5	49	1	5	50	5	5	49	14
5	45	14	5	46	5	5	48	1	4	45	2	4	46	9	4	45	18
4	41	9	4	42	2	4	43	0	3	38	6	3	41	14	3	39	23
3	33	7	3	34	1	3	36	1	2	29	10	2	32	18	2	30	27
2	24	3	2	24	0	2	26	3	1	19	14	1	22	23	1	19	31
1	12	1	1	13	1	1	16	6									
culm.	0	0	culm.	1	3	culm.	2	11	culm.	8	19	culm.	11	28	culm.	8	35
1	12	1	1	11	6	1	8	15									
2	24	3	2	22	10	2	17	20	1	3	24	1	1	32	1	4	37
3	33	7	3	31	15	3	26	25	2	13	28	2	11	35	2	15	38
4	41	9	4	37	20	4	32	29	3	22	32	3	21	37	3	25	37
5	45	14	5	40	24	5	35	32	4	29	35	4	28	38	4	33	36
6	46	19	6	40	29	6	37	36	5	33	37	5	34	37	5	39	33
6^h 45′	46	23	6^h 38′	40	31	6^h 21′	36	36	6	34	36	5^h 39′	36	36	5^h 22′	40	31

f. 200,r.

Hours of Capricorn.	Parallax of the moon. Long.	Lat. south.	Hours of Aquarius.	Parallax of the moon. Long.	Lat. south.	Hours of Pisces.	Parallax of the moon. Long.	Lat. south.	Hours of Aries.	Parallax of the moon. Long.	Lat. south.	Hours of Taurus.	Parallax of the moon. Long.	Lat. south.	Hours of Gemini.	Parallax of the moon. Long.	Lat. south.
5^h 15′	46′	23′	5^h 22′	40′	31′	5^h 39′	36′	36′	6^h	34′	36′	6^h 21′	36′	36′	6^h 38′	40′	31′
5	44	23	5	39	33	5	34	37	5	33	37	6	37	36	6	40	29
4	40	28	4	33	36	4	28	38	4	29	35	5	35	32	5	40	24
3	33	32	3	25	37	3	21	37	3	22	32	4	32	29	4	37	20
2	23	38	2	15	38	2	11	35	2	13	28	3	26	25	3	31	15
1	12	37	1	4	37	1	1	32	1	3	24	2	17	20	2	22	10
												1	8	15	1	11	6
culm.	0	37	culm.	8	35	culm.	11	28	culm.	8	19	culm.	2	11	culm.	1	3
												1	16	6	1	13	1
1	12	37	1	19	31	1	22	23	1	19	14	2	26	3	2	24	0
2	23	38	2	30	27	2	32	18	2	29	10	3	36	1	3	34	1
3	33	32	3	39	23	3	41	14	3	38	6	4	43	0	4	42	2
4	40	28	4	45	18	4	46	9	4	45	2	5	48	1	5	46	5
5	44	23	5	49	14	5	50	5	5	49	1	6	51	2	6	49	10
5^h 15′	46	23	5^h 22′	50	11	5^h 39′	51	3	6	51	0	6^h 21′	51	3	6^h 38′	50	11

f. 200,v.

جداول اختلاف منظر القمر في الطول والعرض للاقليم الثالث وعرضه ل م وساعاته يد

Tables of the parallaxes of the moon in longitude and latitude, in the third clime, whose latitude is 30° 40′ and whose hours are 14^h.

Hours of Cancer.	Parallax of the moon. Long.	Lat. south	Hours of Leo.	Parallax of the moon. Long.	Lat. south	Hours of Virgo.	Parallax of the moon. Long.	Lat. south	Hours of Libra.	Parallax of the moon. Long.	Lat. south	Hours of Scorpio.	Parallax of the moon. Long.	Lat. south	Hours of Sagittarius.	Parallax of the moon. Long.	Lat. south
7^h	43′	28′	6^h 51′	48′	17′	6^h 28′	50′	9′	6^h	51′	6′	5^h 32′	50′	9′	5^h 9′	48′	17′
6	44	23	6	48	14	6	51	8	5	49	7	5	49	10	5	48	19
5	43	19	5	46	11	5	48	6	4	45	8	4	46	14	4	45	23
4	38	15	4	41	8	4	42	6	3	39	11	3	40	18	3	38	28
3	31	11	3	33	6	3	35	7	2	31	14	2	33	23	2	30	32
2	22	8	2	24	6	2	27	8	1	21	19	1	23	28	1	19	36
1	11	7	1	13	7	1	17	11									
culm.	0	6	culm.	2	9	culm.	6	15	culm.	11	24	culm.	12	32	culm.	8	39
1	11	7	1	9	11	1	5	20									
2	21	8	2	20	16	2	14	24	1	0	28	1	1	36	1	3	41
3	31	11	3	28	20	3	22	29	2	10	33	2	9	39	2	14	42
4	38	15	4	34	24	4	28	33	3	18	37	3	18	40	3	23	41
5	43	19	5	37	29	5	32	36	4	25	39	4	25	41	4	30	40
6	44	23	6	37	33	6	32	40	5	29	41	5	29	40	5	35	37
7	43	28	6^h 51′	36	36	6^h 28′	31	40	6	30	41	5^h 32′	31	40	5^h 9′	36	36

f. 201,r.

Hours of Capricorn.	Parallax of the moon. Long.	Lat. south	Hours of Aquarius.	Parallax of the moon. Long.	Lat. south	Hours of Pisces.	Parallax of the moon. Long.	Lat. south	Hours of Aries.	Parallax of the moon. Long.	Lat. south	Hours of Taurus.	Parallax of the moon. Long.	Lat. south	Hours of Gemini.	Parallax of the moon. Long.	Lat. south
5^h	43′	28′	5^h 9′	36′	36′	5^h 32′	31′	40′	6^h	30′	41′	6^h 28′	31′	40′	6^h 51′	36′	36′
4	38	32	5	35	37	5	29	40	5	29	41	6	32	40	6	37	33
3	31	36	4	30	40	4	25	41	4	25	39	5	32	36	5	37	29
2	22	39	3	23	41	3	18	40	3	18	37	4	28	33	4	34	24
1	11	41	2	14	42	2	9	39	2	10	33	3	22	29	3	28	20
			1	3	41	1	1	36	1	0	28	2	14	24	2	20	16
												1	5	20	1	9	11
culm.	0	42	culm.	8	39	culm.	12	32	culm.	11	24	culm.	6	15	culm.	2	9
												1	17	11	1	13	7
			1	19	36	1	23	28	1	21	19	2	27	8	2	24	6
1	11	41	2	30	32	2	33	23	2	31	14	3	35	7	3	33	6
2	22	39	3	38	28	3	40	18	3	39	11	4	42	6	4	41	8
3	31	36	4	45	23	4	46	14	4	45	8	5	48	6	5	46	11
4	38	32	5	48	19	5	49	10	5	49	7	6	51	8	6	48	14
5	43	28	5^h 9′	48	17	5^h 32′	50	9	6	51	6	6^h 28′	50	9	6^h 51′	48	17

f. 201,v.

جداول اختلاف منظر القمر في الطول والعرض للاقليم الرابع وعرضه لو كب وساعاته يد ل

Table of the parallaxes of the moon in longitude and latitude, in the fourth clime, whose latitude is 36° 22′ and whose hours are $14^h\ 30^m$.

Hours of Cancer.	Parallax of the moon. Long. Lat. south.	Hours of Leo.	Parallax of the moon. Long. Lat. south.	Hours of Virgo.	Parallax of the moon. Long. Lat. south.	Hours of Libra.	Parallax of the moon. Long. Lat. south.	Hours of Scorpio.	Parallax of the moon. Long. Lat. south.	Hours of Sagittarius.	Parallax of the moon. Long. Lat. south.
7^h 15′	39′ 37′	7^h 4′	45′ 23′	6^h 35′	49′ 14′	6^h	50′ 11′	5^h 25′	49′ 14′	4^h 56′	45′ 23′
7	40 32	7	46 23	6	48 12	5	49 12	5	48 15	4	42 27
6	41 27	6	46 18	5	47 11	4	45 13	4	45 18	3	37 31
5	40 23	5	45 15	4	43 11	3	39 15	3	40 22	2	29 35
4	36 19	4	40 13	3	36 12	2	31 19	2	32 27	1	19 38
3	29 16	3	32 12	2	28 14	1	22 23	1	23 31		
2	20 13	2	24 11	1	18 16						
1	11 12	1	14 12								
culm.	0 11	culm.	3 14	culm.	8 20	culm.	12 28	culm.	14 35	culm.	9 42
1	11 12	1	8 16								
2	20 13	2	17 20	1	2 24						
3	29 16	3	24 24	2	12 29	1	2 32	1	4 39		
4	36 19	4	31 28	3	19 33	2	7 36	2	6 42	1	1 43
5	40 23	5	34 33	4	24 37	3	15 39	3	15 44	2	11 44
6	41 27	6	34 37	5	27 40	4	21 42	4	21 44	3	20 44
7	40 32	7	32 40	6	28 42	5	25 44	5	26 44	4	27 42
7^h 15′	39 37	7^h 4′	32 40	6^h 35′	27 43	6	26 44	5^h 25′	27 43	4^h 56′	32 40

f. 202,r.

Hours of Capricorn.	Parallax of the moon. Long. Lat. south.	Hours of Aquarius.	Parallax of the moon. Long. Lat. south.	Hours of Pisces.	Parallax of the moon. Long. Lat. south.	Hours of Aries.	Parallax of the moon. Long. Lat. south.	Hours of Taurus.	Parallax of the moon. Long. Lat. south.	Hours of Gemini.	Parallax of the moon. Long. Lat. south.
4^h 45′	39′ 33′	4^h 56′	32′ 40′	5^h 25′	27′ 43′	6^h	26′ 44′	6^h 35′	27′ 43′	7^h 4′	32′ 40′
4	35 35	4	27 42	5	26 44	5	25 44	6	28 42	7	32 40
3	29 39	3	20 44	4	21 44	4	21 42	5	27 40	6	34 37
2	21 42	2	11 44	3	15 44	3	15 39	4	24 37	5	34 33
1	11 44	1	1 43	2	6 42	2	7 36	3	19 33	4	31 28
				1	4 39	1	2 32	2	12 29	3	24 24
								1	2 24	2	17 20
										1	8 16
culm.	0 44	culm.	9 42	culm.	14 35	culm.	12 28	culm.	8 20	culm.	3 14
										1	14 12
								1	18 16	2	24 11
				1	23 31	1	22 23	2	28 14	3	32 12
1	11 44	1	19 38	2	32 27	2	31 19	3	36 12	4	40 13
2	21 42	2	29 35	3	40 22	3	39 15	4	43 11	5	45 15
3	29 39	3	37 31	4	45 18	4	45 13	5	47 11	6	46 18
4	35 35	4	42 27	5	48 15	5	49 12	6	48 12	7	46 23
4^h 45′	39 33	4^h 56′	45 23	5^h 25′	49 14	6	50 11	6^h 35′	49 14	7^h 4′	45 23

f. 202,v.

جداول اختلاف منظر القمر في الطول والعرض للاقليم الخامس وعرضه ما يه وساعاته يه

Table of the parallaxes of the moon in longitude and latitude, in the fifth clime, whose latitude is 41° 15′ and whose hours are 15^h.

Hours of Cancer.	Parallax of the moon. Long.	Lat. south	Hours of Leo.	Parallax of the moon. Long.	Lat. south	Hours of Virgo.	Parallax of the moon. Long.	Lat. south	Hours of Libra.	Parallax of the moon. Long.	Lat. south	Hours of Scorpio.	Parallax of the moon. Long.	Lat. south	Hours of Sagittarius.	Parallax of the moon. Long.	Lat. south
7^h 30′	36′	36′	7^h 16′	43′	27′	6^h 42′	48′	18′	6^h	49′	15′	5^h 18′	48′	18′	4^h 45′	43′	27′
7	37	35	7	44	26	6	47	15	5	47	15	5	47	19	4	40	30
6	39	31	6	45	22	5	46	15	4	44	17	4	44	23	3	36	34
5	38	27	5	42	19	4	41	15	3	38	19	3	39	26	2	28	38
4	34	23	4	38	17	3	36	16	2	32	23	2	32	30	1	19	41
3	28	20	3	32	16	2	27	17	1	23	27	1	24	34			
2	20	17	2	24	15	1	18	20									
1	10	16	1	14	16												
culm.	0	15	culm.	4	18	culm.	9	23	culm.	14	31	culm.	15	38	culm.	10	43
1	10	16	1	6	20												
2	20	17	2	15	24	1	0	27									
3	28	20	3	22	28	2	9	31	1	5	35	1	5	42			
4	34	23	4	27	32	3	16	36	2	4	39	2	4	44	1	0	45
5	38	27	5	30	35	4	21	39	3	11	42	3	12	45	2	9	46
6	39	31	6	31	38	5	24	42	4	17	44	4	17	46	3	18	46
7	37	35	7	28	42	6	24	44	5	21	45	5	23	46	4	24	44
7^h 30′	36	36	7^h 16′	28	43	6^h 42′	23	45	6	22	46	5^h 18′	23	45	4^h 45′	28	43

f. 203,r.

Hours of Capricorn.	Parallax of the moon. Long.	Lat. south	Hours of Aquarius.	Parallax of the moon. Long.	Lat. south	Hours of Pisces.	Parallax of the moon. Long.	Lat. south	Hours of Aries.	Parallax of the moon. Long.	Lat. south	Hours of Taurus.	Parallax of the moon. Long.	Lat. south	Hours of Gemini.	Parallax of the moon. Long.	Lat. south
4^h 30′	36′	36′	4^h 45′	28′	43′	5^h 18′	23′	45′	6^h	22′	46′	6^h 42′	23′	45′	7^h 16′	28′	43′
4	34	38	4	24	44	5	23	46	5	21	45	6	24	44	7	28	42
3	27	42	3	18	46	4	17	46	4	17	44	5	24	42	6	31	38
2	19	44	2	9	46	3	12	45	3	11	42	4	21	39	5	30	35
1	10	46	1	0	45	2	4	44	2	4	39	3	16	36	4	27	32
						1	5	42	1	5	35	2	9	31	3	22	28
												1	0	27	2	15	24
															1	6	20
culm.	0	46	culm.	10	43	culm.	15	38	culm.	14	31	culm.	9	23	culm.	4	18
															1	14	16
												1	18	20	2	24	15
						1	24	34	1	23	27	2	27	17	3	32	16
1	10	46	1	19	41	2	32	30	2	32	23	3	36	16	4	38	17
2	19	44	2	28	38	3	39	26	3	38	19	4	41	15	5	42	19
3	27	42	3	36	34	4	44	23	4	44	17	5	46	15	6	45	22
4	34	38	4	40	30	5	47	19	5	47	15	6	47	15	7	44	26
4^h 30′	36	36	4^h 45′	43	27	5^h 18′	48	18	6	49	15	6^h 42′	48	18	7^h 16′	43	27

f. 203,v.

جداول اختلاف منظر القمر في الطول والعرض للاقليم السادس وعرضه مه كب وساعاته يه ل

Tables of the parallaxes of the moon in longitude and latitude, in the sixth clime, whose latitude is 45° 22′ and whose hours are $15^h\ 30^m$.

Hours of Cancer.	Parallax of the moon. Long.	Lat. south.	Hours of Leo.	Parallax of the moon. Long.	Lat. south.	Hours of Virgo.	Parallax of the moon. Long.	Lat. south.	Hours of Libra.	Parallax of the moon. Long.	Lat. south.	Hours of Scorpio.	Parallax of the moon. Long.	Lat. south.	Hours of Sagittarius.	Parallax of the moon. Long.	Lat. south.
7^h 45′	32′	38′	7^h 28′	41′	30′	6^h 48′	46′	22′	6^h	48′	18′	5^h 12′	46′	22′	4^h 32′	41′	30′
7	35	36	7	42	29	6	46	20	5	46	18	5	46	22	4	39	33
6	36	33	6	42	25	5	44	19	4	44	20	4	43	25	3	34	36
5	35	29	5	40	22	4	41	19	3	38	23	3	38	28	2	27	39
4	32	26	4	37	20	3	35	19	2	31	25	2	32	32	1	18	42
3	25	23	3	31	19	2	27	20	1	23	29	1	24	36			
2	18	20	2	23	18	1	15	23									
1	9	19	1	14	19												
culm.	0	19	culm.	5	20	culm.	10	26	culm.	15	33	culm.	15	40	culm.	10	45
1	9	19	1	4	23												
2	18	20	2	13	26	1	1	30									
3	25	23	3	20	30	2	7	34	1	6	36	1	6	44			
4	32	26	4	25	34	3	13	38	2	2	40	2	2	45	1	1	46
5	35	29	5	27	37	4	18	42	3	9	43	3	9	47	2	7	47
6	36	33	6	28	40	5	20	44	4	14	46	4	15	48	3	16	47
7	35	36	7	26	44	6	21	46	5	18	46	5	19	47	4	22	46
7^h 45′	32	38	7^h 28′	24	45	6^h 48′	20	47	6	20	47	5^h 12′	20	47	4^h 32′	24	45

f. 204,r.

Hours of Capricorn.	Parallax of the moon. Long.	Lat. south.	Hours of Aquarius.	Parallax of the moon. Long.	Lat. south.	Hours of Pisces.	Parallax of the moon. Long.	Lat. south.	Hours of Aries.	Parallax of the moon. Long.	Lat. south.	Hours of Taurus.	Parallax of the moon. Long.	Lat. south.	Hours of Gemini.	Parallax of the moon. Long.	Lat. south.
4^h 15′	32′	38′	4^h 32′	24′	45′	5^h 12′	20′	47′	6^h	20′	47′	6^h 48′	20′	47′	7^h 28′	24′	45′
4	31	40	4	22	46	5	19	47	5	18	46	6	21	46	7	26	44
3	25	43	3	16	47	4	15	48	4	14	46	5	20	44	6	28	40
2	18	46	2	7	47	3	9	47	3	9	43	4	18	42	5	27	37
1	9	47	1	1	46	2	2	45	2	2	40	3	13	38	4	25	34
						1	6	44	1	6	36	2	7	34	3	20	30
												1	1	30	2	13	26
															1	4	23
culm.	0	48	culm.	10	45	culm.	15	40	culm.	15	33	culm.	10	26	culm.	5	20
															1	14	19
												1	15	23	2	23	18
						1	24	36	1	23	29	2	27	20	3	31	19
1	9	47	1	18	42	2	32	32	2	31	25	3	35	19	4	37	20
2	18	46	2	27	39	3	38	28	3	38	23	4	41	19	5	40	22
3	25	43	3	34	36	4	43	25	4	44	20	5	44	19	6	42	25
4	31	40	4	39	33	5	46	22	5	46	18	6	46	20	7	42	29
4^h 15′	32	38	4^h 32′	41	30	5^h 12′	46	22	6	48	18	6^h 48′	46	22	7^h 28′	41	30

f. 204,v.

جداول اختلاف منظر القمر في الطول والعرض للاقليم السابع وعرضه مح يج وساعاته يو

Tables of the parallaxes of the moon in longitude and latitude, in the seventh clime, whose latitude is 48° 13′ and whose hours are 16^h.

Hours of Cancer.	Parallax of the moon. Long. Lat. south	Hours of Leo.	Parallax of the moon. Long. Lat. south	Hours of Virgo.	Parallax of the moon. Long. Lat. south	Hours of Libra.	Parallax of the moon. Long. Lat. south	Hours of Scorpio.	Parallax of the moon. Long. Lat. south	Hours of Sagittarius.	Parallax of the moon. Long. Lat. south
8^h	30′ 42′	7^h 40′	39′ 33′	6^h 54′	45′ 24′	6^h	46′ 21′	5^h 6′	45′ 24′	4^h 20′	39′ 33′
7	33 38	7	40 32	6	45 23	5	45 22	5	45 25	4	37 34
6	34 35	6	40 28	5	43 22	4	42 23	4	42 27	3	33 38
5	32 31	5	39 25	4	40 21	3	37 25	3	37 31	2	27 41
4	29 28	4	35 23	3	35 22	2	31 26	2	31 34	1	19 44
3	24 25	3	30 22	2	27 24	1	23 28	1	24 38		
2	17 23	2	22 22	1	19 26						
1	9 22	1	14 22								
culm.	0 21	culm.	5 23	culm.	11 29	culm.	15 31	culm.	17 41	culm.	10 47
1	9 22	1	3 26								
2	17 23	2	11 29	1	3 32						
3	24 25	3	18 32	2	5 36	1	7 35	1	8 44		
4	29 28	4	22 36	3	11 40	2	0 38	2	0 46	1	2 48
5	32 31	5	25 39	4	15 43	3	6 42	3	7 48	2	6 49
6	34 35	6	25 42	5	18 45	4	11 45	4	13 48	3	14 48
7	33 38	7	23 45	6	18 47	5	14 48	5	17 48	4	20 47
8	30 42	7^h 40′	21 46	6^h 54′	17 48	6	16 48	5^h 6′	17 48	4^h 20′	21 46

f. 205,r.

Hours of Capricorn.	Parallax of the moon. Long. Lat. south	Hours of Aquarius.	Parallax of the moon. Long. Lat. south	Hours of Pisces.	Parallax of the moon. Long. Lat. south	Hours of Aries.	Parallax of the moon. Long. Lat. south	Hours of Taurus.	Parallax of the moon. Long. Lat. south	Hours of Gemini.	Parallax of the moon. Long. Lat. south
4^h	30′ 42′	4^h 20′	21′ 46′	5^h 6′	17′ 48′	6^h	16′ 48′	6^h 54′	17′ 48′	7^h 40′	21′ 46′
3	24 45	4	20 47	5	17 48	5	14 48	6	18 47	7	23 45
2	17 47	3	14 48	4	13 48	4	11 45	5	18 45	6	25 42
1	9 48	2	6 49	3	7 48	3	6 42	4	15 43	5	25 39
		1	2 48	2	0 46	2	0 38	3	11 40	4	22 36
				1	8 44	1	7 35	2	5 36	3	18 32
								1	3 32	2	11 29
										1	3 26
culm.	0 49	culm.	10 47	culm.	17 41	culm.	15 31	culm.	11 29	culm.	5 23
										1	14 22
								1	19 26	2	22 22
				1	24 38	1	23 28	2	27 24	3	30 22
		1	19 44	2	31 34	2	31 26	3	35 22	4	35 23
1	9 48	2	27 41	3	37 31	3	37 25	4	40 21	5	39 25
2	17 47	3	33 38	4	42 27	4	42 23	5	43 22	6	40 28
3	24 45	4	37 34	5	45 25	5	45 22	6	45 23	7	40 32
4	30 42	4^h 20′	39 33	5^h 6′	45 24	6	46 21	6^h 54′	45 24	7^h 40′	39 33

f. 205,v.

جداول حركات الخمسة الكواكب المتحيرة في السنين الرومية المجموعة
Tables of the motions of the five planets in collected Roman years [from the era].

سنو الروم المجموعة Collected Roman years [from the era].	وسط زحل Mean longitude of Saturn.	وسط المشتري Mean longitude of Jupiter.	وسط المريخ Mean longitude of Mars.	حاصة الزهرة Anomaly of Venus.	حاصة عطارد Anomaly of Mercury.
931	87° 15′	260° 4′	117° 58′	231° 11′	263° 0′
951	331 57	147 21	346 20	54 54	277 28
971	216 40	34 39	214 42	238 37	291 55
991	101 23	281 56	83 4	62 20	306 23
1011	346 6	169 14	311 26	246 3	320 50
1031	230 48	56 32	179 48	69 46	335 18
1051	115 31	303 49	48 11	253 29	349 45
1071	0 14	191 7	276 33	77 12	4 13
1091	244 56	78 24	144 55	260 55	18 41
1111	129 39	325 42	13 17	84 38	33 8
1131	14 21	212 59	241 39	268 21	47 36
1151	259 4	100 17	110 2	92 4	62 4
1171	143 47	347 34	338 24	275 47	76 32
1191	28 30	234 52	206 46	99 30	91 0
1211	273 13	122 10	75 8	283 13	105 28
1231	157 55	9 27	303 30	106 56	119 55
1251	42 38	256 45	171 53	290 39	134 23
1271	287 21	144 2	40 15	114 22	148 51
1291	172 4	31 20	268 37	298 5	163 19
1311	56 47	278 37	136 59	121 48	177 46
1331	301 29	165 55	5 21	305 31	192 14
1351	186 12	53 12	233 43	129 14	206 42
1371	70 55	300 30	102 6	312 57	221 9
1391	315 37	187 48	330 28	136 40	235 37
1411	200 20	75 5	198 50	320 23	250 5
1431	85 2	322 23	67 12	144 6	264 33
1451	329 45	209 40	295 34	327 49	279 0
1471	214 28	96 58	163 57	151 32	293 28
1491	99 11	344 16	32 19	335 15	307 56
1511	343 54	231 33	260 41	158 58	322 24
1531	228 36	118 51	129 3	342 41	336 52
1551	113 19	6 8	357 25	166 24	351 19
1571	358 2	253 26	225 48	350 7	5 47
1591	212 45	140 44	94 10	173 50	20 15

f. 206,r.

جدول حركات الكواكب الخمسة المتحيرة في السنين الرومية المبسوطة
Table of the motions of the five planets in single Roman years.

سنو الروم المبسوطة Roman years one by one.	وسط زحل Mean motion of Saturn.	وسط المشتري Mean motion of Jupiter.	وسط المريخ Mean motion of Mars.	حاصة الزهرة Motion of the anomaly of Venus.	حاصة عطارد Motion of the anomaly of Mercury.
1	12° 14′	30° 21′	191° 17′	225° 2′	53° 57′
2	24 27	60 41	22 34	90 5	107 53
3	36 41	91 2	213 52	315 7	161 49
4	48 56	121 28	45 40	180 45	218 52
5	61 10	151 48	236 57	45 47	272 49
6	73 23	182 9	68 15	270 50	326 46
7	85 37	212 30	259 32	135 51	20 43
8	97 53	242 55	91 21	1 30	77 46
9	110 6	273 16	282 38	226 31	131 43
10	122 20	303 36	113 55	91 33	185 37
11	134 34	333 57	305 12	316 35	239 34
12	146 50	4 23	137 1	182 14	296 37
13	159 4	34 44	328 18	47 15	350 34
14	171 17	65 4	159 35	272 17	44 31
15	183 31	95 25	350 53	137 19	98 28
16	195 47	125 50	182 42	2 58	155 30
17	208 0	156 11	13 59	228 0	209 29
18	220 14	186 32	205 16	93 2	263 27
19	232 27	216 52	36 34	318 4	317 25
20	244 43	247 18	228 22	183 43	14 28
	244° 42′ 44″ 20^{III}	247° 17′ 34″ 26^{III}	228° 22′ 10″ 49^{III}	183° 43′ 2″	14° 27′ 43″ 20^{III}
	Sums of years (سنون مجموعة).				
40	129° 25′	134° 35′	96° 44′	7° 26′	28° 55′
60	14 8	21 53	325 6	191 9	43 23
80	258 51	269 10	193 28	14 52	57 51
100	143 34	156 28	61 51	198 35	72 19
200	287 8	312 56	123 42	37 10	144 37
400	214 15	265 51	247 24	74 21	289 14
600	141 23	218 47	11 5	111 31	73 52

f. 206,v.

جداول حركات الكواكب الخمسة المتحيرة في الشهور الرومية

Tables of the motions of the five planets in the Roman months.

اسماء الشهور الرومية Names of the Roman months.	زحل [Mean motion] of Saturn.	المشتري [Mean motion] of Jupiter.	المريخ [Mean motion] of Mars.	الزهرة [Motion of the anomaly] of Venus.	عطارد [Motion of the anomaly] of Mercury.
ādhār [March].	1° 2′	2° 35′	16° 15′	19° 7′	96° 18′
nīsān [April].	2 2	5 4	31 58	37 37	189 30
ayyār [May].	3 4	7 39	48 13	56 43	285 49
ḥazīrān [June].	4 5	10 8	63 56	75 13	19 1
tammūz [July].	5 7	12 43	80 11	94 20	115 19
āb [August].	6 9	15 17	96 26	113 26	211 38
aylūl [September].	7 10	17 45	112 9	131 56	304 50
tishrīn I [October].	8 12	20 22	128 24	151 2	41 9
tishrīn II [November].	9 13	22 51	144 7	169 32	134 21
kānūn I [December].	10 15	25 26	160 22	188 39	230 39
kānūn II [January].	11 17	28 0	176 36	207 45	326 57
subāṭ [February].					
common.	12 14	30 21	191 17	225 2	53 57
leap.	12 16	30 26	191 49	225 39	57 3

f. 207,r.

جداول حركات الكواكب الخمسة المتحيرة في الساعات المعتدلة
Tables of the motions of the five planets in equal hours.

الساعات المستوية Equal hours.	زحل [Mean motion] of Saturn.	المشتري [Mean motion] of Jupiter.	المريخ [Mean motion] of Mars.	حاصة الزهرة Motion of the anomaly of Venus.	حاصة عطارد Motion of the anomaly of Mercury.
1	0° 0′	0° 0′	0° 1′	0° 2′	0° 8′
2	0 0	0 0	0 3	0 3	0 16
3	0 0	0 1	0 4	0 5	0 23
4	0 0	0 1	0 5	0 6	0 31
5	0 0	0 1	0 7	0 8	0 39
6	0 0	0 1	0 8	0 9	0 47
7	0 0	0 1	0 9	0 11	0 54
8	0 1	0 2	0 10	0 12	1 2
9	0 1	0 2	0 12	0 14	1 10
10	0 1	0 2	0 13	0 15	1 18
11	0 1	0 2	0 14	0 17	1 25
12	0 1	0 3	0 16	0 18	1 33
13	0 1	0 3	0 17	0 20	1 41
14	0 1	0 3	0 18	0 22	1 49
15	0 1	0 3	0 20	0 23	1 57
16	0 1	0 3	0 21	0 25	2 4
17	0 1	0 4	0 22	0 26	2 12
18	0 2	0 4	0 24	0 28	2 20
19	0 2	0 4	0 25	0 29	2 28
20	0 2	0 4	0 26	0 31	2 35
21	0 2	0 4	0 28	0 32	2 43
22	0 2	0 5	0 29	0 34	2 50
23	0 2	0 5	0 30	0 36	2 58
24	0 2	0 5	0 31	0 37	3 6

f. 207,v.

جداول حركات الكواكب الخمسة المتحيرة في الايام للشهر الرومي

Tables of the motions of the five planets in the days of the Roman month.

عدد الايام Number of days.	[Mean motion] of Saturn.	[Mean motion] of Jupiter.	[Mean motion] of Mars.	Motion of the anomaly of Venus.	Motion of the anomaly of Mercury.
1	0° 2′	0° 5′	0° 31′	0° 37′	3° 6′
2	0 4	0 10	1 3	1 14	6 13
3	0 6	0 15	1 34	1 51	9 19
4	0 8	0 20	2 6	2 28	12 26
5	0 10	0 25	2 37	3 5	15 32
6	0 12	0 30	3 9	3 42	18 39
7	0 14	0 35	3 40	4 19	21 45
8	0 16	0 40	4 12	4 56	24 51
9	0 18	0 45	4 43	5 33	27 58
10	0 20	0 50	5 15	6 10	31 4
11	0 22	0 55	5 46	6 47	34 11
12	0 24	1 0	6 18	7 24	37 17
13	0 26	1 5	6 49	8 1	40 23
14	0 28	1 10	7 20	8 38	43 30
15	0 30	1 15	7 52	9 15	46 36
16	0 32	1 20	8 23	9 52	49 42
17	0 34	1 25	8 54	10 29	52 49
18	0 36	1 30	9 26	11 6	55 55
19	0 38	1 35	9 57	11 43	59 2
20	0 40	1 40	10 29	12 20	62 8
21	0 42	1 45	11 0	12 57	65 14
22	0 44	1 50	11 32	13 34	68 21
23	0 46	1 55	12 3	14 11	71 27
24	0 48	2 0	12 35	14 48	74 34
25	0 50	2 5	13 6	15 25	77 40
26	0 52	2 10	13 38	16 2	80 46
27	0 54	2 15	14 9	16 39	83 53
28	0 56	2 20	14 41	17 16	86 59
29	0 58	2 25	15 12	17 53	90 6
30	1 0	2 30	15 43	18 30	93 12

f. 208,r.

جداول حركات الكواكب الثابتة وهي حركة واحدة للجميع

Tables of the motion of the fixed stars, which is one and the same for them all.

سنو الروم المجموعة Collected Roman years.	مسيرها المجموع Motion in collected Roman years.
20	0° 18′ 11″
40	0 36 22
60	0 54 33
80	1 12 44
100	1 30 55
120	1 49 5
140	2 7 16
160	2 25 27
180	2 43 38
200	3 1 49
220	3 20 0
240	3 38 11
260	3 56 22
280	4 14 33
300	4 32 44
320	4 50 55
340	5 9 6
360	5 27 16
380	5 45 27
400	6 3 38
420	6 21 49
440	6 40 0

[Motion in single Roman years.]

سنو الروم المبسوطة Single years.	مسيرها المبسوط Motion in single years.
1	0′ 54″ 33‴
2	1 49
3	2 44
4	3 38
5	4 33
6	5 27
7	6 22
8	7 16
9	8 11
10	9 5
11	10 0
12	10 55
13	11 49
14	12 44
15	13 38
16	14 33
17	15 27
18	16 22
19	17 16
20	18′ 11″ 24‴

[Motion in the Roman months.]

اسماء الشهور الرومية Names of the months.	مسيرها في الشهور Motion in the months.
ādhār [Martius]	0′ 5″
nīsān [Aprilis]	0 9
ayyār [Maius]	0 14
ḥazīrān [Iunius]	0 18
tammūz [Iulius]	0 23
āb [Augustus]	0 28
aylūl [September]	0 32
tishrīn I [October]	0 37
tishrīn II [November]	0 41
kānūn I [December]	0 46
kānūn II [Ianuarius]	0 50
subāṭ [Februarius]	0 54

ايام الشهور Days of the month.	في الايام [Motion] in days.
1	0′ 0″
2	0 0
3	0 0
4	0 1
5	0 1
6	0 1
7	0 1
8	0 1
9	0 1
10	0 1
11	0 2
12	0 2
13	0 2
14	0 2
15	0 2
16	0 2
17	0 3
18	0 3
19	0 3
20	0 3
21	0 3
22	0 3
23	0 3
24	0 4
25	0 4
26	0 4
27	0 4
28	0 4
29	0 4
30	0′ 4″ 30‴

f. 208,v.

بعد زحل الابعد سنة اقضا لذي القرنين رمد كح مسير حاصة زحل ليوم ه نز

[Longitude] of the apogee of Saturn in the year 1191 of Dhū 'l-Qarnayn 244° 28′; daily motion of the anomaly of Saturn 0° 57′.

ابتداء جداول تعديل زحل المجموع الكلي بابعاده الثلثة

Beginning of the tables of the equation of Saturn, always compounded with its three distances.

سطرا العدد Columns of numbers.		الثالث تعديل الحاصة والمركز III. Equation of the anomaly and the centre.	الرابع حصص البعدين IV. Shares of the two distances.	الخامس البعد الابعد V. Farthest distance.	السادس البعد الاوسط تعديل الحاصة VI. Mean distance; equation of the anomaly.	السابع البعد الاقرب VII. Nearest distance.
الاول I.	الثاني II.					
1°	359°	0° 7′	decreasing 60′	0° 0′	0° 6′	0° 0′
2	358	0 14	60	0 1	0 12	0 1
3	357	0 20	60	0 1	0 18	0 1
4	356	0 27	60	0 1	0 24	0 1
5	355	0 33	60	0 2	0 30	0 2
6	354	0 39	60	0 2	0 36	0 2
7	353	0 46	60	0 2	0 42	0 2
8	352	0 52	60	0 3	0 48	0 3
9	351	0 58	59	0 3	0 54	0 3
10	350	1 5	59	0 3	1 0	0 3
11	349	1 11	59	0 4	1 6	0 4
12	348	1 17	59	0 4	1 11	0 4
13	347	1 24	58	0 4	1 17	0 5
14	346	1 30	58	0 4	1 23	0 5
15	345	1 36	57	0 5	1 28	0 6
16	344	1 43	57	0 5	1 34	0 6
17	343	1 49	56	0 5	1 40	0 7
18	342	1 55	56	0 5	1 45	0 7
19	341	2 1	56	0 6	1 51	0 8
20	340	2 7	55	0 6	1 57	0 8
21	339	2 13	55	0 6	2 2	0 8
22	338	2 19	55	0 6	2 8	0 9
23	337	2 25	54	0 7	2 13	0 9
24	336	2 31	54	0 7	2 18	0 9
25	335	2 37	53	0 7	2 24	0 10
26	334	2 43	53	0 7	2 29	0 10
27	333	2 49	52	0 7	2 34	0 10
28	332	2 55	52	0 8	2 40	0 11
29	331	3 1	51	0 8	2 45	0 11
30	330	3 6	51	0 8	2 50	0 11

f. 209,r.

من جداول تعديل زحل المجموع الكلي بابعاده الثلثة

From the tables of the equation of Saturn, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
31°	329°	3° 12′	decreasing 50′	0° 8′	2° 55′	0° 12′
32	328	3 18	50	0 9	3 0	0 12
33	327	3 23	49	0 9	3 5	0 12
34	326	3 29	49	0 9	3 10	0 13
35	325	3 34	48	0 9	3 15	0 13
36	324	3 39	48	0 10	3 20	0 13
37	323	3 45	47	0 10	3 25	0 14
38	322	3 50	47	0 10	3 30	0 14
39	321	3 55	46	0 10	3 35	0 14
40	320	4 0	46	0 10	3 40	0 15
41	319	4 5	45	0 11	3 45	0 15
42	318	4 10	44	0 11	3 50	0 15
43	317	4 15	44	0 11	3 55	0 16
44	316	4 20	43	0 11	3 59	0 16
45	315	4 25	42	0 11	4 3	0 16
46	314	4 30	42	0 12	4 8	0 17
47	313	4 35	41	0 12	4 13	0 17
48	312	4 39	40	0 12	4 17	0 17
49	311	4 44	39	0 12	4 22	0 18
50	310	4 48	38	0 13	4 26	0 18
51	309	4 52	37	0 13	4 30	0 18
52	308	4 57	36	0 13	4 34	0 18
53	307	5 1	35	0 13	4 38	0 19
54	306	5 5	35	0 14	4 42	0 19
55	305	5 9	34	0 14	4 46	0 19
56	304	5 13	33	0 14	4 50	0 19
57	303	5 17	32	0 14	4 53	0 19
58	302	5 21	32	0 15	4 57	0 20
59	301	5 25	31	0 15	5 1	0 20
60	300	5 29	30	0 15	5 5	0 20

f. 209,v.

من جداول تعديل زحل المجموع الكلي بابعاده الثلثة

From the tables of the equation of Saturn, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
			decreasing			
61°	299°	5° 33′	29′	0° 15′	5° 8′	0° 20′
62	298	5 37	28	0 16	5 12	0 20
63	297	5 41	27	0 16	5 15	0 20
64	296	5 44	26	0 16	5 19	0 20
65	295	5 47	25	0 16	5 22	0 20
66	294	5 50	24	0 17	5 25	0 20
67	293	5 53	23	0 17	5 28	0 21
68	292	5 56	22	0 17	5 31	0 21
69	291	5 59	21	0 17	5 34	0 21
70	290	6 2	20	0 17	5 37	0 21
71	289	6 5	19	0 17	5 40	0 21
72	288	6 7	18	0 18	5 42	0 21
73	287	6 10	16	0 18	5 45	0 21
74	286	6 12	15	0 18	5 47	0 21
75	285	6 14	14	0 18	5 49	0 21
76	284	6 16	13	0 18	5 51	0 21
77	283	6 18	12	0 18	5 53	0 21
78	282	6 19	11	0 18	5 55	0 22
79	281	6 21	9	0 18	5 57	0 22
80	280	6 22	8	0 19	5 59	0 22
81	279	6 23	7	0 19	6 0	0 22
82	278	6 25	6	0 19	6 2	0 22
83	277	6 26	5	0 19	6 4	0 22
84	276	6 27	4	0 19	6 5	0 22
85	275	6 28	3	0 19	6 7	0 22
86	274	6 28	2	0 19	6 8	0 22
87	273	6 29	1	0 19	6 9	0 22
			increasing			
88	272	6 30	1	0 19	6 10	0 22
89	271	6 30	2	0 19	6 11	0 23
90	270	6 31	3	0 19	6 12	0 23

f. 210,r.

من جداول تعديل زحل المجموع الكلي بابعاده الثلثة

From the tables of the equation of Saturn, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
			increasing			
91°	269°	6° 31′	4′	0° 19′	6° 12′	0° 23′
92	268	6 31	5	0 20	6 12	0 23
93	267	6 31	6	0 20	6 12	0 23
94	266	6 31	7	0 20	6 13	0 24
95	265	6 30	8	0 20	6 13	0 24
96	264	6 30	9	0 20	6 13	0 24
97	263	6 29	10	0 20	6 13	0 24
98	262	6 29	11	0 20	6 13	0 24
99	261	6 28	12	0 20	6 13	0 24
100	260	6 28	13	0 21	6 12	0 24
101	259	6 27	14	0 21	6 12	0 24
102	258	6 26	15	0 21	6 12	0 25
103	257	6 25	15	0 21	6 11	0 25
104	256	6 24	16	0 21	6 11	0 25
105	255	6 22	17	0 21	6 9	0 25
106	254	6 21	18	0 21	6 8	0 25
107	253	6 19	19	0 21	6 7	0 25
108	252	6 17	20	0 20	6 6	0 25
109	251	6 16	20	0 20	6 4	0 25
110	250	6 14	21	0 20	6 2	0 24
111	249	6 12	22	0 20	6 0	0 24
112	248	6 10	23	0 20	5 59	0 24
113	247	6 8	24	0 20	5 57	0 24
114	246	6 6	24	0 20	5 55	0 24
115	245	6 4	25	0 20	5 53	0 24
116	244	6 1	26	0 19	5 51	0 24
117	243	5 58	27	0 19	5 48	0 24
118	242	5 55	28	0 19	5 47	0 23
119	241	5 52	29	0 19	5 43	0 23
120	240	5 49	30	0 19	5 41	0 23

f. 210,v.

من جداول تعديل زحل المجموع الكلي بابعاده الثلثة

From the tables of the equation of Saturn, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
121°	239°	5° 46′	increasing 30′	0° 19′	5° 37′	0° 23′
122	238	5 43	31	0 19	5 35	0 23
123	237	5 40	32	0 19	5 32	0 23
124	236	5 36	32	0 19	5 28	0 23
125	235	5 32	33	0 18	5 25	0 22
126	234	5 28	34	0 18	5 21	0 22
127	233	5 24	35	0 18	5 18	0 22
128	232	5 20	36	0 18	5 14	0 22
129	231	5 16	37	0 18	5 10	0 22
130	230	5 12	37	0 17	5 6	0 21
131	229	5 8	38	0 17	5 2	0 21
132	228	5 3	39	0 17	4 58	0 21
133	227	4 58	40	0 17	4 54	0 20
134	226	4 53	41	0 17	4 50	0 20
135	225	4 48	42	0 17	4 45	0 20
136	224	4 43	42	0 16	4 41	0 19
137	223	4 38	43	0 16	4 36	0 19
138	222	4 33	44	0 16	4 31	0 19
139	221	4 28	44	0 16	4 26	0 18
140	220	4 23	45	0 15	4 21	0 18
141	219	4 17	46	0 15	4 16	0 18
142	218	4 12	46	0 15	4 11	0 17
143	217	4 6	47	0 15	4 6	0 17
144	216	4 0	48	0 14	4 0	0 17
145	215	3 54	49	0 14	3 55	0 16
146	214	3 48	49	0 14	3 49	0 16
147	213	3 42	50	0 13	3 43	0 15
148	212	3 36	50	0 13	3 37	0 15
149	211	3 30	51	0 12	3 31	0 14
150	210	3 24	51	0 12	3 25	0 14

f. 211,r.

تمام جداول تعديل زحل المجموع الكلي بابعاده الثلثة

End of the tables of the equation of Saturn, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
151°	209°	3° 18′	increasing 52′	0° 12′	3° 19′	0° 13′
152	208	3 12	53	0 11	3 13	0 13
153	207	3 5	53	0 11	3 7	0 13
154	206	2 59	54	0 11	3 1	0 12
155	205	2 53	54	0 10	2 55	0 12
156	204	2 46	55	0 10	2 48	0 12
157	203	2 40	55	0 9	2 42	0 11
158	202	2 34	56	0 9	2 36	0 11
159	201	2 27	56	0 8	2 29	0 11
160	200	2 21	57	0 8	2 23	0 10
161	199	2 14	57	0 7	2 16	0 10
162	198	2 7	58	0 7	2 9	0 10
163	197	2 0	58	0 6	2 2	0 9
164	196	1 53	58	0 6	1 55	0 9
165	195	1 46	59	0 6	1 48	0 8
166	194	1 39	59	0 5	1 41	0 8
167	193	1 32	59	0 5	1 34	0 7
168	192	1 25	59	0 5	1 27	0 7
169	191	1 18	59	0 4	1 20	0 6
170	190	1 11	59	0 4	1 13	0 6
171	189	1 4	60	0 4	1 6	0 5
172	188	0 57	60	0 3	0 59	0 5
173	187	0 50	60	0 3	0 52	0 4
174	186	0 43	60	0 3	0 45	0 4
175	185	0 36	60	0 2	0 38	0 3
176	184	0 29	60	0 2	0 31	0 3
177	183	0 22	60	0 2	0 23	0 2
178	182	0 15	60	0 1	0 16	0 1
179	181	0 8	60	0 1	0 8	0 1
180	180	0 0	60	0 0	0 0	0 0

f. 211,v.

بعد المشتري الابعد لسنة اقضا لذي القرنين قصد كح مسير حاصة المشتري ليوم ٥ ند

[Longitude] of the apogee of Jupiter in the year 1191 of Dhū 'l-qarnayn 164° 28′; daily motion of the anomaly of Jupiter 0° 54′.

ابتداء جداول تعديل المشتري المجموع الكلي بابعاده الثلثة

Beginning of the tables of the equation of Jupiter, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.		decreasing			
1°	359°	0° 6′	60′	0° 0′	0° 10′	0° 0′
2	358	0 11	60	0 1	0 20	0 1
3	357	0 16	60	0 1	0 29	0 1
4	356	0 21	60	0 1	0 39	0 1
5	355	0 27	60	0 2	0 49	0 2
6	354	0 32	60	0 2	0 58	0 2
7	353	0 37	60	0 2	1 8	0 2
8	352	0 42	60	0 3	1 18	0 3
9	351	0 47	60	0 3	1 28	0 3
10	350	0 53	60	0 3	1 37	0 3
11	349	0 58	60	0 4	1 47	0 4
12	348	1 3	59	0 4	1 56	0 4
13	347	1 8	59	0 4	2 6	0 5
14	346	1 13	59	0 5	2 15	0 5
15	345	1 18	59	0 5	2 24	0 6
16	344	1 23	59	0 5	2 34	0 6
17	343	1 28	58	0 6	2 43	0 6
18	342	1 33	58	0 6	2 52	0 7
19	341	1 38	58	0 6	3 2	0 7
20	340	1 43	57	0 7	3 11	0 8
21	339	1 48	57	0 7	3 20	0 8
22	338	1 53	57	0 7	3 30	0 8
23	337	1 58	56	0 8	3 39	0 9
24	336	2 3	56	0 8	3 48	0 9
25	335	2 8	55	0 8	3 57	0 9
26	334	2 13	55	0 9	4 6	0 10
27	333	2 17	54	0 9	4 15	0 10
28	332	2 22	54	0 9	4 24	0 10
29	331	2 26	53	0 10	4 33	0 11
30	330	2 31	53	0 10	4 42	0 11

f. 212,r.

من جداول تعديل المشتري المجموع الكلي بابعاده الثلثة

From the tables of the equation of Jupiter, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
31°	329°	2° 36′	decreasing 52′	0° 10′	4° 51′	0° 11′
32	328	2 41	52	0 11	5 0	0 12
33	327	2 45	51	0 11	5 8	0 12
34	326	2 50	50	0 11	5 17	0 12
35	325	2 55	50	0 12	5 26	0 13
36	324	2 59	49	0 12	5 34	0 13
37	323	3 4	49	0 12	5 43	0 13
38	322	3 8	48	0 13	5 52	0 14
39	321	3 12	47	0 13	6 1	0 14
40	320	3 16	47	0 13	6 9	0 14
41	319	3 20	46	0 14	6 17	0 15
42	318	3 24	45	0 14	6 25	0 15
43	317	3 28	45	0 14	6 33	0 15
44	316	3 32	44	0 15	6 41	0 16
45	315	3 36	43	0 15	6 49	0 16
46	314	3 40	43	0 15	6 57	0 17
47	313	3 44	42	0 16	7 5	0 17
48	312	3 47	41	0 16	7 12	0 18
49	311	3 51	40	0 16	7 20	0 18
50	310	3 55	39	0 17	7 28	0 19
51	309	3 58	38	0 17	7 36	0 19
52	308	4 2	37	0 17	7 43	0 19
53	307	4 5	36	0 18	7 50	0 20
54	306	4 8	35	0 18	7 57	0 20
55	305	4 12	35	0 18	8 4	0 20
56	304	4 15	34	0 19	8 11	0 21
57	303	4 18	33	0 19	8 17	0 21
58	302	4 21	32	0 19	8 24	0 21
59	301	4 24	31	0 20	8 31	0 22
60	300	4 27	30	0 20	8 37	0 22

f. 212,v.

من جداول تعديل المشتري المجموع الكلي بابعاده الثلثة

From the tables of the equation of Jupiter, always compounded with its three distances,

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
			decreasing			
61°	299°	4° 30′	29′	0° 20′	8° 44′	0° 22′
62	298	4 33	28	0 21	8 50	0 23
63	297	4 36	27	0 21	8 56	0 23
64	296	4 39	26	0 21	9 2	0 23
65	295	4 42	25	0 22	9 8	0 24
66	294	4 44	24	0 22	9 14	0 24
67	293	4 46	22	0 22	9 20	0 24
68	292	4 48	21	0 23	9 26	0 25
69	291	4 50	20	0 23	9 31	0 25
70	290	4 52	19	0 23	9 36	0 25
71	289	4 54	18	0 24	9 41	0 26
72	288	4 56	17	0 24	9 46	0 26
73	287	4 58	16	0 24	9 51	0 26
74	286	5 0	15	0 25	9 56	0 27
75	285	5 1	14	0 25	10 0	0 27
76	284	5 3	13	0 25	10 5	0 27
77	283	5 5	12	0 25	10 9	0 28
78	282	5 6	11	0 26	10 13	0 28
79	281	5 7	9	0 26	10 17	0 28
80	280	5 8	8	0 26	10 21	0 29
81	279	5 9	7	0 26	10 25	0 29
82	278	5 10	6	0 26	10 29	0 29
83	277	5 11	5	0 27	10 32	0 29
84	276	5 12	4	0 27	10 35	0 30
85	275	5 13	3	0 27	10 38	0 30
86	274	5 13	2	0 27	10 41	0 30
87	273	5 14	1	0 27	10 44	0 30
88	272	5 14	0	0 27	10 47	0 30
			increasing			
89	271	5 14	1	0 27	10 49	0 31
90	270	5 14	2	0 27	10 51	0 31

f. 213,r.

من جداول تعديل المشتري المجموع الكلي بابعاده الثلثة

From the tables of the equation of Jupiter, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
			increasing			
91°	269°	5° 14′	3′	0° 27′	10° 53′	0° 31′
92	268	5 15	4	0 28	10 56	0 31
93	267	5 15	5	0 28	10 59	0 31
94	266	5 15	5	0 28	11 0	0 31
95	265	5 15	6	0 28	11 1	0 31
96	264	5 15	7	0 28	11 2	0 31
97	263	5 15	8	0 28	11 2	0 31
98	262	5 14	9	0 28	11 2	0 31
99	261	5 14	10	0 29	11 3	0 31
100	260	5 13	11	0 29	11 3	0 32
101	259	5 13	12	0 29	11 3	0 32
102	258	5 12	13	0 29	11 3	0 32
103	257	5 11	14	0 29	11 2	0 32
104	256	5 10	15	0 29	11 2	0 32
105	255	5 9	16	0 29	11 2	0 32
106	254	5 8	17	0 29	11 1	0 32
107	253	5 6	18	0 30	11 0	0 32
108	252	5 4	19	0 30	10 59	0 32
109	251	5 3	20	0 30	10 57	0 33
110	250	5 1	21	0 30	10 55	0 33
111	249	4 59	22	0 30	10 53	0 33
112	248	4 57	22	0 30	10 51	0 33
113	247	4 55	23	0 30	10 48	0 33
114	246	4 53	24	0 30	10 45	0 33
115	245	4 51	25	0 30	10 42	0 33
116	244	4 49	26	0 30	10 39	0 33
117	243	4 47	27	0 30	10 35	0 33
118	242	4 45	28	0 30	10 31	0 33
119	241	4 43	29	0 30	10 27	0 33
120	240	4 40	30	0 29	10 23	0 33

f. 213,v.

من جداول تعديل المشتري المجموع الكلي بابعاده الثلثة

From the tables of the equation of Jupiter, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
121°	239°	4° 35′	increasing 31′	0° 29′	10° 19′	0° 32′
122	238	4 35	32	0 29	10 15	0 32
123	237	4 32	33	0 29	10 10	0 32
124	236	4 29	34	0 29	10 5	0 32
125	235	4 26	35	0 29	10 0	0 32
126	234	4 23	36	0 29	9 54	0 32
127	233	4 20	36	0 28	9 48	0 32
128	232	4 16	37	0 28	9 42	0 31
129	231	4 12	38	0 28	9 36	0 31
130	230	4 9	39	0 28	9 30	0 31
131	229	4 5	40	0 28	9 24	0 31
132	228	4 1	41	0 28	9 17	0 31
133	227	3 58	41	0 27	9 10	0 31
134	226	3 54	42	0 27	9 2	0 30
135	225	3 50	43	0 27	8 55	0 30
136	224	3 46	43	0 26	8 47	0 30
137	223	3 42	44	0 26	8 39	0 30
138	222	3 38	45	0 26	8 31	0 29
139	221	3 34	45	0 25	8 22	0 29
140	220	3 30	46	0 25	8 13	0 29
141	219	3 25	47	0 23	8 4	0 28
142	218	3 21	47	0 24	7 55	0 27
143	217	3 16	48	0 24	7 46	0 27
144	216	3 11	49	0 24	7 36	0 26
145	215	3 7	49	0 23	7 26	0 25
146	214	3 2	50	0 23	7 16	0 25
147	213	2 57	50	0 22	7 5	0 24
148	212	2 53	51	0 22	6 55	0 24
149	211	2 48	51	0 21	6 45	0 23
150	210	2 43	52	0 21	6 34	0 22

f. 214,r.

تمام جداول تعديل المشتري المجموع الكلي بابعاده الثلثة

End of the tables of the equation of Jupiter, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
151°	209°	2° 39′	increasing 52′	0° 21′	6° 23′	0° 22′
152	208	2 34	53	0 20	6 12	0 21
153	207	2 29	53	0 19	6 0	0 20
154	206	2 24	53	0 19	5 48	0 20
155	205	2 19	54	0 18	5 36	0 19
156	204	2 14	54	0 17	5 24	0 18
157	203	2 9	55	0 17	5 12	0 18
158	202	2 4	55	0 16	5 0	0 17
159	201	1 58	56	0 15	4 47	0 16
160	200	1 53	56	0 15	4 35	0 16
161	199	1 48	57	0 14	4 22	0 15
162	198	1 42	57	0 13	4 9	0 14
163	197	1 36	57	0 13	3 56	0 14
164	196	1 31	58	0 12	3 42	0 13
165	195	1 25	58	0 11	3 29	0 12
166	194	1 20	58	0 11	3 16	0 12
167	193	1 15	59	0 10	3 3	0 11
168	192	1 9	59	0 9	2 49	0 10
169	191	1 4	59	0 9	2 35	0 10
170	190	0 58	60	0 8	2 21	0 9
171	189	0 52	60	0 7	2 7	0 8
172	188	0 47	60	0 7	1 53	0 7
173	187	0 42	60	0 6	1 39	0 6
174	186	0 36	60	0 5	1 25	0 5
175	185	0 30	60	0 5	1 11	0 5
176	184	0 24	60	0 4	0 57	0 4
177	183	0 18	60	0 3	0 43	0 3
178	182	0 12	60	0 2	0 29	0 2
179	181	0 6	60	0 1	0 15	0 1
180	180	0 0	60	0 0	0 0	0 0

f. 214,v.

بعد المريخ الابعد لسنة اقضا لذي القرنين قكو نح مسير حاصة المريخ ليوم ٥ كح

[Longitude] of the apogee of Mars in the year 1191 of Dhū 'l-qarnayn 126° 58′; daily motion of the anomaly of Mars 0° 28′.

ابتداء جداول تعديل المريخ المجموع الكلي بابعاده الثلثة

Beginning of the tables of the equation of Mars, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.		decreasing			
1°	359°	0° 11′	60′	0° 2′	0° 24′	0° 2′
2	358	0 22	60	0 3	0 48	0 3
3	357	0 33	60	0 4	1 12	0 4
4	356	0 44	60	0 6	1 36	0 6
5	355	0 55	60	0 7	2 0	0 7
6	354	1 5	59	0 8	2 24	0 8
7	353	1 16	59	0 10	2 48	0 10
8	352	1 27	59	0 11	3 12	0 12
9	351	1 38	59	0 12	3 35	0 13
10	350	1 49	59	0 14	3 59	0 15
11	349	2 0	58	0 15	4 23	0 17
12	348	2 10	58	0 16	4 47	0 18
13	347	2 21	58	0 18	5 11	0 20
14	346	2 32	58	0 19	5 34	0 21
15	345	2 42	58	0 20	5 57	0 22
16	344	2 53	57	0 22	6 21	0 24
17	343	3 3	57	0 23	6 45	0 26
18	342	3 13	57	0 24	7 8	0 27
19	341	3 24	56	0 26	7 32	0 29
20	340	3 35	56	0 27	7 56	0 31
21	339	3 45	56	0 28	8 19	0 32
22	338	3 56	55	0 30	8 43	0 34
23	337	4 6	55	0 32	9 7	0 35
24	336	4 16	55	0 33	9 30	0 36
25	335	4 26	54	0 35	9 54	0 38
26	334	4 36	54	0 37	10 18	0 40
27	333	4 46	53	0 38	10 41	0 41
28	332	4 56	53	0 40	11 5	0 43
29	331	5 6	52	0 41	11 28	0 45
30	330	5 16	52	0 42	11 51	0 46

f. 215,r.

من جداول تعديل المريخ المجموع الكلي بابعاده الثلثة

From the tables of the equation of Mars, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
31°	329°	5° 26′	decreasing 51′	0° 44′	12° 15′	0° 48′
32	328	5 36	51	0 46	12 38	0 50
33	327	5 45	50	0 47	13 1	0 51
34	326	5 55	50	0 49	13 25	0 53
35	325	6 4	49	0 51	13 48	0 55
36	324	6 13	49	0 52	14 11	0 57
37	323	6 22	48	0 54	14 34	0 58
38	322	6 31	47	0 55	14 57	1 0
39	321	6 40	47	0 57	15 20	1 1
40	320	6 49	46	0 58	15 43	1 3
41	319	6 58	45	0 59	16 6	1 5
42	318	7 7	45	1 1	16 29	1 6
43	317	7 16	44	1 2	16 52	1 8
44	316	7 24	44	1 4	17 15	1 10
45	315	7 32	43	1 5	17 38	1 11
46	314	7 41	42	1 6	18 1	1 13
47	313	7 49	41	1 8	18 24	1 15
48	312	7 57	40	1 9	18 46	1 16
49	311	8 5	40	1 11	19 9	1 18
50	310	8 13	39	1 12	19 31	1 20
51	309	8 20	38	1 13	19 53	1 22
52	308	8 28	37	1 15	20 16	1 24
53	307	8 35	36	1 16	20 38	1 26
54	306	8 42	35	1 17	21 0	1 28
55	305	8 50	35	1 19	21 23	1 30
56	304	8 57	34	1 21	21 45	1 32
57	303	9 5	33	1 22	22 7	1 34
58	302	9 11	32	1 24	22 29	1 36
59	301	9 18	31	1 26	22 51	1 38
60	300	9 24	30	1 27	23 13	1 40

f. 215,v.

من جداول تعديل المريخ المجموع الكلي بابعاده الثلثة

From the tables of the equation of Mars, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
			decreasing			
61°	299°	9° 31′	29′	1° 29′	23° 35′	1° 42′
62	298	9 37	28	1 31	23 57	1 44
63	297	9 43	27	1 32	24 18	1 46
64	296	9 49	26	1 34	24 40	1 48
65	295	9 55	25	1 36	25 1	1 50
66	294	10 0	24	1 37	25 22	1 53
67	293	10 5	23	1 39	25 44	1 55
68	292	10 10	22	1 41	26 5	1 57
69	291	10 15	21	1 43	26 26	1 59
70	290	10 20	20	1 45	26 47	2 1
71	289	10 25	19	1 47	27 8	2 3
72	288	10 29	17	1 49	27 29	2 6
73	287	10 34	16	1 51	27 50	2 8
74	286	10 38	15	1 53	28 11	2 10
75	285	10 42	14	1 55	28 31	2 12
76	284	10 46	13	1 57	28 52	2 15
77	283	10 50	12	1 59	29 12	2 17
78	282	10 53	11	2 1	29 32	2 19
79	281	10 57	10	2 3	29 52	2 21
80	280	11 0	9	2 5	30 12	2 23
81	279	11 3	8	2 8	30 32	2 26
82	278	11 6	7	2 10	30 52	2 28
83	277	11 9	6	2 12	31 11	2 30
84	276	11 12	4	2 14	31 30	2 33
85	275	11 15	3	2 16	31 49	2 35
86	274	11 17	2	2 18	32 8	2 38
87	273	11 19	1	2 21	32 27	2 41
			increasing			
88	272	11 21	1	2 23	32 46	2 43
89	271	11 22	2	2 25	33 4	2 46
90	270	11 23	3	2 28	33 22	2 49

f. 216,r.

من جداول تعديل المريخ المجموع الكلي بابعاده الثلثة

From the tables of the equation of Mars, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
			increasing			
91°	269°	11° 24′	3′	2° 30′	33° 40′	2° 52′
92	268	11 24	4	2 32	33 58	2 55
93	267	11 25	5	2 35	34 15	2 58
94	266	11 25	6	2 37	34 32	3 1
95	265	11 25	7	2 39	34 49	3 4
96	264	11 25	8	2 42	35 6	3 7
97	263	11 24	9	2 44	35 23	3 10
98	262	11 24	10	2 46	35 40	3 13
99	261	11 23	11	2 49	35 56	3 16
100	260	11 22	12	2 51	36 12	3 19
101	259	11 21	13	2 54	36 28	3 22
102	258	11 20	14	2 57	36 43	3 25
103	257	11 19	14	2 59	36 58	3 28
104	256	11 17	15	3 1	37 13	3 31
105	255	11 15	16	3 4	37 27	3 35
106	254	11 13	17	3 6	37 41	3 38
107	253	11 11	18	3 9	37 55	3 42
108	252	11 9	19	3 12	38 9	3 46
109	251	11 6	20	3 15	38 23	3 49
110	250	11 3	21	3 18	38 36	3 53
111	249	11 0	22	3 21	38 49	3 57
112	248	10 57	22	3 24	39 1	4 1
113	247	10 53	23	3 27	39 13	4 5
114	246	10 49	24	3 31	39 24	4 9
115	245	10 45	25	3 34	39 35	4 13
116	244	10 41	26	3 37	39 45	4 17
117	243	10 36	27	3 41	39 55	4 22
118	242	10 32	28	3 45	40 4	4 26
119	241	10 27	29	3 48	40 13	4 31
120	240	10 22	30	3 52	40 21	4 36

f. 216,v.

من جداول تعديل المريخ المجموع الكلي بابعاده الثلثة

From the tables of the equation of Mars, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
121°	239°	10° 17′	increasing 30′	3° 55′	40° 29′	4° 40′
122	238	10 12	31	3 59	40 36	4 45
123	237	10 6	32	4 3	40 43	4 50
124	236	10 0	33	4 7	40 48	4 55
125	235	9 54	34	4 11	40 53	5 0
126	234	9 48	35	4 14	40 58	5 5
127	233	9 41	35	4 17	41 2	5 10
128	232	9 34	36	4 21	41 6	5 15
129	231	9 27	37	4 25	41 8	5 21
130	230	9 20	38	4 28	41 9	5 26
131	229	9 13	39	4 31	41 9	5 32
132	228	9 5	40	4 35	41 9	5 38
133	227	8 57	40	4 38	41 7	5 44
134	226	8 49	41	4 41	41 4	5 49
135	225	8 41	42	4 45	41 0	5 55
136	224	8 32	42	4 48	40 55	6 1
137	223	8 23	43	4 51	40 50	6 7
138	222	8 14	44	4 55	40 45	6 14
139	221	8 5	44	4 59	40 39	6 20
140	220	7 56	45	5 3	40 31	6 26
141	219	7 46	46	5 7	40 21	6 32
142	218	7 37	46	5 11	40 8	6 38
143	217	7 27	47	5 15	39 53	6 44
144	216	7 17	48	5 19	39 37	6 51
145	215	7 7	48	5 22	39 20	6 57
146	214	6 57	49	5 25	39 1	7 4
147	213	6 47	50	5 28	38 40	7 11
148	212	6 37	50	5 30	38 16	7 17
149	211	6 27	51	5 32	37 51	7 24
150	210	6 16	51	5 34	37 25	7 31

f. 217,r.

تمام جداول تعديل المريخ المجموع الكلي بابعاده الثلثة

End of the tables of the equation of Mars, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
151°	209°	6° 5′	increasing 52′	5° 36′	36° 57′	7° 36′
152	208	5 54	52	5 37	36 26	7 41
153	207	5 43	53	5 38	35 52	7 45
154	206	5 32	53	5 38	35 15	7 50
155	205	5 20	54	5 38	34 35	7 54
156	204	5 9	55	5 38	33 53	7 58
157	203	4 57	55	5 37	33 7	8 0
158	202	4 45	56	5 36	32 20	8 2
159	201	4 32	56	5 34	31 30	8 3
160	200	4 20	57	5 30	30 36	8 2
161	199	4 8	57	5 25	29 38	8 0
162	198	3 55	57	5 18	28 35	7 59
163	197	3 43	58	5 10	27 28	7 57
164	196	3 31	58	5 1	25 16	7 53
165	195	3 18	58	4 52	24 3	7 47
166	194	3 5	58	4 41	23 45	7 37
167	193	2 52	58	4 30	22 24	7 23
168	192	2 39	59	4 18	21 0	7 6
169	191	2 26	59	4 4	19 0	6 45
170	190	2 13	59	3 48	18 1	6 22
171	189	1 59	59	3 32	16 25	5 57
172	188	1 46	59	3 12	14 45	5 30
173	187	1 33	59	2 51	13 1	5 0
174	186	1 20	59	2 26	11 14	4 26
175	185	1 7	59	2 3	9 26	3 47
176	184	0 54	60	1 40	7 37	3 7
177	183	0 40	60	1 17	5 45	2 23
178	182	0 27	60	0 51	3 52	1 35
179	181	0 14	60	0 26	1 57	0 48
180	180	0 0	60	0 0	0 0	0 0

f. 217,v.

بعد الزهرة الابعد لسنة اقضا لذي القرنين قب يد

[Longitude] of the apogee of Venus in the year 1191 of Dhū 'l-Qarnayn 82° 14′; [daily motion of the anomaly of Venus 0° 37′.]

ابتداء جداول تعديل الزهرة المجموع الكلي بابعادها الثلثة

Beginning of the tables of the equation of Venus, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.		decreasing			
1°	359°	0° 2′	60′	0° 0′	0° 26′	0° 0′
2	358	0 4	60	0 1	0 51	0 1
3	357	0 6	60	0 1	1 16	0 1
4	356	0 8	60	0 1	1 41	0 1
5	355	0 10	60	0 1	2 6	0 2
6	354	0 12	60	0 1	2 31	0 2
7	353	0 14	60	0 1	2 56	0 2
8	352	0 16	59	0 2	3 21	0 3
9	351	0 18	59	0 2	3 46	0 3
10	350	0 19	59	0 2	4 11	0 3
11	349	0 21	59	0 3	4 36	0 4
12	348	0 23	59	0 3	5 1	0 4
13	347	0 25	58	0 3	5 26	0 4
14	346	0 27	58	0 4	5 51	0 5
15	345	0 29	58	0 4	6 16	0 5
16	344	0 31	58	0 4	6 41	0 5
17	343	0 33	57	0 5	7 6	0 6
18	342	0 35	57	0 5	7 31	0 6
19	341	0 37	57	0 5	7 56	0 6
20	340	0 39	57	0 6	8 21	0 7
21	339	0 41	56	0 6	8 46	0 7
22	338	0 43	56	0 6	9 11	0 7
23	337	0 45	56	0 7	9 36	0 8
24	336	0 47	55	0 7	10 1	0 8
25	335	0 49	55	0 7	10 26	0 8
26	334	0 51	55	0 8	10 51	0 9
27	333	0 52	54	0 8	11 16	0 9
28	332	0 54	54	0 8	11 41	0 9
29	331	0 56	53	0 9	12 6	0 10
30	330	0 58	52	0 9	12 30	0 10

f. 218,r.

من جداول تعديل الزهرة المجموع الكلي بابعادها الثلثة

From the tables of the equation of Venus, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
31°	329°	0° 59′	decreasing 51′	0° 9′	12° 55′	0° 10′
32	328	1 1	51	0 10	13 20	0 11
33	327	1 3	50	0 10	13 44	0 11
34	326	1 5	50	0 10	14 9	0 11
35	325	1 6	49	0 11	14 34	0 12
36	324	1 8	49	0 11	14 58	0 12
37	323	1 10	48	0 11	15 23	0 12
38	322	1 11	48	0 12	15 48	0 12
39	321	1 13	47	0 12	16 12	0 13
40	320	1 15	46	0 12	16 37	0 13
41	319	1 16	46	0 13	17 1	0 13
42	318	1 18	45	0 13	17 25	0 13
43	317	1 19	44	0 13	17 50	0 14
44	316	1 21	44	0 14	18 14	0 14
45	315	1 22	43	0 14	18 38	0 14
46	314	1 24	42	0 14	19 3	0 15
47	313	1 25	42	0 15	19 27	0 15
48	312	1 26	41	0 15	19 51	0 15
49	311	1 28	40	0 15	20 15	0 16
50	310	1 29	40	0 16	20 39	0 16
51	309	1 31	39	0 16	21 3	0 16
52	308	1 32	38	0 16	21 27	0 17
53	307	1 33	37	0 17	21 51	0 17
54	306	1 34	36	0 17	22 15	0 17
55	305	1 36	35	0 17	22 39	0 18
56	304	1 37	34	0 18	23 3	0 18
57	303	1 38	33	0 18	23 27	0 18
58	302	1 39	32	0 18	23 51	0 19
59	301	1 40	31	0 19	24 15	0 19
60	300	1 41	30	0 19	24 38	0 19

f. 218,v.

من جداول تعديل الزهرة المجموع الكلي بابعادها الثلثة

From the tables of the equation of Venus, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
61°	299°	1° 42′	decreasing 29′	0° 19′	25° 2′	0° 19′
62	298	1 43	28	0 20	25 25	0 20
63	297	1 44	27	0 20	25 48	0 20
64	296	1 45	26	0 21	26 11	0 21
65	295	1 46	25	0 21	26 34	0 21
66	294	1 47	24	0 21	26 57	0 22
67	293	1 48	23	0 22	27 20	0 22
68	292	1 49	22	0 22	27 43	0 23
69	291	1 50	21	0 23	28 6	0 24
70	290	1 51	20	0 23	28 29	0 24
71	289	1 51	19	0 24	28 52	0 25
72	288	1 52	18	0 24	29 14	0 26
73	287	1 53	16	0 24	29 37	0 26
74	286	1 54	15	0 25	29 59	0 27
75	285	1 55	14	0 25	30 21	0 28
76	284	1 55	13	0 25	30 43	0 28
77	283	1 56	12	0 26	31 5	0 29
78	282	1 56	11	0 26	31 27	0 30
79	281	1 57	10	0 27	31 49	0 30
80	280	1 57	9	0 27	32 11	0 31
81	279	1 58	8	0 28	32 33	0 31
82	278	1 58	7	0 28	32 55	0 31
83	277	1 58	6	0 29	33 17	0 32
84	276	1 59	5	0 30	33 38	0 32
85	275	1 59	4	0 30	34 0	0 32
86	274	1 59	3	0 31	34 21	0 33
87	273	1 59	2	0 32	34 42	0 33
88	272	1 59	1	0 32	35 3	0 34
89	271	1 59	increasing 1	0 33	35 24	0 34
90	270	1 59	2	0 34	35 44	0 35

f. 219,r.

من جداول تعديل الزهرة المجموع الكلي بابعادها الثلثة

From the tables of the equation of Venus, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
			increasing			
91°	269°	1° 59′	3′	0° 34′	36° 4′	0° 36′
92	268	1 59	4	0 35	36 24	0 36
93	267	1 59	5	0 35	36 44	0 37
94	266	1 59	5	0 36	37 4	0 37
95	265	1 59	6	0 36	37 24	0 38
96	264	1 59	7	0 37	37 43	0 38
97	263	1 59	8	0 37	38 2	0 39
98	262	1 58	9	0 38	38 21	0 40
99	261	1 58	10	0 38	38 40	0 40
100	260	1 58	11	0 39	38 59	0 41
101	259	1 58	12	0 39	39 17	0 42
102	258	1 57	13	0 40	39 35	0 43
103	257	1 57	14	0 40	39 53	0 43
104	256	1 56	15	0 41	40 11	0 44
105	255	1 56	16	0 42	40 29	0 45
106	254	1 55	17	0 43	40 46	0 45
107	253	1 55	18	0 43	41 3	0 46
108	252	1 54	19	0 44	41 20	0 47
109	251	1 54	20	0 45	41 37	0 47
110	250	1 53	21	0 45	41 53	0 48
111	249	1 53	22	0 46	42 9	0 49
112	248	1 52	23	0 47	42 24	0 50
113	247	1 51	24	0 48	42 39	0 51
114	246	1 50	25	0 49	42 53	0 52
115	245	1 50	26	0 50	43 7	0 52
116	244	1 49	27	0 51	43 21	0 53
117	243	1 48	28	0 52	43 35	0 54
118	242	1 47	29	0 53	43 48	0 55
119	241	1 46	30	0 54	44 1	0 56
120	240	1 45	31	0 55	44 13	0 57

f. 219,v.

من جداول تعديل الزهرة المجموع الكلي بابعادها الثلثة

From the tables of the equation of Venus, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
121°	239°	1° 44′	increasing 31′	0° 55′	44° 25′	0° 58′
122	238	1 43	32	0 57	44 36	0 59
123	237	1 42	33	0 58	44 47	1 0
124	236	1 41	33	0 59	44 57	1 1
125	235	1 40	34	1 0	45 6	1 2
126	234	1 38	35	1 1	45 15	1 4
127	233	1 37	36	1 1	45 21	1 5
128	232	1 36	37	1 2	45 27	1 6
129	231	1 35	38	1 3	45 33	1 8
130	230	1 33	39	1 4	45 39	1 9
131	229	1 32	39	1 5	45 45	1 10
132	228	1 31	40	1 6	45 50	1 11
133	227	1 29	40	1 7	45 54	1 12
134	226	1 28	41	1 9	45 57	1 13
135	225	1 27	42	1 10	45 59	1 15
136	224	1 25	42	1 11	45 59	1 16
137	223	1 23	43	1 12	45 58	1 17
138	222	1 22	44	1 13	45 57	1 19
139	221	1 20	44	1 15	45 55	1 20
140	220	1 19	45	1 17	45 51	1 21
141	219	1 17	46	1 19	45 46	1 23
142	218	1 15	46	1 21	45 39	1 24
143	217	1 14	47	1 23	45 31	1 25
144	216	1 12	48	1 25	45 21	1 27
145	215	1 10	48	1 27	45 9	1 28
146	214	1 9	49	1 28	44 55	1 30
147	213	1 7	50	1 29	44 39	1 32
148	212	1 5	50	1 31	44 21	1 34
149	211	1 3	51	1 33	44 1	1 36
150	210	1 1	52	1 35	43 39	1 38

f. 220,r.

تمام جداول تعديل الزهرة المجموع الكلي بابعادها الثلثة

End of the tables of the equation of Venus, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
151°	209°	0° 59′	increasing 52′	1° 35′	43° 15′	1° 40′
152	208	0 58	53	1 36	42 48	1 42
153	207	0 56	54	1 37	42 18	1 45
154	206	0 54	54	1 38	41 45	1 46
155	205	0 52	55	1 39	41 8	1 47
156	204	0 50	55	1 39	40 28	1 48
157	203	0 48	56	1 40	39 46	1 49
158	202	0 46	56	1 40	38 58	1 50
159	201	0 44	56	1 41	38 7	1 51
160	200	0 42	57	1 41	37 12	1 51
161	199	0 40	57	1 42	36 12	1 52
162	198	0 38	57	1 42	35 7	1 52
163	197	0 36	57	1 41	33 58	1 52
164	196	0 34	58	1 40	32 44	1 51
165	195	0 32	58	1 38	31 24	1 50
166	194	0 30	58	1 36	29 58	1 48
167	193	0 28	58	1 34	28 25	1 45
168	192	0 26	58	1 31	26 46	1 43
169	191	0 24	59	1 28	25 2	1 38
170	190	0 21	59	1 24	23 12	1 33
171	189	0 19	59	1 19	21 15	1 27
172	188	0 17	59	1 12	19 11	1 21
173	187	0 15	59	1 4	17 2	1 14
174	186	0 13	59	0 56	14 47	1 7
175	185	0 11	60	0 48	12 27	0 57
176	184	0 9	60	0 40	10 5	0 46
177	183	0 6	60	0 31	7 38	0 35
178	182	0 4	60	0 21	5 9	0 24
179	181	0 2	60	0 11	2 36	0 12
180	180	0 0	60	0 0	0 0	0 0

f. 220,v.

بعد عطارد الابعد لسنة اقضا لذي القرنين را كح مسير حاصة عطارد ليوم ج و

[Longitude] of the apogee of Mercury in the year 1191 of Dhū 'l-qarnayn 201° 28′; daily motion of the anomaly of Mercury 3° 6′.

ابتداء جداول تعديل عطارد المجموع الكلي بابعاده الثلثة

Beginning of the tables of the equation of Mercury, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.		decreasing			
1°	359°	0° 3′	60′	0° 2′	0° 17′	0° 1′
2	358	0 6	60	0 4	0 33	0 2
3	357	0 9	60	0 5	0 49	0 3
4	356	0 12	59	0 7	1 6	0 4
5	355	0 15	59	0 9	1 22	0 5
6	354	0 17	59	0 10	1 38	0 5
7	353	0 20	58	0 12	1 55	0 6
8	352	0 23	58	0 14	2 11	0 7
9	351	0 25	58	0 15	2 27	0 8
10	350	0 28	57	0 17	2 44	0 9
11	349	0 30	57	0 19	3 0	0 10
12	348	0 32	57	0 20	3 16	0 11
13	347	0 35	56	0 22	3 32	0 12
14	346	0 38	56	0 23	3 48	0 13
15	345	0 40	55	0 24	4 4	0 14
16	344	0 43	55	0 26	4 21	0 15
17	343	0 45	54	0 28	4 37	0 16
18	342	0 47	54	0 29	4 53	0 17
19	341	0 50	53	0 31	5 9	0 18
20	340	0 53	53	0 33	5 25	0 19
21	339	0 55	52	0 35	5 41	0 20
22	338	0 58	51	0 37	5 57	0 21
23	337	1 0	51	0 38	6 13	0 22
24	336	1 2	50	0 39	6 29	0 23
25	335	1 5	49	0 41	6 45	0 24
26	334	1 8	48	0 43	7 1	0 24
27	333	1 10	47	0 45	7 17	0 25
28	332	1 13	46	0 46	7 33	0 26
29	331	1 15	45	0 48	7 49	0 27
30	330	1 17	44	0 49	8 5	0 28

f. 221,r.

من جداول تعديل عطارد المجموع الكلي بابعاده الثلثة

From the tables of the equation of Mercury, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
31°	329°	1° 20′	decreasing 43′	0° 51′	8° 20′	0° 29′
32	328	1 23	42	0 53	8 35	0 30
33	327	1 25	41	0 54	8 50	0 31
34	326	1 28	40	0 56	9 5	0 32
35	325	1 31	39	0 58	9 20	0 33
36	324	1 33	38	0 59	9 36	0 34
37	323	1 36	36	1 1	9 51	0 35
38	322	1 38	35	1 2	10 6	0 36
39	321	1 40	34	1 3	10 21	0 37
40	320	1 43	33	1 5	10 36	0 38
41	319	1 45	32	1 7	10 51	0 39
42	318	1 47	31	1 8	11 6	0 40
43	317	1 50	29	1 10	11 20	0 41
44	316	1 52	28	1 12	11 35	0 42
45	315	1 54	27	1 13	11 49	0 43
46	314	1 57	25	1 15	12 4	0 44
47	313	1 59	24	1 17	12 18	0 45
48	312	2 1	23	1 18	12 33	0 45
49	311	2 4	21	1 20	12 48	0 46
50	310	2 6	20	1 22	13 2	0 47
51	309	2 8	19	1 23	13 16	0 48
52	308	2 10	17	1 25	13 30	0 49
53	307	2 12	16	1 27	13 44	0 49
54	306	2 14	15	1 28	13 58	0 50
55	305	2 16	13	1 30	14 12	0 51
56	304	2 18	12	1 32	14 26	0 52
57	303	2 19	11	1 34	14 39	0 53
58	302	2 21	9	1 36	14 52	0 54
59	301	2 23	8	1 38	15 5	0 55
60	300	2 25	7	1 39	15 18	0 56

f. 221,v.

من جداول تعديل عطارد المجموع الكلي بابعاده الثلثة

From the tables of the equation of Mercury, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
61°	299°	2° 27′	decreasing 5′	1° 41′	15° 31′	0° 57′
62	298	2 29	4	1 42	15 44	0 59
63	297	2 31	2	1 44	15 56	1 0
64	296	2 33	1	1 46	16 9	1 1
65	295	2 35	increasing 1	1 48	16 21	1 2
66	294	2 36	2	1 49	16 33	1 4
67	293	2 38	4	1 51	16 45	1 5
68	292	2 40	6	1 53	16 57	1 6
69	291	2 41	8	1 54	17 9	1 7
70	290	2 43	10	1 56	17 21	1 8
71	289	2 44	12	1 58	17 32	1 9
72	288	2 45	15	1 59	17 43	1 11
73	287	2 47	16	2 1	17 54	1 12
74	286	2 48	18	2 3	18 5	1 13
75	285	2 49	20	2 4	18 16	1 14
76	284	2 50	22	2 6	18 27	1 15
77	283	2 51	24	2 8	18 37	1 16
78	282	2 52	25	2 9	18 47	1 17
79	281	2 53	26	2 11	18 57	1 18
80	280	2 54	29	2 13	19 7	1 19
81	279	2 55	30	2 14	19 16	1 20
82	278	2 56	32	2 16	19 26	1 21
83	277	2 57	34	2 18	19 35	1 22
84	276	2 57	35	2 19	19 44	1 23
85	275	2 58	37	2 21	19 53	1 24
86	274	2 59	38	2 23	20 2	1 25
87	273	2 59	40	2 24	20 10	1 26
88	272	3 0	41	2 26	20 18	1 27
89	271	3 0	43	2 28	20 26	1 28
90	270	3 1	44	2 29	20 33	1 29

f. 222,r.

من جداول تعديل عطارد المجموع الكلي بابعاده الثلثة

From the tables of the equation of Mercury, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
			increasing			
91°	269°	3° 1′	46′	2° 31′	20° 40′	1° 30′
92	268	3 1	47	2 33	20 47	1 31
93	267	3 2	48	2 34	20 54	1 32
94	266	3 2	49	2 36	21 1	1 33
95	265	3 2	50	2 38	21 7	1 34
96	264	3 2	50	2 39	21 13	1 35
97	263	3 2	51	2 41	21 19	1 36
98	262	3 1	52	2 43	21 24	1 37
99	261	3 1	53	2 44	21 29	1 38
100	260	3 1	54	2 46	21 34	1 39
101	259	3 0	55	2 48	21 38	1 40
102	258	3 0	56	2 49	21 42	1 41
103	257	2 59	56	2 50	21 46	1 42
104	256	2 59	57	2 52	21 49	1 43
105	255	2 58	57	2 53	21 52	1 44
106	254	2 58	58	2 55	21 55	1 45
107	253	2 57	58	2 57	21 57	1 45
108	252	2 56	59	2 58	21 59	1 46
109	251	2 55	59	3 0	22 0	1 47
110	250	2 54	59	3 1	22 1	1 48
111	249	2 53	59	3 2	22 2	1 49
112	248	2 52	59	3 2	22 2	1 50
113	247	2 51	60	3 3	22 1	1 51
114	246	2 50	60	3 4	22 0	1 52
115	245	2 49	60	3 5	21 59	1 53
116	244	2 48	60	3 5	21 58	1 54
117	243	2 46	60	3 6	21 56	1 55
118	242	2 45	60	3 6	21 53	1 56
119	241	2 43	60	3 7	21 50	1 56
120	240	2 41	60	3 8	21 47	1 57

f. 222,v.

من جداول تعديل عطارد المجموع الكلي بابعاده الثلثة

From the tables of the equation of Mercury, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
121°	239°	2° 39′	decreasing 60′	3° 8′	21° 43′	1° 58′
122	238	2 37	60	3 9	21 38	1 58
123	237	2 35	60	3 9	21 33	1 59
124	236	2 34	59	3 9	21 27	1 59
125	235	2 32	59	3 10	21 21	2 0
126	234	2 30	59	3 10	21 15	2 0
127	233	2 28	58	3 11	21 8	2 0
128	232	2 26	58	3 11	21 1	2 0
129	231	2 24	58	3 12	20 53	2 0
130	230	2 22	57	3 12	20 44	2 1
131	229	2 20	57	3 12	20 35	2 1
132	228	2 18	57	3 12	20 25	2 1
133	227	2 16	56	3 11	20 14	2 1
134	226	2 14	56	3 10	20 2	2 1
135	225	2 11	56	3 9	19 50	2 1
136	224	2 9	55	3 8	19 37	2 1
137	223	2 7	55	3 7	19 24	2 1
138	222	2 5	55	3 6	19 10	2 1
139	221	2 2	54	3 5	18 55	2 1
140	220	2 0	54	3 4	18 40	2 0
141	219	1 57	53	3 2	18 24	2 0
142	218	1 55	53	3 1	18 7	1 59
143	217	1 52	52	2 59	17 50	1 58
144	216	1 49	51	2 57	17 32	1 58
145	215	1 47	51	2 55	17 14	1 57
146	214	1 44	50	2 53	16 55	1 55
147	213	1 41	50	2 51	16 35	1 53
148	212	1 38	49	2 48	16 14	1 51
149	211	1 34	49	2 45	15 53	1 49
150	210	1 32	48	2 42	15 31	1 47

f. 223,r.

تمام جداول تعديل عطارد المجموع الكلي بابعاده الثلثة

End of the tables of the equation of Mercury, always compounded with its three distances.

Columns of numbers.		III. Equation of the anomaly and the centre.	IV. Shares of the two distances.	V. Farthest distance.	VI. Mean distance; equation of the anomaly.	VII. Nearest distance.
I.	II.					
151°	209°	1° 30′	decreasing 48′	2° 39′	15° 10′	1° 45′
152	208	1 27	47	2 36	14 45	1 43
153	207	1 24	47	2 32	14 20	1 41
154	206	1 21	46	2 29	13 55	1 39
155	205	1 18	46	2 25	13 29	1 37
156	204	1 15	45	2 21	13 3	1 35
157	203	1 12	45	2 17	12 36	1 32
158	202	1 9	44	2 13	12 9	1 29
159	201	1 6	43	2 9	11 41	1 26
160	200	1 3	43	2 5	11 12	1 23
161	199	1 0	43	2 0	10 43	1 20
162	198	0 57	42	1 55	10 13	1 17
163	197	0 54	42	1 50	9 43	1 14
164	196	0 51	42	1 44	9 12	1 11
165	195	0 48	41	1 38	8 40	1 7
166	194	0 45	41	1 32	8 7	1 4
167	193	0 42	41	1 26	7 34	1 0
168	192	0 39	41	1 19	7 1	0 56
169	191	0 35	40	1 13	6 27	0 52
170	190	0 32	40	1 7	5 53	0 47
171	189	0 28	40	1 1	5 19	0 42
172	188	0 25	40	0 55	4 45	0 38
173	187	0 22	40	0 49	4 10	0 33
174	186	0 19	40	0 42	3 35	0 28
175	185	0 16	40	0 35	3 0	0 24
176	184	0 13	40	0 28	2 24	0 19
177	183	0 9	40	0 21	1 48	0 14
178	182	0 6	40	0 14	1 12	0 10
179	181	0 3	40	0 7	0 36	0 5
180	180	0 0	40	0 0	0 0	0 0

f. 223,v.

جداول مقامات الكواكب الثلثة العلوية لمعرفة الرجوع والاستقامة
Tables of the stations of the three upper planets, for finding their retrogradations and their direct motions.

سطري العدد المتفاضلة و و Columns of numbers, at intervals of six degrees.		لزحل Saturn.		للمشتري Jupiter.		للمريخ Mars.	
		المقام الاول First station.	المقام الثاني Second station.	First station.	Second station.	First station.	Second station.
6°	354°	112° 45′	247° 15′	124° 5′	235° 55′	157° 33′	202° 27′
12	348	112 46	247 14	124 5	235 55	157 37	202 23
18	342	112 49	247 11	124 8	235 52	157 47	202 13
24	336	112 52	247 8	124 12	235 48	157 57	202 3
30	330	112 56	247 4	124 17	235 43	158 15	201 45
36	324	113 1	246 59	124 22	235 38	158 33	201 27
42	318	113 7	246 53	124 28	235 32	158 57	201 3
48	312	113 14	246 46	124 34	235 26	159 21	200 39
54	306	113 21	246 39	124 43	235 17	159 51	200 9
60	300	113 28	246 32	124 52	235 8	160 21	199 39
66	294	113 36	246 24	125 2	234 58	160 56	199 4
72	288	113 44	246 16	125 12	234 48	161 31	198 29
78	282	113 53	246 7	125 22	234 38	162 9	197 51
84	276	114 2	245 58	125 32	234 28	162 47	197 13
90	270	114 11	245 49	125 42	234 18	163 22	196 38
96	264	114 19	245 41	125 50	234 10	163 57	196 3
102	258	114 27	245 33	125 58	234 2	164 31	195 29
108	252	114 35	245 25	126 6	233 54	165 5	194 55
114	246	114 43	245 17	126 14	233 46	165 39	194 21
120	240	114 50	245 10	126 24	233 36	166 11	193 49
126	234	114 57	245 3	126 34	233 26	166 41	193 19
132	228	115 3	244 57	126 42	233 18	167 11	192 49
138	222	115 9	244 51	126 48	233 12	167 37	192 23
144	216	115 14	244 46	126 54	233 6	168 1	191 59
150	210	115 19	244 41	127 0	233 0	168 21	191 39
156	204	115 23	244 37	127 4	232 56	168 41	191 19
162	198	115 26	244 34	127 8	232 52	168 53	191 7
168	192	115 28	244 32	127 11	232 49	169 5	190 55
174	186	115 29	244 31	127 11	232 49	169 10	190 50
180	180	115 29	244 31	127 11	232 49	169 14	190 46

f. 224,r.

جداول مقامات الكوكبين السفليين لمعرفة الرجوع والاستقامة
Tables of the stations of the two lower planets, for finding their retrogradations and their direct motions.

سطري العدد متفاضلة و و Columns of numbers, at intervals of six degrees.		للزهرة Venus.		لعطارد Mercury.	
		First station.	Second station.	First station.	Second station.
6°	354°	165° 53′	194° 7′	147° 13′	212° 47′
12	348	165 54	194 6	147 8	212 52
18	342	165 56	194 4	147 0	213 0
24	336	165 58	194 2	146 49	213 11
30	330	166 2	193 58	146 36	213 24
36	324	166 6	193 54	146 21	213 39
42	318	166 11	193 49	146 6	213 54
48	312	166 17	193 43	145 50	214 10
54	306	166 23	193 37	145 34	214 26
60	300	166 30	193 30	145 18	214 42
66	294	166 38	193 22	145 5	214 55
72	288	166 46	193 14	144 55	215 5
78	282	166 54	193 6	144 49	215 11
84	276	167 2	192 58	144 43	215 17
90	270	167 10	192 50	144 38	215 22
96	264	167 17	192 43	144 34	215 26
102	258	167 24	192 36	144 32	215 28
108	252	167 32	192 28	144 30	215 30
114	246	167 40	192 20	144 30	215 30
120	240	167 46	192 14	144 29	215 31
126	234	167 51	192 9	144 29	215 31
132	228	167 56	192 4	144 30	215 30
138	222	168 2	191 58	144 32	215 28
144	216	168 6	191 54	144 34	215 26
150	210	168 10	191 50	144 36	215 24
156	204	168 14	191 46	144 37	215 23
162	198	168 17	191 43	144 38	215 22
168	192	168 19	191 41	144 39	215 21
174	186	168 20	191 40	144 40	215 20
180	180	168 21	191 39	144 40	215 20

f. 224,v.

جدول عروض الثلثة الكواكب العلوية
Table of the latitudes of the three upper planets.

سطري العدد متفاضلة و و Columns of numbers, at intervals of six degrees.		عرض زحل يزاد على مركزه ن خمسون Latitude of Saturn, to whose centre 50° are to be added.		عرض المشتري ينقص من مركزه ك عشرون Latitude of Jupiter, from whose centre 20° are to be subtracted.		عرض المريخ لا ينقص من مركزه ولا يزاد عليه Latitude of Mars, to whose centre nothing is to be added or subtracted.	
		شمال North.	جنوب South.	North.	South.	North.	South.
6°	354°	2° 4′	2° 2′	1° 7′	1° 5′	0° 8′	0° 4′
12	348	2 5	2 3	1 8	1 6	0 9	0 4
18	342	2 6	2 3	1 8	1 6	0 11	0 5
24	336	2 7	2 4	1 9	1 7	0 13	0 6
30	330	2 8	2 5	1 10	1 8	0 14	0 7
36	324	2 10	2 7	1 11	1 9	0 15	0 9
42	318	2 11	2 8	1 12	1 10	0 18	0 12
48	312	2 12	2 10	1 13	1 11	0 21	0 15
54	306	2 14	2 12	1 14	1 13	0 24	0 18
60	300	2 16	2 15	1 16	1 16	0 28	0 22
66	294	2 18	2 18	1 18	1 18	0 32	0 26
72	288	2 21	2 21	1 21	1 21	0 36	0 30
78	282	2 24	2 24	1 24	1 24	0 41	0 36
84	276	2 27	2 27	1 27	1 27	0 46	0 42
90	270	2 30	2 30	1 30	1 30	0 52	0 49
96	264	2 33	2 33	1 33	1 33	0 59	0 56
102	258	2 36	2 36	1 36	1 36	1 6	1 4
108	252	2 39	2 39	1 39	1 39	1 14	1 13
114	246	2 42	2 42	1 42	1 42	1 23	1 24
120	240	2 45	2 45	1 45	1 45	1 34	1 37
126	234	2 47	2 48	1 47	1 48	1 48	1 51
132	228	2 50	2 51	1 50	1 51	2 1	2 10
138	222	2 53	2 54	1 52	1 54	2 16	2 32
144	216	2 55	2 56	1 55	1 57	2 34	2 56
150	210	2 57	2 58	1 58	2 0	2 55	3 29
156	204	2 59	3 0	2 0	2 3	3 16	4 9
162	198	3 0	3 2	2 2	2 5	3 38	4 55
168	192	3 1	3 3	2 3	2 6	4 0	5 52
174	186	3 2	3 4	2 4	2 7	4 14	6 36
180	180	3 2	3 5	2 4	2 8	4 21	7 7

« ἀπόγειον is the north, περίγειον the south ». افيثجون هو الشمال فرنجيون هو الجنوب

f. 225,r.

جدول عروض الكوكبين السفليين
Table of the latitudes of the two lower planets.

سطري العدد متفاضلة و و Columns of numbers, at intervals of six degrees.		عرض الزهرة Latitude of Venus.		عرض عطارد Latitude of Mercury,		حصص العروض لجميع الكواكب Shares of the latitudes, common to all the planets.
		الميل Inclination.	الانحراف Obliquity.	Inclination.	Obliquity.	
6°	354°	1° 2′	0° 8′	1° 45′	0° 11′	59° 36′
12	348	1 1	0 16	1 44	0 22	58 36
18	342	1 0	0 25	1 43	0 33	57 0
24	336	0 59	0 33	1 40	0 44	54 36
30	330	0 57	0 41	1 36	0 55	52 0
36	324	0 55	0 49	1 30	1 6	48 24
42	318	0 51	0 57	1 23	1 16	44 24
48	312	0 46	1 5	1 16	1 26	40 0
54	306	0 41	1 13	1 8	1 35	35 12
60	300	0 35	1 20	0 59	1 44	30 0
66	294	0 29	1 28	0 49	1 52	24 24
72	288	0 23	1 35	0 38	2 0	18 24
78	282	0 16	1 42	0 26	2 7	12 24
84	276	0 8	1 50	0 16	2 14	6 24
90	270	0 0	1 57	0 0	2 20	0 0
96	264	0 10	2 3	0 15	2 25	6 24
102	258	0 20	2 9	0 31	2 28	12 24
108	252	0 32	2 15	0 48	2 29	18 24
114	246	0 44	2 20	1 6	2 30	24 24
120	240	0 59	2 24	1 25	2 29	30 0
126	234	1 18	2 27	1 45	2 26	35 12
132	228	1 38	2 30	2 6	2 20	40 0
138	222	1 59	2 30	2 27	2 11	44 24
144	216	2 23	2 28	2 47	2 0	48 24
150	210	3 3	2 22	3 7	1 46	52 0
156	204	3 44	2 12	3 26	1 29	54 36
162	198	4 26	1 55	3 42	1 10	57 0
168	192	5 13	1 27	3 54	0 48	58 36
174	186	5 52	0 48	4 2	0 24	59 36
180	180	6 22	0 0	4 5	0 0	60 0

f. 225,v.

جدول ابعاد الكواكب الثلثة العلوية للرؤية والاختفاء في الشرق والغرب لعرض لو

Table of the elongations of the three upper planets, for finding their appearances and disappearances in the east and in the west, for the latitude 36° 0′.

اسماء البروج Names of the signs of the zodiac.	رؤية زحل Appearance of Saturn.	اختفاء زحل Disappearance of Saturn.	رؤية المشتري Appearance of Jupiter.	اختفاء المشتري Disappearance of Jupiter.	رؤية المريخ Appearance of Mars.	اختفاء المريخ Disappearance of Mars.
	التشريق Morning rising.	التغريب Evening setting.	Morning rising.	Evening setting.	Morning rising.	Evening setting.
Aries.	29° 28′	13° 46′	19° 13′	9° 28′	29° 0′	14° 37′
Taurus.	27 26	14 3	18 21	9 38	27 51	15 8
Gemini.	22 10	15 5	14 55	10 16	22 54	16 7
Cancer.	17 18	17 9	11 41	11 44	18 55	18 15
Leo.	14 8	19 48	9 44	13 32	16 7	22 54
Virgo.	13 8	22 0	9 7	15 23	15 8	27 51
Libra.	12 55	22 32	9 0	16 7	14 52	29 0
Scorpio.	13 1	21 20	9 7	15 23	15 8	27 51
Sagittarius.	13 47	18 35	9 44	13 32	16 4	22 54
Capricorn.	16 36	16 6	11 44	11 44	18 35	18 15
Aquarius.	21 16	14 40	14 54	10 16	22 34	16 7
Pisces.	26 46	14 0	18 11	9 38	27 10	15 8

f. 226,r.

جدول ابعاد الكوكبين السفليين للروية والاختفاء في الشرق والغرب لعرض لو

Table of the elongations of the two lower planets, for finding their appearances and disappearances in the east and in the west, for the latitude 36° 0′.

اسماء البروج Names of the signs of the zodiac.	رؤية الزسرة Appearance of Venus.		اختفاء الزهرة Disappearance of Venus.		رؤية عطارد Appearance of Mercury.		اختفاء عطارد Disappearance of Mercury.	
	طلوع العشيات من ♉ الى قلز **Evening rising from 0° to 137°.**	غروب الغدوات من رجح الى سص **Morning setting from 223° to 360°.**	طلوع الغدوات من قف الى رجح **Morning rising from 180° to 223°.**	غروب العشيات من قلز الى قف **Evening setting from 137° to 180°.**	طلوع العشيات من ♉ الى قيب **Evening rising from 0° to 112°.**	غروب الغدوات من رمح الى سص **Morning setting from 248° to 360°.**	طلوع الغدوات من قف الى رمح **Morning rising from 180° to 248°.**	غروب العشيات من قيب الى قف **Evening setting from 112° to 180°.**
Aries.	15° 0′	7° 25′	3° 36′	7° 27′	24° 10′	12° 20′	22° 43′	11° 19′
Taurus.	13 48	7 25	4 9	8 3	21 15	12 8	25 23	12 12
Gemini.	10 59	7 38	5 34	8 47	17 10	12 36	22 28	14 44
Cancer.	8 28	8 38	10 42	10 44	14 9	14 9	18 48	19 48
Leo.	7 5	9 39	17 25	9 30	12 33	16 39	15 8	23 29
Virgo.	6 13	10 46	23 17	7 43	12 8	20 23	13 15	26 36
Libra.	6 7	11 19	22 27	6 40	12 10	23 7	12 0	25 38
Scorpio.	6 31	11 26	15 14	6 17	12 41	23 49	11 35	20 35
Sagittarius.	7 26	10 47	8 1	6 12	14 3	20 44	11 56	16 41
Capricorn.	9 38	9 38	5 18	6 18	16 59	16 59	12 36	12 30
Aquarius.	12 47	8 29	3 59	6 25	21 35	14 7	14 25	11 32
Pisces.	14 28	7 43	3 43	6 51	24 38	12 54	18 22	11 7

f. 239,v.

جدول زيادات الكواكب التي يزاد اوساطها في الاصل للتحاويل

Table of the increments by which the mean positions of the planets at the epoch are to be increased for [computing] the revolutions [of the years].

Number of complete Roman years.	زيادة وسط القمر Increment of the mean position of the moon.	زيادة حاصة القمر Increment of the anomaly of the moon.	زيادة العقد الشمالي Increment of the northern node.	زيادة وسط زحل ونقصان حاصته Increment of the mean position and decrement of the anomaly of Saturn.	زيادة وسط المشتري ونقصان حاصته Increment of the mean position and decrement of the anomaly of Jupiter.	زيادة وسط المريخ ونقصان حاصته Increment of the mean position and decrement of the anomaly of Mars.	زيادة حاصة الزهرة Increment of the anomaly of Venus.	زيادة حاصة عطارد Increment of the anomaly of Mercury.
1	132° 33′	91° 52′	19° 20′	12° 14′	30° 22′	191° 25′	225° 11′	54° 42′
2	265 7	183 43	38 41	24 28	60 44	22 50	90 22	109 23
3	37 40	275 35	58 1	36 42	91 6	214 15	315 33	164 4
4	170 13	7 27	77 22	48 57	121 27	45 40	180 43	218 45
5	302 47	99 19	96 42	61 11	151 49	237 5	45 54	273 27
6	75 20	191 10	116 3	73 25	182 11	68 30	271 5	327 8
7	207 53	283 2	135 23	85 39	212 33	259 55	136 16	22 50
8	340 27	14 54	154 44	97 53	242 55	91 20	1 27	77 32
9	113 1	106 45	174 4	110 7	273 17	282 45	226 38	132 13
10	245 34	198 37	193 25	122 21	303 38	114 9	91 48	186 54
11	18 7	290 29	212 45	134 35	334 0	305 33	316 59	241 35
12	150 40	22 20	232 6	146 50	4 22	136 57	182 10	296 16

f. 240,r.

جدول تحاويل السنين الرومية
Table of the revolutions of the [tropical] Roman years.

السنون المجموعة للتحاويل Collected years for the revolutions.	ازمان المطالع Times of the ascensions.	السنون البسوطة للتحاويل Single years for the revolutions.	ازمان المطالع Times of the ascensions.
20	292° 0′	1	86° 36′
		2	173 12
40	224 0	3	259 48
		4	346 24
60	156 0	5	73 0
		6	159 36
80	88 0	7	246 12
		8	332 48
100	20 0	9	59 24
		10	146 0
120	312 0	11	232 36
		12	319 12
140	244 0	13	45 48
		14	132 24
160	176 0	15	219 0
		16	305 36
180	108 0	17	32 12
		18	118 48
200	40 0	19	205 24
		20	292 0

جداول الساعات
Tables of the hours.

الساعات Hours.	الارتفاع Altitude. ساعات الجدي Hours of Capricorn.	الارتفاع Altitude. ساعات السرطان Hours of Cancer.
1	8° 3′	13° 17′
2	15 18	27 1
3	21 28	41 56
4	26 18	56 29
5	29 23	70 3
6	30 25	77 35
7	29 23	70 3
8	26 18	56 29
9	21 28	41 56
10	15 18	27 1
11	8 3	13 17
12	0 0	0 0

جدول سموت الساعات
Table of the directions of the hours.

عدد الساعات Number of the hours.	ساعات راس السرطان Hours of the beginning of Cancer. سمت الساعات Direction of the hours.	ساعات راس الجدي Hours of the beginning of Capricorn. سمت الساعات Direction of the hours.
1	19° 40′ 0″ [north]	37° 2′ 30″ [south]
2	10 23 40 [ditto]	45 22 20 [ditto]
3	0 41 13 [ditto]	54 56 30 [ditto]
4	11 40 44 [south]	65 37 30 [ditto]
5	33 29 30 [ditto]	77 27 30 [ditto]
6	90 0 0 [ditto]	90 0 0 [ditto]